



Towards distinguishing cybersecurity attacks and safety faults in distributed energy resources-rich smart grids: a systematic literature review

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Abstract

This study presents a Systematic Literature Review (SLR) of methods and challenges related to distinguishing cybersecurity attacks and safety faults in Distributed Energy Resources (DER)-rich smart grids. The increasing integration of DERs has improved grid flexibility and sustainability, but has also introduced greater operational complexity and expanded the attack surface of smart grid infrastructures. In such environments, cyber-attacks and safety faults may produce similar anomalies, making accurate distinction a critical requirement for resilient and secure grid operation. This review examines the literature on anomaly detection and event classification in DER-rich smart grids towards distinguishing these anomalies and events, with a particular focus on approaches that could support the differentiation of malicious and non-malicious events. It analyzes the main categories of data used in the literature, including sensor data, smart meter data, network traffic, phasor measurements, and weather-related information. It also reviews machine learning and artificial intelligence techniques, including supervised, unsupervised, and deep learning approaches, and discusses their applicability, strengths, and limitations within different contexts. Furthermore, the review synthesizes evaluation practices, operational application domains, and key open challenges, including limited dataset realism, data quality issues, explainability, scalability, and model generalization. Existing testbeds are highlighted as essential for reproducing realistic grid conditions and validating approaches; however, in their current state, no testbed framework is directly applicable to distinguishing cybersecurity attacks from operational safety faults in an informed manner. Importantly, only 6 of the 25 included primary studies explicitly attempt to distinguish cyber-attacks from physical safety faults within a single evaluation setting. The review is therefore explicitly positioned as a combined systematic mapping and gap analysis: rather than demonstrating a mature solution space, it documents how sparsely the central distinction problem is addressed, characterizes the few existing attempts in depth, and derives a research agenda from this gap. Overall, the review identifies major research gaps and outlines directions for developing more safe and secure solutions for DER-rich smart grids.

Keywords Systematic literature review · Anomaly detection · DER-rich smart grids · Cybersecurity attacks · Safety faults · Machine learning · Transparency

1 Introduction

1.1 Background and context: integration and impact of DERs in smart grids

The integration of Distributed Energy Resources (DERs) into modern Smart Grids (SGs) has been comprehensively reviewed by Ohanu, Rufai, and Oluchi [1], highlighting the transformative impact of renewable energy. The study

addresses both the potential and challenges of renewable integration, including the intermittent nature of solar power and its spatial demands. Despite these hurdles, solar energy significantly cuts greenhouse gas emissions and improves the efficiency of systems involving renewable energy sources and electric vehicles. The study emphasizes the need for advanced energy storage and robust communication systems to ensure the sustainable, efficient operation of future SGs, which are pivotal for enhanced energy management and reduced power losses.

Extended author information available on the last page of the article

This integration of DERs into the power grid brings multiple benefits, significantly enhancing the overall system performance:

- Efficiency: DERs improve the efficiency of the energy grid by reducing energy wastage and optimizing the distribution and generation of electricity closer to the point of use [2].
- Flexibility: They potentially increase grid flexibility when managed well, allowing for more dynamic responses to changes in electricity demand and supply, thereby stabilizing the grid during peak loads or generation fluctuations [3].
- Sustainability: DERs contribute to sustainability by integrating renewable energy sources like solar and wind, reducing reliance on fossil fuels and lowering carbon emissions [4].

These advantages are crucial for developing a resilient, adaptive, and environmentally friendly energy system. However, the integration of DERs introduces several challenges, particularly system level challenges. According to recent research [5], it exist challenges that include a broader integration of DERs demands advanced regulatory frameworks and technological upgrades to address issues of integration, operational coordination, and economic impacts on traditional power systems.

Particularly such integration impacts grid security and reliability:

- Grid Security: As outlined in Qi et al. [6], the increased complexity and decentralized nature of DERs enhance vulnerability to both cyber and physical threats, making grid security a significant concern.
- Grid Reliability: Study [7] highlights how the intermittency of renewable sources in DER systems can lead to instability, affecting the reliability of power delivery and grid operations.

These references collectively emphasize the need for robust solutions and innovative management strategies to tackle the challenges posed by DER integration effectively.

1.2 Problem statement: challenges in anomaly detection for DER-rich environments

The integration of Distributed Energy Resources (DERs) into Smart Grids has significantly increased the complexity of grid operations, making them more susceptible to various anomalies. Such anomalies can be due to safety faults for instance faults happening at the protection systems in the physical layer of architecture. Anomalies can also arise from

cybersecurity attacks or they can be attributed to interoperability issues. As illustrated in Fig. 1, potential vulnerabilities span both the physical and cyber domains, highlighting the need for robust anomaly detection and mitigation.

Traditional anomaly detection methods, designed for simpler, centralized grids, are insufficient for these advanced systems due to their inability to handle the scale, variability, and data intensity inherent in DER-rich environments.

One critical vulnerability in SGs is the susceptibility to cybersecurity attacks such as False Data Injection (FDI) as illustrated by Lin et al. [8] where they investigate the false data injection attacks against the distributed energy routing process.

On the other hand, safety faults can occur due to random equipment failures or external factors affecting grid components. Pathan et al. [9] employed a non-sequential Monte Carlo simulation approach to model the random occurrence of faults in the system. This method accounts for customer interruptions and the corresponding repair times required to restore supply, highlighting the grid's vulnerability to unpredictable physical faults.

Interdependencies between the safe operation of physical protection systems and the safe operation of cyber systems are crucial. Failure of any of them can cause physical and data damage, combined with the use of tools and procedures like monitoring software and protection systems, blur the line between cybersecurity and safety responses, a challenge further compounded by the limitations of traditional anomaly detection methods in handling the complexity of DER-rich environments. Therefore the importance of distinguishing between these safety and cybersecurity anomalies is crucial to ensure grid stability and operational safety.

There exist many different types of anomalies that can be detected however some are harder to distinguish such as unexpected voltage drops [10], equipment misbehaviour [11], or unusual operational data patterns [12], can be caused either by physical faults or by cyber-attacks. Without further investigation, it may not be immediately clear whether the anomaly is a safety issue (e.g., equipment failure) or a cybersecurity breach (e.g., a malicious manipulation of control systems). Based on these examples, we can confirm that there exist limitations of traditional anomaly detection methods in handling the complexity of DER-rich environments.

Machine Learning (ML) and Artificial Intelligence (AI) have emerged as promising techniques for anomaly detection in SGs [13]. They leverage vast amounts of real-time and historical data to detect and classify anomalies [14]. However, existing methods often rely on framework-based approaches, such as bow-tie analysis [15], which may lack the robustness needed to accurately differentiate between cybersecurity incidents and safety faults. Given these examples, we can confirm that there is a pressing need for advanced models and algorithms that can effectively handle

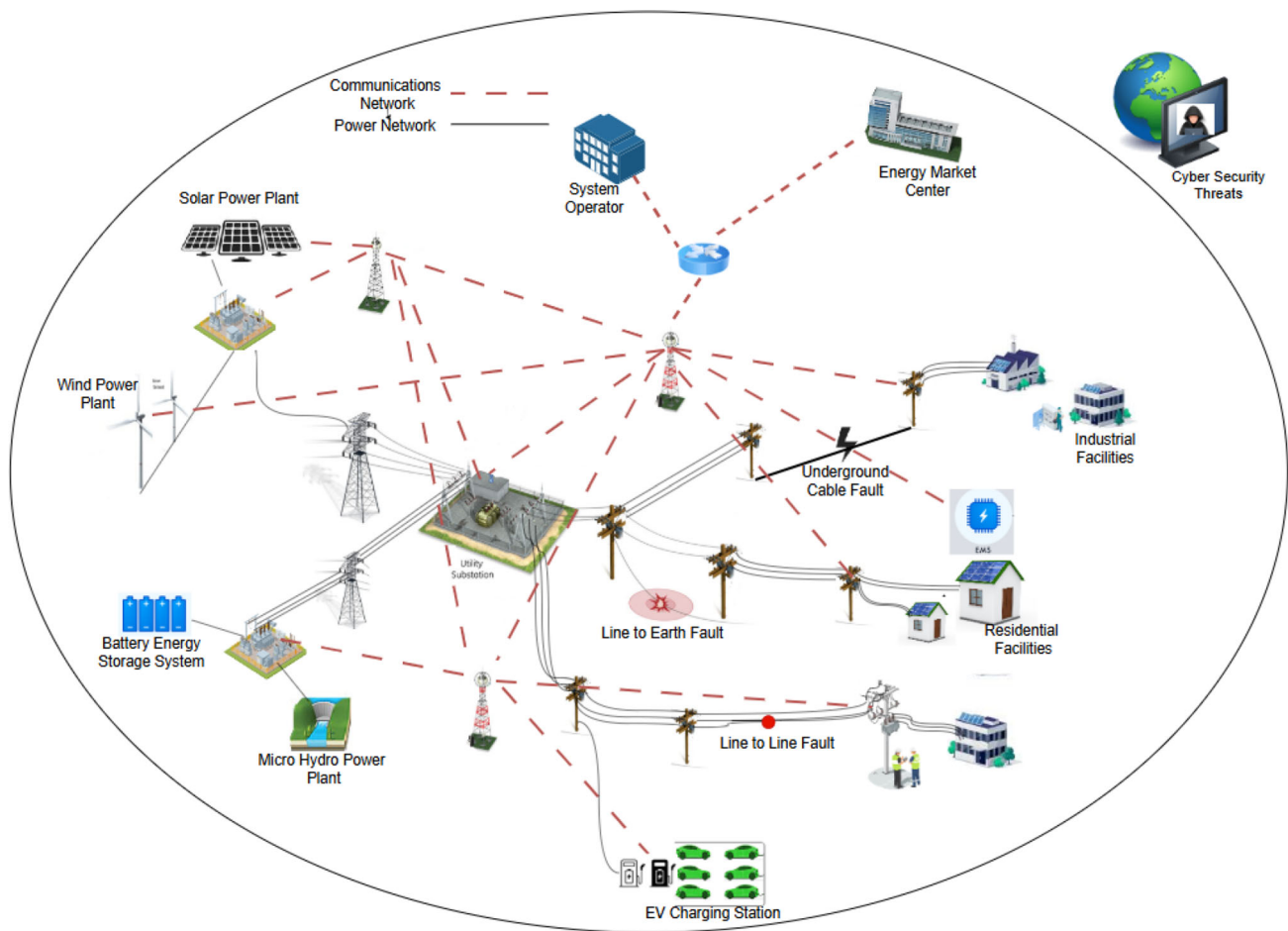


Fig. 1 Conceptual Diagram of Smart Grid Vulnerabilities and Cybersecurity Challenges. Illustrates potential security risks and fault scenarios in a DER-rich Smart Grid environment

the complexity of DER-rich environments and provide precise anomaly classification. Such solutions should provide precise anomaly classification, enabling grid operators to distinguish accurately between safety faults and cybersecurity attacks. Developing robust, intelligent anomaly detection methods is essential for maintaining grid stability, ensuring operational safety, and protecting against both physical and cyber threats in modern SGs.

1.3 Motivation for the study: need for advanced anomaly detection systems

The increasing complexity of SGs, driven by the integration of Distributed Energy Resources (DERs), advanced sensors, automation, and bidirectional energy flows, demands more sophisticated anomaly detection systems. Traditional detection methods, which were designed for simpler, centralized grids, are no longer sufficient to handle the scale, variability, and data intensity of modern SGs. As the grid evolves, so do the challenges of monitoring and ensuring its security and efficiency, necessitating the development of more advanced

systems capable of identifying both physical faults and cyber anomalies.

Various test environments have been established to validate Smart Grid technologies. Pacific Northwest National Laboratory's (PNNL) GridLAB-D [16] offers complex power system modeling capabilities, while the National Renewable Energy Laboratory's (NREL) Energy Systems Integration Facility (ESIF) [17] provides real-world testing of DERs and cybersecurity systems. The PowerCyber Testbed [18] at Iowa State University specializes in testing cyber-attacks and defense strategies, and the Real-Time Digital Simulator (RTDS) enables real-time digital simulation for hardware-in-the-loop testing. Despite these resources, a comprehensive test environment capable of distinguishing between cyber-attacks and physical faults in large-scale grid operations remains elusive. Replicating the dynamic interactions of physical and cyber systems at scale, while ensuring the safety of infrastructure during cybersecurity simulations, presents a significant challenge.

To address this gap, a structured review of existing research is needed to inform the development of an effective

testbed for anomaly detection in DER-rich SGs. By analyzing current methods and environments, this review will identify both the strengths and limitations of existing anomaly detection systems in complex, decentralized grids. It will explore challenges such as the integration of DERs, the interplay between physical and cyber threats, and the gaps in real-time detection capabilities. This foundation will guide the creation of a testbed that better reflects real-world conditions and enhances the resilience and effectiveness of anomaly detection in SGs.

Intended Audience and Multidisciplinary Scope.

Although the lead author's primary expertise resides in software engineering, data science, and cybersecurity/cyber risk engineering-not advanced Smart Grid or DER systems; this study results from close collaboration with co-authors who possess deep domain knowledge in energy infrastructure. Together, we aim to serve a broad range of professionals and researchers by addressing four key viewpoints:

- (1) **Software Engineering:** System integration, lifecycle methodologies (e.g., TDD, Agile, DevSecOps), and architectural considerations for anomaly detection frameworks.
- (2) **Data Science:** Design and evaluation of machine learning models, performance metrics, and data handling (both real-time and historical) tailored for grid operations.
- (3) **Cybersecurity/Cyber Risk Engineering:** Threat modeling, resilient architecture strategies, and risk mitigation in the face of modern cyber-attacks, including false data injection and Distributed Denial-of-Service (DDoS) scenarios.
- (4) **Smart Grid Professionals:** Targeting roles including Distribution System Operators (DSOs), Transmission System Operators (TSOs), grid operators, system integrators, and infrastructure managers. This includes addressing operational challenges and practical considerations informed by recent literature and best practices from relevant domain-specific journals and industry standards.

By combining these perspectives with domain-specific insights from Smart Grid and DER specialists, we aim to advance the reliability, security, and scalability of anomaly detection solutions in modern power systems.

1.4 Objectives of the study

To conduct a Systematic Literature Review (SLR) to evaluate the current state of anomaly detection in DER-rich SGs. The review aims to analyze existing approaches and technologies, identifying their effectiveness in detecting anomalies within decentralized grid environments, and providing a clear understanding of the advances made in this area.

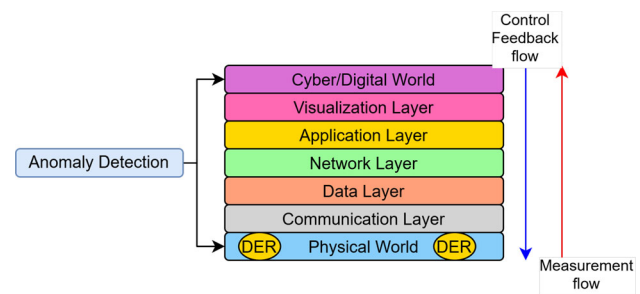


Fig. 2 Two-Layer Anomaly Detection Model in a DER-rich Smart Grid

The study will focus specifically on how anomaly detection systems can distinguish between cybersecurity attacks and physical safety faults. With the increasing complexity of SGs and the growing integration of DERs, it is crucial to differentiate between these two types of anomalies to ensure grid stability and security. By examining current research, the review will identify the strengths and limitations of existing solutions in addressing both cybersecurity threats and physical grid faults, such as presented in the Fig. 2 where anomaly Detection in DER-rich SGs shows that we emphasize the two key layers of the CPS, cyber digital world with the cybersecurity and the physical world with the physical operational fault among all the layers.

The Fig. 2 emphasizes the two layers that are most central to the review problem: the cyber layer, where control logic, communication, and data manipulation can be compromised, and the physical layer, where equipment faults, operational disturbances, and protection events are observed. By explicitly showing the interface between these two layers, the figure supports the paper's central argument that meaningful distinction between cybersecurity attacks and safety faults requires joint interpretation of cyber and physical evidence rather than isolated anomaly flags.

The final objective is to gather insights that will contribute to the development of a comprehensive test environment. This environment will serve as a platform for validating advanced anomaly detection systems, simulating real-world conditions in SGs, and helping to improve the resilience and accuracy of anomaly detection, particularly in distinguishing between cyber and physical faults.

Positioning as a gap analysis. It should be stated upfront that the body of primary research directly addressing the cyber-attack versus safety-fault distinction is small: as the results of this review show, only 6 of the 25 included studies attempt the distinction explicitly within a single evaluation setting. We deliberately retained the inclusion criteria registered in the review protocol rather than broadening them post hoc, because such broadening would have changed the research questions after data collection and weakened the validity of the review. Consequently, this study should be read as a combined systematic mapping and *gap analysis*: its

primary contributions are (i) a transparent map of the adjacent anomaly-detection literature in DER-rich smart grids, (ii) an in-depth characterization of the few studies that do attempt the distinction (Sect. 3.7), and (iii) a research agenda derived from the documented gap, rather than a synthesis of a mature solution space.

1.5 Distinguishing information and communications technology engineering versus electrical engineering perspectives

While this study addresses anomaly detection in DER-rich SGs at a high level, it is important to distinguish the roles and challenges that fall within Information and communications technology engineering versus those within electrical engineering. From the Information and communications technology engineering standpoint, tasks such as data ingestion, pipeline construction, data cleaning, feature engineering, and scalable storage architectures must be addressed to ensure ML models receive high-quality, real-time data. Meanwhile, from the electrical engineering perspective, core challenges involve designing grid infrastructure for stable integration of DERs, managing real-time power flows, understanding fault behaviors, and ensuring physical equipment reliability. These complementary domains must work in tandem to build robust anomaly detection solutions.

1.6 Research questions

To guide the literature review and investigation, the following research questions have been formulated:

- RQ1: What types of data (e.g., operational, environmental, system logs) are crucial for anomaly detection in DER-rich Smart Grids, and how are they sourced and processed?
- RQ2: Which specific ML models and AI techniques have shown promise in accurately detecting anomalies within DER-rich Smart Grid environments, and what are the reasons for their effectiveness?
- RQ3: In which operational contexts (e.g., real-time monitoring, predictive maintenance) within DER-rich Smart Grids are anomaly detection systems most commonly applied, and with what outcomes?
- RQ4: What benchmarks and performance metrics (e.g., detection accuracy, false positive rates) are most effective in evaluating the success of anomaly detection systems in Smart Grids?
- RQ5: What are the primary technical and operational challenges in deploying ML models and AI techniques for anomaly detection in Smart Grids, and what future opportunities exist for overcoming these hurdles?

- RQ6: What are the characteristics of effective testbeds for evaluating and validating anomaly detection systems in DER-rich Smart Grids, and how do these environments simulate real-world conditions?

2 Methodology

2.1 Systematic literature review (SLR)

The methodology for this study follows a systematic literature review (SLR) approach to ensure a comprehensive and structured evaluation of existing research. The SLR will be conducted in line with the guidelines outlined by Kitchenham and Charters (2007) [19]. The SLR process involves clearly defined steps for searching, screening, and synthesizing relevant studies to answer specific research questions. The review focuses on studies related to anomaly detection, cybersecurity, and machine learning in the context of industrial control systems (ICS), cyber-physical systems (CPS), and distributed energy resources (DER).

In addition, and in line with Kitchenham and Charters, the review process is structured around three major phases: review planning, review conduct, and review reporting. During the planning phase, the need for the review, the research questions, the search scope, and the eligibility criteria were defined. During the conduct phase, studies were retrieved, screened, assessed for relevance, and analyzed using a structured extraction form. During the reporting phase, the findings were synthesized narratively and organized according to the six analytical groups introduced later in this section.

To improve transparency and reproducibility, the review protocol was defined before the final synthesis stage. The protocol specified: (i) the scientific databases to be queried, (ii) the search terms and search logic, (iii) the inclusion and exclusion criteria, (iv) the study screening strategy, (v) the data extraction structure, and (vi) the synthesis approach.

The review was designed to be concept-centric rather than author-centric. Accordingly, the selected studies were not only catalogued bibliographically but also analyzed according to recurring methodological, operational, and evaluation themes. Because the primary studies are heterogeneous in terms of datasets, infrastructures, anomaly types, and evaluation metrics, this review adopts a qualitative and thematic narrative synthesis rather than a statistical meta-analysis.

2.2 Research questions and review scope

The scope of this SLR is limited to studies addressing anomaly detection in DER-rich smart grids and related industrial cyber-physical environments, with particular attention to the distinction between cybersecurity attacks and

safety faults. The review considers studies situated in or closely related to ICS, CPS, SCADA, smart grids, and DER-integrated energy systems.

The review is guided by six research questions, which are operationalized later through six extraction groups. These research questions aim to identify: the data used in anomaly detection, the AI/ML methods applied, the application domains and operational contexts, the evaluation strategies and metrics, the main challenges and opportunities, and the role of testbeds in validation.

Given that the screening results later show that only a minority of eligible studies (6 of 25) explicitly address the cyber-attack versus safety-fault distinction, the review scope is interpreted as a gap analysis with respect to this central objective: the six research questions map the surrounding evidence base, while the distinction question itself is answered primarily by characterizing how few and how limited the direct attempts are. The protocol-defined inclusion criteria were intentionally not broadened after data collection, in order to preserve the internal validity of the review; the implications of this choice are discussed in the threats to validity (Sect. 2.13).

2.3 Review protocol and governance

Following Kitchenham-style [19] SLR good practices, the review protocol was designed to reduce bias and support repeatability. The protocol included the search sources, eligibility criteria, screening decisions, and extraction dimensions. The protocol was iteratively refined during pilot searches in order to better align the search strings with the terminology used in the DER-rich smart grid and industrial cybersecurity literature. Such refinement is consistent with the iterative nature of search design in software engineering and interdisciplinary review practice, provided that the modifications are reported transparently.

A governance mechanism was also considered during the review process to improve consistency in study selection and extraction. Screening decisions were guided by the predefined criteria, and doubtful cases were examined in relation to the research questions and the study scope.

2.4 Search strategy

A comprehensive search strategy has been devised to locate relevant studies from high-quality scientific databases. The search strings were designed by combining keywords from several domains, including cybersecurity, anomaly detection, machine learning, and industrial systems. The databases targeted for this review include:

- IEEE Xplore
- ACM Digital Library

- Scopus
- Web of Science
- SpringerLink
- Wiley Online Library

The three iterative search strings, reported here verbatim from the review protocol, are as follows. They share the same six keyword blocks (security concern; anomaly type; analytical action; system/application context; AI/ML; evaluation/validation), and differ only in the Boolean operators that connect the anomaly-type terms and the system/application terms:

- **Search String 1 (broad):** (“Cybersecurity”OR“Cyber Security”OR“Cyber Incidents”) AND ((“Attacks”OR “Cyber-attacks”OR “Cyber Incidents”) OR (“Faults”OR “Failures”)) AND (“Distinguish”OR“Classify”OR “Detection”OR“Anomaly Detection”) AND ((“Industrial Control Systems”OR“ICS”OR“Cyber Physical Systems”OR“CPS”OR“SCADA”) OR (“Distributed Energy Resources”OR“DER”OR“Der-rich”OR“Smart-Grid”OR “DSO”OR“Distribution System Operator”)) AND (“Machine Learning”OR“ML”OR “Artificial Intelligence” OR“AI”) AND (“Testbeds”OR“Simulation”OR“Digital Twin”OR “Evaluation Metrics”OR“Performance Evaluation”OR “Explainable AI”OR“XAI”)
- **Search String 2 (system-focused refinement):** identical to Search String 1, except that the industrial-system block and the DER/smart-grid block are connected by **AND** instead of OR, i.e., ((“Industrial Control Systems”OR“ICS”OR“Cyber Physical Systems”OR “CPS”OR“SCADA”) **AND** (“Distributed Energy Resources”OR“DER”OR“Der-rich”OR“Smart-Grid”OR“DSO”OR “Distribution System Operator”)), so that retrieved studies must address both an industrial control/cyber-physical context and a DER or smart-grid context.
- **Search String 3 (distinction-focused refinement):** identical to Search String 2, except that the attack block and the fault block are additionally connected by **AND** instead of OR, i.e., ((“Attacks”OR“Cyber-attacks”OR “Cyber Incidents”) **AND** (“Faults”OR “Failures”)), so that retrieved studies must mention both attack-related and fault/failure-related terms, which is the closest lexical proxy for the attack–fault distinction objective of this review.

To further strengthen reproducibility, the search process was conducted iteratively. Broad initial strings were first used to identify the dominant terminology in the literature, after which narrower and more specialized strings were applied. This strategy was intended to balance recall and precision:

Search String 1 was designed to maximize coverage, while Search Strings 2 and 3 were intended to progressively refine the corpus toward studies more directly related to the distinction between cyber incidents and operational faults in DER-rich or industrial cyber-physical settings.

In Kitchenham-style reviews, the exact reporting of search expressions is important for replicability. Therefore, the master Boolean strings should be interpreted as canonical forms that may require small syntax adaptations depending on the database interface (e.g., quotation marks, wildcard support, field restrictions, or Boolean nesting rules).

Search Provenance and Per-Database Adaptation. The searches were executed on 26 February 2024. For each database and search string, the review protocol records the exact query URL, the applied content-type and date filters (2014–2024; journal articles, conference papers, and book chapters), and the number of returned records, so that each row of Table 1 can be traced back to a concrete executed query. For Scopus, an unrestricted all-fields variant of Search String 1 returned 7,058 records and was not exported; the title–abstract–keyword variant (108 records) was used instead and is the value reported in Table 1. Earlier pilot strings explored during protocol refinement (eight variants) are likewise documented in the protocol and were used only to calibrate terminology, not to retrieve the analyzed corpus.

The results of the searches are summarized in Table 1, which presents the number of records retrieved per scientific database for each search string, with columns indicating the Search String Number (Nb), the Scientific Database (Scientific Db), and the Total Number of Records (Records).

The search results from the different databases show substantial variation in the number of retrieved records. This variation is expected due to differences in database indexing policies, coverage, search syntax interpretation, and the degree of overlap between engineering, cybersecurity, and energy-system venues. Consequently, the search process did not rely on a single source but instead combined complementary databases to reduce source selection bias.

2.5 Study selection procedure

After the initial retrieval phase, the identified studies underwent a multi-stage screening process. First, records were screened based on title, keywords, and abstract to remove clearly irrelevant studies. Second, potentially relevant studies were assessed in more detail through full-text or near-full-text inspection when needed. Third, duplicate records and overlapping versions of the same work were removed before the final selection was confirmed.

The study selection process was guided primarily by the predefined inclusion and exclusion criteria. However, because the review topic is interdisciplinary and terminology is not always used consistently across energy, control,

Table 1 Scientific database search results

Search string no.	Scientific database	Total number of records
1	IEEE	362
1	ACM	282
1	SCOPUS	108
1	WebOfScience	275
1	Springer	579
1	Wiley	438
2	IEEE	72
2	ACM	76
2	SCOPUS	22
2	WebOfScience	50
2	Springer	235
2	Wiley	149
3	IEEE	17
3	ACM	40
3	SCOPUS	2
3	WebOfScience	9
3	Springer	209
3	Wiley	86

and cybersecurity communities, special care was taken when screening borderline papers. In such cases, studies were retained when they substantially contributed to at least one of the review objectives and remained within the overall scope of anomaly detection in DER-rich smart grid or closely related industrial cyber-physical environments.

To align more closely with Kitchenham recommendations, the study selection can be understood as moving from identification, to screening, to eligibility assessment, and finally to inclusion. The complete selection pipeline, with the number of records entering and leaving each stage, is summarized in the PRISMA-style flow diagram in Fig. 3 [20], and each stage is explained in detail in Sect. 2.9.

2.6 Inclusion and exclusion criteria

The following inclusion and exclusion criteria were established to filter studies for relevance and quality:

Inclusion Criteria:

- Studies published in the last 10 years (2014–2024).
- Studies that explicitly discuss methods to distinguish/classify attacks and faults in ICS, CPS, SCADA, or DER systems.
- Studies that have proposed a method and applied it to at least one example, use-case, or real system.
- Journal articles, conference papers, or book chapters.
- Studies written in the English language.

Exclusion Criteria:

- Studies irrelevant to anomaly detection or cybersecurity in industrial and energy systems.
- Secondary studies (e.g., literature reviews, surveys, mapping studies).
- Studies written in languages other than English.
- Duplicate studies or extended versions of previously published papers.
- Inaccessible studies or those with fewer than 4 pages.
- Studies that only provide superficial descriptions of the proposed method without substantial analysis or application.

In addition to topical relevance, the eligibility criteria were chosen to preserve a minimum level of methodological substance in the selected studies. For example, papers that merely discussed anomaly detection conceptually without implementation details, evaluation evidence, or application context were excluded, since they did not provide enough empirical or technical material to answer the research questions.

The decision to restrict the corpus to English-language publications and to studies published in the last ten years was motivated by feasibility and relevance. The fast evolution of AI/ML techniques, digital twin technologies, DER integration, and cyber-physical security practices makes recent literature particularly important. Nonetheless, these restrictions may introduce language bias and time-window bias, which are discussed later as threats to validity.

2.7 Quality assessment of primary studies

In addition to inclusion and exclusion screening, a light-quality appraisal was considered in order to assess the robustness and reporting quality of the selected primary studies. This step is consistent with Kitchenham's recommendation to evaluate not only relevance but also the credibility of included studies.

A study can be assessed using the following indicative quality criteria:

- **QA1:** Are the study objectives or research problem clearly stated?
- **QA2:** Is the anomaly detection or classification method described with sufficient technical detail?
- **QA3:** Is the application domain, system context, or dataset adequately described?
- **QA4:** Is there a clear evaluation procedure, experiment, testbed, simulation, or use case?
- **QA5:** Are performance metrics or validation criteria reported clearly?
- **QA6:** Are the limitations, assumptions, or threats discussed?

- **QA7:** Is the study reproducible to some degree based on the reported information?

Each criterion may be scored, for example, as 0 (no), 0.5 (partially), or 1 (yes), allowing a total quality score to be computed for each primary study.

The purpose of this quality appraisal is not necessarily to exclude all lower-scoring studies, but rather to support a more nuanced interpretation of the evidence base. Studies with stronger experimental descriptions, clearer datasets, and better-reported evaluation procedures can be given greater interpretive weight during synthesis.

2.8 Duplicate handling and version selection

Duplicate handling was performed after the initial retrieval and during the screening stages. Records retrieved from multiple databases were compared based on title, authorship, publication year, venue, and study content. When multiple versions of substantially the same work were identified, preference was given to the most complete and relevant version, typically the journal extension when it significantly expanded the conference publication.

This procedure is important because interdisciplinary studies in smart grids, CPS, and cybersecurity may appear in multiple indexing services and sometimes in both conference and extended journal forms.

2.9 Screened studies

The complete selection funnel is summarized in the PRISMA-style flow diagram in Fig. 3 [20]; all counts are reproduced from the review protocol workbook and the structured analysis workbook. The three iterative search strings returned 2,044, 604, and 363 exported records respectively, i.e., 3,011 records in total across the six databases (the unrestricted Scopus variant of String 1, with 7,058 hits, was inspected online but not exported, as documented in the protocol). The combined records were uploaded to the Rayyan screening platform, where duplicate detection and resolution left 1,271 unique records; the duplicate-handling log recorded 376 flagged candidate pairs, of which 173 records were deleted and 22 were confirmed as distinct works. The unique records were then screened on title and abstract against the predefined inclusion and exclusion criteria (**Stage A**); borderline cases were discussed by the review team as described in Sect. 2.12.

The two refined searches (Strings 2 and 3), which by construction nest within the scope of String 1 and concentrate the records most relevant to the research questions, were exported in full and consolidated in a structured analysis workbook: 542 exported rows, reduced to 499 records once remaining cross-database duplicates were marked. Each

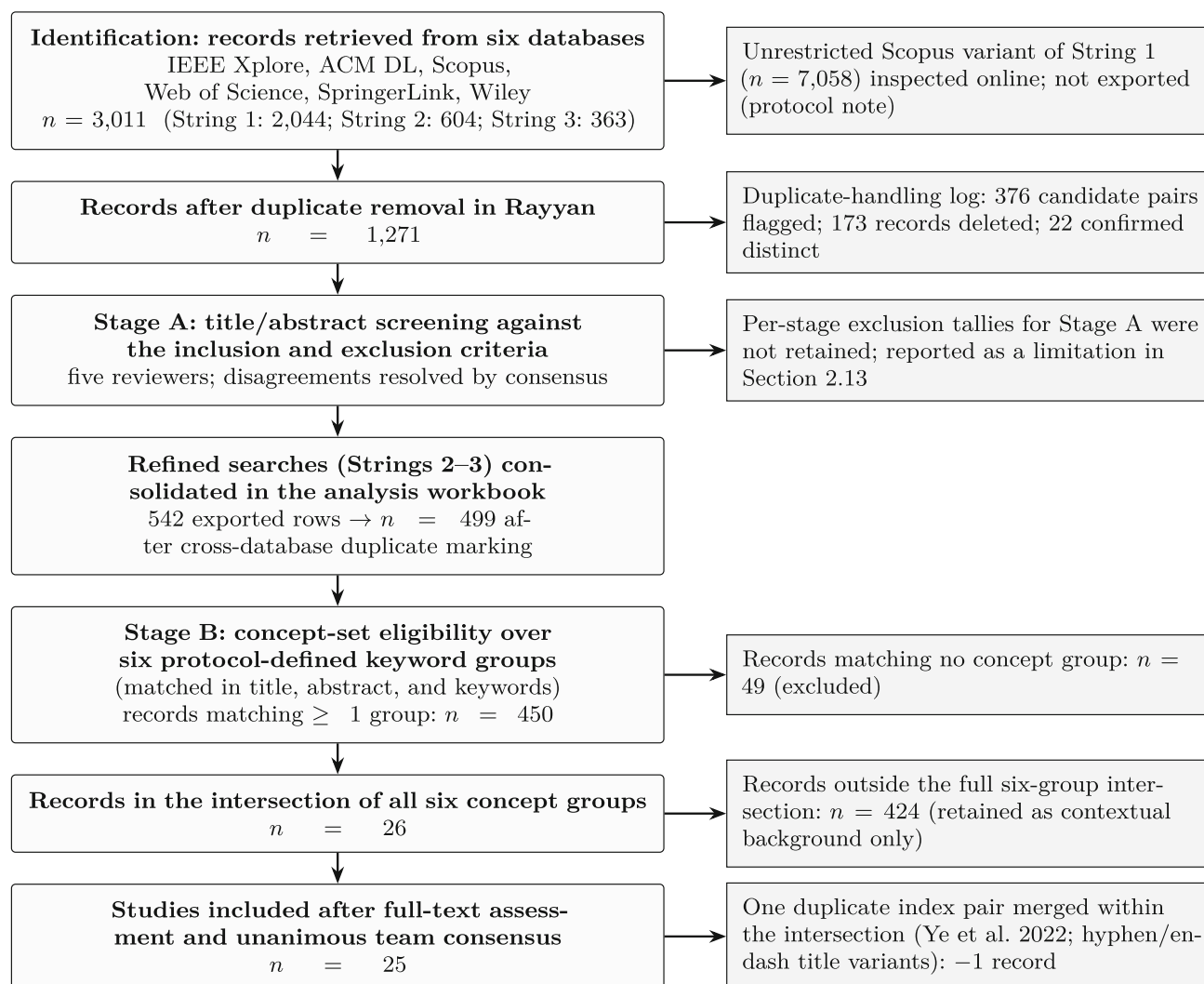


Fig. 3 PRISMA-style flow diagram of the study selection process, following the reporting logic of the PRISMA 2020 statement [20]. Stage A applies the formal inclusion and exclusion criteria; Stage B applies

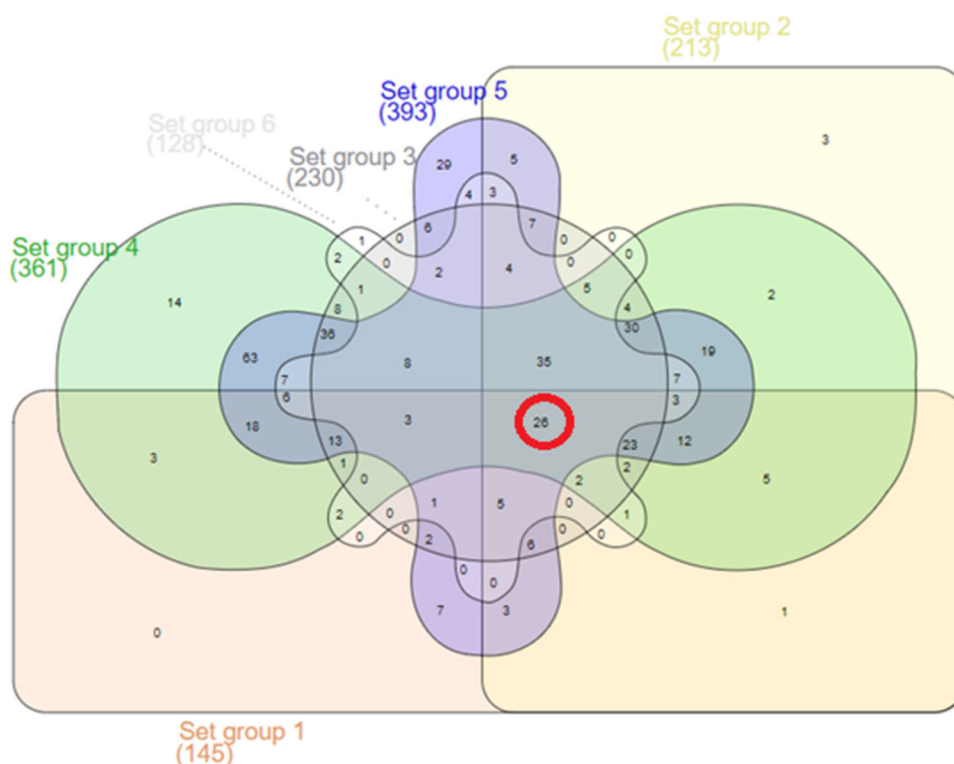
the protocol-defined concept-set eligibility analysis that operationalizes inclusion criterion I2. All counts are reproduced from the review protocol workbook and the structured analysis workbook

record was then automatically annotated, over its title, abstract, and author keywords, against the six protocol-defined concept groups (**Stage B**): Group 1, cybersecurity concerns; Group 2, anomaly-related terms covering both security threats (attacks) and safety (faults, failures); Group 3, analytical approaches; Group 4, systems and applications (ICS/CPS/SCADA systems and DER/smart-grid applications); Group 5, AI/ML methodologies; and Group 6, explainability, evaluation, and validation. Of the 499 records, 450 matched at least one concept group and 49 matched none. The figure of 450 records therefore denotes the *concept-matched pool entering the set-group analysis*, not the direct output of the criteria-based screening; the previous version of this subsection conflated the two stages, which is corrected here.

The overlap structure of the six concept groups over the concept-matched pool was explored with the InteractiVenn tool [21] and is shown in Fig. 4. Exactly 26 records lie in the central intersection of all six sets, i.e., they touch every thematic dimension of the research questions simultaneously. Within these 26 records, one study was indexed twice under marginally different titles (the review of cyber-physical security for photovoltaic systems by Ye et al. [22], appearing with a hyphen and an en-dash variant); merging this pair yields the 25 unique candidate studies that proceeded to full-text assessment, all of which were confirmed for inclusion by unanimous consensus of the review team (Sect. 2.12).

The studies listed in Table 2 encompass a diverse range of research contributions aimed at enhancing anomaly detection in Distributed Energy Resource (DER)-rich SG. Collectively,

Fig. 4 Venn diagram of the six protocol-defined concept groups over the concept-matched pool of 450 records (Stage B), created with InteractiVenn [21]. Set sizes: Group 1 Cybersecurity Concerns (145); Group 2 Anomaly–Security Threat/Safety (213); Group 3 Analytical Approaches (230); Group 4 Systems and Applications (361); Group 5 Methodologies and AI Technologies (393); Group 6 Explainability, Evaluation and Validation (128). The central intersection of all six sets contains 26 records; merging one duplicate index pair (Ye et al. 2022) yields the 25 included studies. The sets depict the concept-set eligibility analysis, not the three database search strings



the final set of 25 papers, screened without duplicates, represents key advancements in the field, offering insights into both theoretical frameworks and practical applications for enhancing grid resilience against diverse threats.

To make the relationship between the two stages explicit: Stage A (formal screening) determined whether a record was admissible at all, using the predefined inclusion and exclusion criteria. Stage B (the set-group analysis) is an *eligibility* step defined in the review protocol that operationalizes inclusion criterion I2—the requirement that studies explicitly relate to distinguishing or classifying attacks and faults in ICS, CPS, SCADA, or DER systems—in a transparent and auditable way: a record qualified for the final corpus only when all six thematic dimensions of the research questions (cybersecurity concerns; attack- and fault-related anomalies; an analytical approach; a system or application context; AI/ML methodology; and evaluation, validation, or explainability aspects) were simultaneously present in its title, abstract, or author keywords. The set-group analysis is therefore neither a replacement for the formal criteria nor an after-the-fact thematic prioritization; it is the documented mechanism by which the multi-dimensional eligibility requirement was applied uniformly to every concept-matched record.

The strictness of requiring all six groups simultaneously explains the sharp reduction from 450 concept-matched records to 25 included studies and reflects the specificity of the research questions: many studies address anomaly

detection in smart grids or cyber-physical systems in general, but only a small subset connects security and safety anomalies, an explicit analytical approach, an operational system or application context, AI/ML methods, and evaluation or validation concerns within a single contribution. Records satisfying fewer than six groups remain visible in Fig. 4 and informed the contextual discussion, but were not extracted.

The Venn representation should accordingly be read as an exploratory and documentary device for Stage B: it makes the eligibility logic auditable by exposing the size of each thematic set and of their overlaps, and it shows that the final corpus was not drawn from a single query or a single theme. It complements, rather than replaces, the formal selection narrative summarized in Fig. 3.

2.10 Data extraction

A systematic data extraction process was implemented to capture key information from the selected studies, ensuring consistent and comprehensive collection across all studies. A structured form was developed to extract specific elements aligning with six main groups, which emerged logically from recurring themes in the literature.

Through iterative search refinement and thematic analysis, six groups were identified, representing the critical areas of anomaly detection in DER-rich SG. These groups address the

Table 2 List of research papers and their authors

Refs.	Author	Title
[23]	Santorsola, Alessandro; Maci, Antonio; Delvecchio, Piero; Coscia, Antonio	A Reinforcement-Learning-based Agent to discover Safety-Critical States in Smart Grid Environments
[22]	Ye, Jin; Giani, Annarita; Elasser, Ahmed; Mazumder, Sudip K.; Farnell, Chris; Mantooth, Homer Alan; Kim, Taesic; Liu, Jianzhe; Chen, Bo; Seo, Gab-Su; Song, Wenzhan; Greidanus, Mateo D. Roig; Sahoo, Subham; Blaabjerg, Frede; Zhang, Jinan; Guo, Lulu; Ahn, Bohyun; Shadmand, Mohammad B.; Gajanur, Nanditha R.; Abbaszadeh, Mohammad Ali	A Review of Cyber-Physical Security for Photovoltaic Systems
[24]	Shukla, Saurabh; Thakur, Subhasis; Breslin, John G	Anomaly Detection in Smart Grid Network Using FC-Based Blockchain Model and Linear SVM
[25]	Reddy Shabad, Prem Kumar; Alrashide, Abdulmueen; Mohammed, Osama	Anomaly Detection in SG using Machine Learning
[26]	Yan, Weizhong; Mestha, Lalit K.; Abbaszadeh, Masoud	Attack Detection for Securing Cyber Physical Systems
[27]	G, Amritha; Kh, Vishakh; C, Jishnu Shankar V; Nair, Manjula G	Autoencoder Based FDI Attack Detection Scheme For Smart Grid Stability
[28]	Mylrea, Michael; Fracchia, Charles; Grimes, Howard; Austad, Wayne; Shannon, Gregory; Reid, Bill; Case, Nathan	BioSecure Digital Twin: Manufacturing Innovation and Cybersecurity Resilience
[29]	Naeem, Hamad; Ullah, Farhan; Srivastava, Gautam	Classification of intrusion cyber-attacks in smart power grids using deep ensemble learning with metaheuristic-based optimization
[30]	Suprabhath Koduru, Sriranga; Machina, Venkata Siva Prasad; Madichetty, Sreedhar	Cyber Attacks in Cyber-Physical Microgrid Systems: A Comprehensive Review
[31]	Appiah-Kubi, Jennifer; Liu, Chen-Ching	Cyberattack Correlation and Mitigation for Distribution Systems via Machine Learning
[32]	Al Homoud, Leen; Reghunath, Rinith; Bayes, Safin; Peerzada, Aaqib; Davis, Katherine; Balog, Robert S	Cyber-Physical Defense in Smart Distribution Networks
[33]	Czekster, Ricardo M.; Morisset, Charles; Moorsel, Aad van; Mace, John C.; Bassage, Walter A.; Clark, John A	Cybersecurity Roadmap for Active Buildings
[34]	Jahromi, Martiya Zare; Jahromi, Amir Abiri; Kundur, Deepa; Sanner, Scott; Kassouf, Marthe	Data analytics for cybersecurity enhancement of transformer protection
[35]	Fournaris, Apostolos P.; Komninos, Andreas; Lalos, Aris S.; Kalogeras, Athanasios P.; Koulamas, Christos; Serpanos, Dimitrios	Design and Run-Time Aspects of Secure Cyber-Physical Systems
[36]	Tertytchny, Georgios; Nicolaou, Nicolas; Michael, Maria K	Differentiating Attacks and Faults in Energy Aware Smart Home System using Supervised Machine Learning
[37]	Shen, Zhongjie; Xu, Wenqing; Li, Weikai; Shi, Yaoyao; Gao, Fan	Digital twin application for attack detection and mitigation of PV-based smart systems using fast and accurate hybrid machine learning algorithm
[38]	Mylrea, Michael; Nielsen, Matt; John, Justin; Abbaszadeh, Masoud	Digital Twin Industrial Immune System: AI-driven Cybersecurity for Critical Infrastructures
[39]	Oyucu, Saadin; Polat, Onur; Türkoğlu, Muammer; Polat, Hüseyin; Aksöz, Ahmet; Ağdaş, Mehmet Tevfik	Ensemble Learning Framework for DDoS Detection in SDN-Based SCADA Systems
[40]	Johannesen, Nils Jakob; Kolhe, Mohan Lal; Goodwin, Morten	Evaluating Anomaly Detection Algorithms through different Grid scenarios using k-Nearest Neighbor, iforest and Local Outlier Factor

Table 2 continued

Refs.	Author	Title
[41]	Husnoo, Muhammad Akbar; Anwar, Adnan; Reda, Haftu Tasew; Hosseinzadeh, Nasser; Islam, Shama Naz; Mahmood, Abdun Naser; Doss, Robin	FedDiSC: A computation-efficient federated learning framework for power systems disturbance and cyber attack discrimination
[42]	Agnew, Dennis; Aljohani, Nader; Mathieu, Reynold; Boamah, Sharon; Nagaraj, Keerthiraj; McNair, Janise; Bretas, Arturo	Implementation Aspects of SG Cyber-Security Cross-Layered Framework for Critical Infrastructure Operation
[43]	Padmapriya, Valluri; Srivenkatesh, Muktevi	IoT Network based Cyber Attack Mitigation in Digital Twin with Multi Level Key Management Using Enhanced KNN Model
[44]	Hyder, Burhan; Ahmed, Arman; Mana, Priya; Edgar, Thomas; Niddodi, Shwetha	Leveraging High-Fidelity Datasets for Machine Learning-based Anomaly Detection in SG
[45]	Yilmaz, Yasin; Uludag, Suleyman	Mitigating IoT-based cyber-attacks on the Smart Grid
[46]	Chhokra, Ajay; Barreto, Carlos; Dubey, Abhishek; Karsai, Gabor; Koutsoukos, Xenofon	Power-attack: a comprehensive tool-chain for modeling and simulating attacks in power systems

complexities and nuances of our research questions. Respectively Group 1 to 6 correspond to research question 1 to 6:

- Group 1 focuses on the types of data utilized, their sources, and how they are processed, which is fundamental to anomaly detection.
- Group 2 delves into the AI and ML methodologies applied, highlighting the models and techniques that have shown promise in this field.
- Group 3 examines the various application domains (where anomaly detection systems are deployed?) and operational contexts (how anomaly detection systems are used?). For more detail, cf. Fig. 6
- Group 4 addresses the evaluation strategies and performance metrics used to assess the effectiveness of these systems.
- Group 5 discusses the technical and operational challenges faced in deploying these technologies and explores future opportunities for improvement.
- Group 6 considers the characteristics of testbeds used for evaluation and validation, emphasizing the importance of simulating real-world conditions.

To avoid ambiguity with the selection stage: these six *extraction groups*, aligned one-to-one with research questions RQ1–RQ6, are distinct from the six *concept groups* (protocol-defined keyword sets) used for the Stage B eligibility analysis in Sect. 2.9, even though both are numbered 1–6. The concept groups are keyword lists matched mechanically against titles, abstracts, and author keywords to decide *which* records enter the corpus; the extraction groups structure *what* information is extracted from the studies once included. Both derive from the same research-question themes but play different roles in the review pipeline.

Compared to other literature reviews, such as Machine Learning for Intrusion Detection in Industrial Control Systems [47], our extraction approach not only addresses the core elements discussed in prior work but also provides additional depth and structure. Specifically, this allows for a broader analysis across various facets of anomaly detection, extending beyond specific challenges in ICS to include diverse applications relevant to DER-rich SG.

The structured extraction form was intended to improve consistency when analyzing the selected studies. In practice, each study was examined with respect to the six groups, and relevant attributes were recorded to support cross-study comparison. This helped reduce purely narrative interpretation and enabled a more systematic mapping between individual papers and the research questions.

To make the extraction logic more explicit, the form can be understood as capturing, at minimum, the following fields where available: bibliographic information, system/application context, anomaly type, cyber-related or safety-related event type, input data characteristics, AI/ML technique, evaluation environment, performance metrics, validation setting, testbed characteristics, reported limitations, and future work directions.

The six groups should therefore be interpreted as analytical synthesis dimensions rather than as mutually exclusive categories. A single study may contribute to several groups simultaneously, which is particularly important in cyber-physical smart grid research, where data, models, applications, and validation environments are often strongly interdependent.

Taxonomy of Detection Targets and System Layers. To complement the six extraction groups with a more explicitly analytical lens, we introduce a two-dimensional taxonomy that maps the reviewed studies according to (i) the

Fig. 5 Analytical taxonomy of the reviewed studies by anomaly type and detection layer

Network / communication Application / control Physical / sensor	Cybersecurity attacks Shukla 2022 Naeem 2024 Oyucu 2024 Yilmaz 2017	Operational faults —	Both / distinction-oriented Husnoo 2023 Agnew 2022
	Appiah-Kubi 2023 Fournaris 2019	Santorsola 2023	Tertytchny 2019 Shen 2023 Mylrea 2021
	Amritha 2022 Jahromi 2022	Johannesen 2022 Al Homoud 2021 Reddy 2021	Yan 2019 Hyder 2023 Chhokra 2021

primary anomaly type addressed (cybersecurity attack, operational fault, or both) and (ii) the main layer at which the detection logic is situated (physical/sensor layer, network/communication layer, or application/control layer). The purpose of this taxonomy is to make visible how the literature is distributed across the cyber–physical stack and, more importantly, how rarely studies perform an explicit distinction between cyber and physical anomalies in the same evaluative setting.

By organizing the literature into these six groups, as presented in Fig. 6, our approach comprehensively addresses our research questions and objectives, enabling meaningful insights and supporting the development of advanced anomaly detection systems capable of distinguishing between cybersecurity attacks and safety faults. The figure should therefore be read not only as a visual summary of themes, but also as the operational bridge between the review questions and the later discussion structure.

The text in *emphasized text* highlights specific research studies that align with each group's criteria, providing real-world examples or theoretical insights that directly support and expand upon the key concepts outlined in each key concepts-group topic.

2.11 Synthesis strategy

The synthesis of the selected studies was primarily narrative and thematic. This choice was motivated by the diversity of the included works with respect to research objectives, infrastructures, data modalities, anomaly definitions, machine learning techniques, and evaluation settings. Because of this heterogeneity, a formal meta-analysis would not have been methodologically appropriate.

Instead, the studies were grouped according to the six extraction dimensions, enabling comparisons across common themes while preserving contextual specificity. This synthesis strategy supports both descriptive mapping of the literature and interpretive analysis of methodological trends,

gaps, and opportunities. It also makes it possible to discuss how different strands of literature contribute differently to the central challenge of distinguishing cybersecurity attacks from safety faults.

The Fig. 5 provides a high-level mapping of representative studies across the cyber–physical stack. It shows that network-layer work is predominantly attack-centric, whereas physical-layer work more often blends operational and cyber evidence. The detailed per-study classification is provided in Table 3.

2.12 Reviewer agreement and bias control

The review was conducted by a team of five reviewers. Screening and extraction decisions were guided by the pre-defined criteria and analytical dimensions described above. The lead author performed the initial retrieval, deduplication in Rayyan, and the first-pass title/abstract screening; candidate studies and all borderline cases were then brought to the full team. Where uncertainty arose regarding the relevance of a paper, the study was examined collectively with respect to the review scope, the research questions, and its empirical contribution, and the decision was settled by discussion until consensus was reached. This procedure helped reduce arbitrary exclusion of borderline but potentially valuable interdisciplinary studies.

For the final corpus, all five reviewers assessed the 25 candidate studies emerging from the selection funnel (Sect. 2.9) and unanimously agreed that each of them should be read in full and included. No study reached the final stage with an unresolved disagreement.

We acknowledge, however, that the screening was organized as a lead-screener-plus-consensus process rather than as fully independent parallel coding by two or more reviewers on a common sample. Because formal inter-rater reliability coefficients such as Cohen's kappa [48] presuppose independent ratings of the same items, no such coefficient was computed, and we refrain from reporting one retrospectively.

Table 3 Taxonomy-based mapping of the 25 reviewed studies.

Study	Primary anomaly type	Detection approach	Grid layer	Cyber/physical distinction	Tier
Santorsola 2023	Safety-critical operational states	RL/white-box simulation	Application/control	N	T3
Ye 2022	Cyber-physical PV attacks	Survey of ML/DL methods	Cross-layer	Partial	T3
Shukla 2022	Network intrusions	Linear SVM + blockchain	Network/communication	N	T4
Reddy 2021	Grid anomalies/outliers	Isolation Forest + PCA	Physical/sensor	N	T2
Yan 2019	Malicious actions (physical-layer)	ELM/SDAE + physics-based modeling	Physical/sensor	Y	T2
Amritha 2022	FDI attacks	Autoencoder	Physical/sensor	N	T4
Mylrea 2021a	Cyber resilience/process anomalies	Digital twin + XAI	Application/control	Partial	T4
Naeem 2024	Intrusion attacks	Deep ensemble learning	Network/communication	N	T3
Suprabhath 2023	Microgrid cyber-attacks	DL/RL survey	Application/control	Partial	T4
Appiah-Kubi 2023	Cyberattack correlation	Multi-agent ML	Application/control	Partial	T3
Al Homoud 2021	Coordinated attacks/abnormal states	LOF (local detection)	Physical/sensor	Partial	T3
Czekster 2022	Active-building cyber risks	RA/simulation roadmap	Application/control	Partial	T4
Jahromi 2022	Relay anomalies/cyber-attacks	Autoencoders (linear/FC/conv), LSTM	Physical/sensor	Partial	T3
Fournaris 2019	Secure CPS runtime anomalies	Knowledge-based/monitoring logic	Network/application	Partial	T1
Tertytchny 2019	Attack–fault differentiation	Supervised ML	Application/control	Y	T3
Shen 2023	Data-integrity attacks	Hybrid ML + digital twin + SHT	Application/control	Y	T3
Mylrea 2021b	Cyber-physical threats	Digital Ghost + AI	Application/control	Partial	T3
Oyucu 2024	DDoS attacks	Ensemble learning	Network/communication	N	T4
Johannesen 2022	Grid anomalies across scenarios	kNN/iForest/LOF	Physical/sensor	Y	T1
Husnoo 2023	Disturbance vs. cyberattack	Federated learning + DAE	Network/communication	Y	T2
Agnew 2022	Cross-layer cyber threats	SDN + ML + state estimation	Network/communication	Partial	T3
Padmapriya 2024	IoT cyber-attacks	EKNN + digital twin	Network/communication	N	T4
Hyder 2023	Cyber-physical anomalies	High-fidelity ML benchmarking	Physical/sensor	Y	T3

Table 3 continued

Study	Primary anomaly type	Detection approach	Grid layer	Cyber/physical distinction	Tier
Yilmaz 2017	IoT cyber-attacks	Hierarchical IDS/ODIT	Network/communication	N	T4
Chhokra 2021	Protection-system attacks	Simulation tool-chain	Physical/sensor	Partial	T3

This table complements the thematic groups by explicitly classifying each study according to anomaly type, methodological family, main grid layer addressed, whether cyber–physical distinction is attempted explicitly, and the realism tier of the evaluation environment. The tier allocation is aligned with the dataset-fidelity analysis reported earlier: 2 studies in T1, 3 in T2, 12 in T3, and 8 in T4

Consensus-based resolution mitigates, but does not quantify, individual screening bias; this is reported transparently as a limitation in Sect. 2.13, and future updates of this review will adopt independent dual screening on at least a sample of records so that a kappa statistic can be reported.

2.13 Threats to validity of the review process

As with any SLR, this review is subject to several threats to validity. First, **search bias** may arise from the choice of keywords and Boolean expressions, especially in an interdisciplinary field where similar concepts may be described differently across energy systems, cybersecurity, and machine learning communities. Although iterative search refinement was used to reduce this risk, it cannot be eliminated completely.

Second, **database and language bias** may affect the coverage of the review. While several major scientific databases were queried, some relevant studies may still have been missed because of indexing limitations, venue-specific coverage differences, or the restriction to English-language publications.

Third, **evaluation data bias** remains an important limitation. A large share of the reviewed studies relies on simulated, synthetic, or otherwise non-operational datasets, which may lead to optimistic estimates of model performance. The realism of the underlying data-generating process therefore constrains the strength of any cross-study inference, especially for claims related to cyber–physical distinction under DER-rich operating conditions.

Fourth, **corpus size and evidence-base limitations** should be acknowledged. The final corpus of 25 studies was intentionally focused on the specific challenge of distinguishing cybersecurity events from physical safety faults in DER-rich grids, but this narrow focus may have excluded relevant work in closely related domains such as building energy systems or industrial microgrids. In addition, only 6 of the 25 studies (24%) explicitly attempt such cyber–physical distinction. Consequently, the evidence base for this central question remains limited, and the corresponding conclusions should be interpreted with caution.

Fifth, a **deployment feasibility gap** exists in the primary literature. Most reviewed studies do not report inference latency, memory footprint, model update cost, communication overhead, or SCADA/EMS integration requirements. As a result, this review can assess experimental effectiveness more reliably than practical deployability.

Sixth, **selection bias** may arise during screening, particularly for borderline studies with partial relevance. The use of explicit inclusion and exclusion criteria helped mitigate this issue, but subjective judgement may still have influenced some decisions.

A related threat concerns **inter-rater reliability**. As reported in Sect. 2.12, the five-person team resolved screening decisions by consensus and was unanimous on the inclusion of the final 25 studies, but screening was not organized as independent parallel coding, and no quantitative agreement coefficient such as Cohen's kappa [48] was computed. The level of chance-corrected agreement among reviewers therefore cannot be reported, and the robustness of borderline screening decisions cannot be quantified; future updates of this review will include independent dual screening on a sample of records to address this.

An additional **process-documentation limitation** is that the per-stage exclusion tallies of the first-level (Stage A) title/abstract screening were not retained in the review log: the protocol workbook records the post-deduplication pool (1,271 records) and the downstream consolidated and concept-matched pools (Fig. 3), but not the exact number of records excluded per criterion at Stage A. The corresponding cells of the PRISMA-style flow are therefore reported at the granularity available, and this gap is acknowledged openly rather than reconstructed retrospectively.

Seventh, **reporting bias** in the primary studies themselves may affect the synthesis. Some studies provide limited details on datasets, testbeds, or evaluation procedures, making it difficult to assess their robustness or reproducibility.

Finally, the review is affected by **time-window bias**. Restricting the study period to the last ten years improves topical relevance in a fast-evolving field, but it may also omit foundational earlier works.

Fig. 6 Classification groups for data extraction in the systematic literature review. This figure summarizes the six analytical synthesis dimensions used in the review: data utilization and sources, AI/ML methodologies, application domains, evaluation strategies, challenges and opportunities, and testbeds for evaluation and validation. Its purpose is not merely descriptive; it also explains how the research questions were operationalized into a structured extraction framework, thereby making the transition from study selection to thematic synthesis more transparent and reproducible

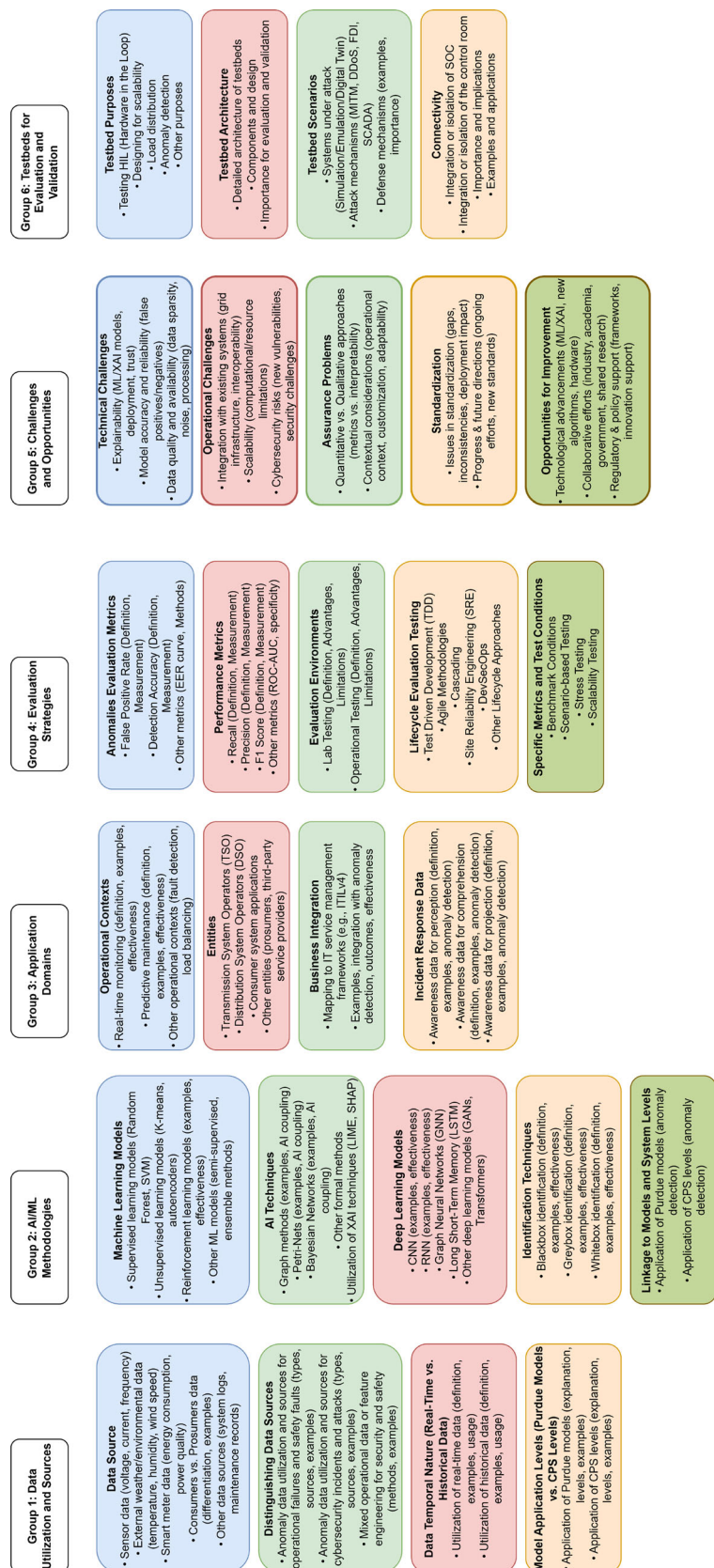




Fig. 7 Topics mapping under data Utilization and Sources category/group 1 shown in Fig. 6

An additional synthesis limitation is the impossibility of a formal statistical meta-analysis. The reviewed papers vary substantially in datasets, anomaly definitions, DER contexts, validation platforms, and reported metrics; in many cases, even the same metric is computed under non-comparable assumptions. For that reason, the revised manuscript strengthens cross-study analytical comparison, but it does not claim quantitative pooled effect sizes or universal method rankings.

2.14 Reproducibility and replication considerations

To facilitate future replication of this review, the methodological elements reported in this section now include the database sources, the exact executed Boolean search strings with the search date (Search Strategy subsection), the eligibility criteria, the two-stage screening rationale with a PRISMA-style flow diagram (Fig. 3), the duplicate-handling record from Rayyan, the concept-group definitions and set sizes underlying the Venn-based eligibility analysis (Fig. 4), and the extraction structure. The principal remaining repro-

ducibility gap is the per-criterion exclusion log of the Stage A title/abstract screening, which was not retained (Sect. 2.13); replication of Stage B, in contrast, is fully mechanical given the concept-group keyword lists and the record pool.

Overall, the methodology was designed to balance breadth and specificity: broad enough to capture the interdisciplinary nature of anomaly detection in DER-rich smart grids, yet specific enough to focus on the key challenge of distinguishing cybersecurity attacks from safety faults in operationally meaningful contexts.

Group 1: Data utilization and sources

The first column in Fig. 6 represents the key concepts-group topic.

- **Data Source:** Understanding and categorizing the types of data (e.g., sensor, environmental) is essential, as these sources provide the raw information that anomaly detection systems rely on. *Gungor et al. [49] present a comprehensive review of communication technologies and standards in SG such as data sources in SG, including sensor data, environmental data emphasizing the role of advanced metering infrastructure (AMI) and information and communication technologies (ICT) in enhancing data communication, which is fundamental to smart grid operations.*
- **Data Temporal Nature (Real-Time vs. Historical Data):** Distinguishing between real-time and historical data usage is critical, as it informs how and when data is processed, impacting detection speed and the nature of response actions. *Fang et al. [50] provide a thorough survey of smart grid technologies, exploring the smart infrastructure system, management, and protection systems, highlighting the importance of real-time and historical data in improving grid reliability and efficiency.*
- **Distinguishing Data Sources:** Differentiating data based on its relevance to operational failures or cybersecurity incidents allows for more accurate detection and classification of anomalies, aligning responses to the specific type of threat. *Sridhar et al. [51] examine cyber-physical system security for the electric power grid, discussing potential cyber-attacks and the importance of securing both physical and cyber infrastructures in SG.*
- **Model Application Levels (Purdue Models vs. CPS Levels):** Applying models like Purdue or CPS levels structures the data according to system hierarchy, making detection methods more efficient by matching them with specific operational layers within the smart grid. *Stouffer [52] provides guidance on securing operational technology (OT), including the application of models like the Purdue Enterprise Reference Architecture to enhance cybersecurity in industrial control systems.*

Group 2: AI/ML methodologies

The second column in Fig. 6 represent the key concepts-group topic.

- **Machine Learning Models:** This is fundamental for detecting anomalies as it encompasses supervised, unsupervised, and reinforcement learning models, each offering different strengths, from accuracy in known conditions to adaptability for new threats. *Chandola et al. [53] offer an extensive survey of anomaly detection techniques across different research areas, detailing machine learning approaches such as supervised, unsupervised, and semi-supervised learning.*
- **Deep Learning Models:** Advanced models like CNNs, RNNs, and LSTMs are critical for recognizing complex patterns in large datasets, which enhances detection capabilities, especially in time-series data. *LeCun, Bengio, and Hinton [54] discuss deep learning architectures, including CNNs and RNNs, highlighting their success in learning intricate structures in large datasets, which is crucial for applications like image and speech recognition.*
- **Graph-Based and Formal AI Methods/Explainable AI (XAI):** Methods such as graph-based approaches, Petri-net-inspired formalisms, Bayesian reasoning, and explainable AI techniques like LIME and SHAP help interpret model decisions, improve transparency, and potentially strengthen anomaly distinction in cyber-physical environments. *Lundberg and Lee [55] present SHAP (SHapley Additive exPlanations), a unified approach to interpreting model predictions, enhancing explainability in machine learning models.*
- **Identification Techniques:** These techniques (black-box, grey-box, white-box) characterize anomaly detectors according to how transparently they model the underlying system, which in turn affects their applicability and reliability across various operational contexts. In power systems or SG, for instance, *black-box* models rely primarily on empirical data, requiring extensive datasets but offering rapid deployment; *grey-box* methods combine partial physical knowledge with data-driven learning, yielding moderate interpretability and potentially improved robustness; and *white-box* approaches leverage thorough system equations or domain physics, maximizing transparency but demanding extensive knowledge of system dynamics. The suitability of each approach varies depending on factors such as data availability, computational constraints, and the degree of domain knowledge required. *Ljung [56] elaborates on these system identification paradigms, detailing the trade-offs and potential applications for each modeling strategy.*

- **Linkage to Models and System Levels:** Aligning machine learning models with operational layers (e.g., Purdue models) tailors detection processes to specific levels of smart grid architecture, improving response accuracy and efficiency. *Lee and Seshia [57] introduce embedded systems in the context of cyber-physical systems (CPS), discussing system design and the interaction between computational and physical processes.*

Group 3: application domains

The third column in Fig. 6 represent the key concepts-group topic.

- **Operational Contexts:** Applying anomaly detection effectively in contexts like real-time monitoring and predictive maintenance requires different detection strategies tailored to each context. *Katipamula and Brambley [58] review methods for fault detection, diagnostics, and prognostics in building systems, focusing on real-time monitoring and predictive maintenance strategies.*
- **Entities:** Identifying specific roles (Transmission System Operators, Distribution System Operators) allows for targeted detection solutions that address unique risks and operational needs of various system operators and prosumers. *Fang et al. [50] explore the smart grid's infrastructure, management, and protection systems, discussing the roles of various entities like Transmission System Operators (TSOs) and Distribution System Operators (DSOs).*
- **Business Integration:** Mapping anomaly detection to IT management frameworks like ITIL ensures that detection aligns with business goals, enhancing both operational and strategic relevance. *Cartlidge [59] provides an introductory overview of ITIL 4, detailing IT service management practices and their relevance to integrating technologies like anomaly detection.*
- **Incident Response Data:** Awareness data for perception, comprehension, and projection provides context for anomalies, making it easier to classify incidents and respond effectively. *Endsley [60] proposes a theory of situation awareness in dynamic systems, emphasizing perception, comprehension, and projection, which are crucial for incident response and decision-making.*

Group 4: evaluation strategies

The fourth column in Fig. 6 represent the key concepts-group topic.

- **Anomalies Evaluation Metrics:** Metrics like false positive rate and detection accuracy are critical for assessing

detector performance, ensuring reliable anomaly detection without excessive false alarms. These metrics are indeed for data anomaly detectors, even though the anomalies can be rooted in real-world (physical or cyber) issues. *Fawcett [61] provides an introduction to ROC (Receiver Operating Characteristic) analysis, discussing evaluation metrics such as false positive rate and detection accuracy in classifier performance.*

- **Performance Metrics:** Metrics such as recall, precision, and F1 score offer nuanced insights into model performance, helping balance detection effectiveness with efficiency. *Sokolova and Lapalme [62] systematically analyze performance measures for classification tasks, elaborating on metrics like recall, precision, and F1 score.*
- **Evaluation Environments:** Distinguishing between lab and operational testing environments reveals limitations in a controlled setting versus real-world applications, ensuring comprehensive evaluation. *Montoya et al. [63] review advanced laboratory testing methods using real-time simulation and hardware-in-the-loop techniques, comparing lab and operational environments for smart grid applications.*
- **Lifecycle Evaluation Testing:** Incorporating Test-Driven Development (TDD), Agile, and other methodologies ensures detectors remain robust and adaptable through continuous testing and refinement. *Humble and Farley [64] introduce continuous delivery practices, discussing techniques like Test-Driven Development (TDD), Agile methodologies, and DevOps for reliable software releases.*
- **Specific Metrics and Test Conditions:** Benchmarking, scenario-based testing, stress, and scalability testing help gauge how well detectors perform under various real-world conditions and workloads. *Jain [65] presents 'The Art of Computer Systems Performance Analysis,' detailing techniques for experimental design, measurement, simulation, and modeling, including benchmarking and stress testing.*

Group 5: challenges and opportunities

The fifth column in Fig. 6 represent the key concepts-group topic.

- **Technical Challenges:** Addressing data and AI model explainability, accuracy, and data quality issues is essential to improve the deployment and reliability of machine learning models in live environments. *Doshi-Velez and Kim [66] advocate for a rigorous science of interpretable machine learning, addressing challenges in*

model explainability and its importance for trust and deployment.

- **Operational Challenges:** Integration with existing systems, interoperability among heterogeneous components, scalability, and cybersecurity risks require adaptive approaches to ensure detectors are resilient and compatible with diverse infrastructure. *Amin and Wollenberg [67] discuss the transition toward a smart grid for 21st-century power delivery, highlighting challenges in security, scalability, and integration with existing infrastructure.*
- **Assurance Problems:** Balancing quantitative metrics with data interpretability allows anomaly detectors to provide accurate and understandable results, which is essential in operational contexts. *Sapori et al. [68] propose a quantitative approach to risk management in critical infrastructures, discussing the integration of quantitative and qualitative assessments in security management.*
- **Standardization:** Gaps in standards and ongoing efforts for harmonization ensure consistency, enabling broader adoption and cross-system compatibility. *The IEC 62351 series [69] focuses on cybersecurity for power systems, providing standards and guidelines to secure communication protocols in the energy sector.*
- **Opportunities for Improvement:** Advances in technology, collaborative initiatives, and regulatory support can drive improvements in anomaly detection, enhancing both security and operational effectiveness. *The H2020 program [70] addresses cybersecurity in the Electrical Power and Energy System (EPES), promoting research and innovation to enhance protection against cyber-attacks and data breaches.*

Group 6: testbeds for evaluation and validation

The sixth column in Fig. 6 represent the key concepts-group topic.

- **Testbed Purposes:** Specific goals, like Hardware-in-the-Loop (HIL) testing or anomaly detection, define the testbed's scope, ensuring it accurately simulates real-world conditions for effective evaluation. *Hahn et al. [71] present the architecture and application of cyber-physical security testbeds for the smart grid, focusing on aspects like Hardware-in-the-Loop (HIL) testing and anomaly detection.*
- **Testbed Architecture:** A detailed architecture highlights testbed components and design, making it easier to assess and validate anomaly detection performance across diverse system setups. *Steurer et al. [72] describe a megawatt-scale power hardware-in-the-loop simula-*

tion setup for motor drives, detailing the testbed's architecture and components for evaluation.

- **Testbed Scenarios:** Simulated scenarios (e.g., DDoS, Man-in-the-Middle attacks) provide realistic testing conditions, essential for evaluating the resilience of detectors against a range of threat vectors. *Wei et al. [73] propose an integrated security system for protecting the smart grid against cyber-attacks, discussing scenarios like Man-in-the-Middle (MITM), DDoS, and False Data Injection attacks.*
- **Connectivity:** Decisions on integrating or isolating systems like Security Operations Centers (SOC) and control rooms impact detection strategy, as connectivity settings affect real-world applicability and the accuracy of simulated responses. *Singh et al. [74] investigate the integration of distributed energy resources, addressing challenges and implications for system operation and control in SG.*

3 Thematic synthesis and discussion

3.1 Data utilization and sources

3.1.1 Introduction

An essential aspect of managing these sophisticated grids is the effective detection of anomalies, which can stem from operational failures, safety faults, or cybersecurity threats. The rapid integration of Distributed Energy Resources (DERs) into SG has transformed the traditional energy landscape, introducing both opportunities and complexities.

To highlight the dual-layer structure of anomaly detection in a DER-rich Smart Grid, we depict in Fig. 2 how the physical layer (sensors, generation, loads) intersects with the cyber layer (control systems, communication).

Central to this detection is the utilization of diverse data types, sourced and processed to provide real-time insights and historical context.

Following Fig. 7, we discuss the different data sources, their temporal nature, their distinguishing criteria, and finally model application levels.

Information and communications technology engineering considerations From an Information and Communications Technology engineering perspective, handling these heterogeneous data sources involves creating robust ETL (Extract, Transform, Load) pipelines, ensuring secure and low-latency ingestion of real-time sensor streams, and managing a high-throughput message bus (e.g., Kafka) for data integration. ICT engineers must also tackle challenges of data redundancy, compression, partitioning strategies, and schema evolution to ensure that real-time data remains consistent and queryable for anomaly detection models.

Electrical Engineering Considerations From an Electrical Engineering perspective, instrumenting the grid with sensors requires detailed knowledge of voltage levels, current ratings, and placement of measurement equipment in strategic feeder lines or substations. Understanding how transient disturbances propagate through transmission and distribution lines is fundamental to ensuring data are collected at the correct sampling rates for accurate fault detection and grid protection coordination.

3.1.2 Data sources

This discussion delves into the critical data types necessary for robust anomaly detection in DER-rich SG. We explore various data sources, including sensor measurements like voltage, current, and frequency, which offer granular operational details. Environmental factors such as temperature, humidity, and wind speed are also considered, given their significant impact on grid performance. Smart meter data provides insights into energy consumption patterns and power quality, highlighting the roles of both consumers and prosumers in the energy ecosystem.

Sensor Data

Based on our extraction criteria, we summarize sensor data utilized by various studies in different contexts, primarily focusing on electrical systems.

Voltage, Current, and Frequency: This is the most common set of data collected, as noted by multiple studies such as [23] and [27]. This data is typically used in power systems including PV systems monitoring [22] and IoT sensors anomaly detection [24].

Specialized Electrical Data: Some studies also consider **active and reactive power** in addition to voltage and current, as indicated by [25, 32]. Similarly, [36] focus specifically on home appliances, and [37] address renewable energy sources like wind turbines and photovoltaics.

Other Sensor Data: Operational conditions of gas turbines are examined by [26], suggesting a more specialized application. Mylrea et al. [28, 38] describe performance behaviors in production and sensor data from various framework nodes, respectively.

Advanced Monitoring Systems: SCADA and integrated sensor systems for grid monitoring are discussed by studies like [29, 31]. Additionally, [39] measures computing resources such as CPU, memory, and register values over Modbus-TCP.

Phasor Measurement Units (PMUs): Data specifically from PMUs are collected by [41, 46], emphasizing its importance in real-time monitoring of power grids. They offer time-synchronized data that, when integrated with SCADA systems, provide enhanced situational awareness. As shown in Fig. 8, this synergy allows for real-time monitoring of critical grid parameters.

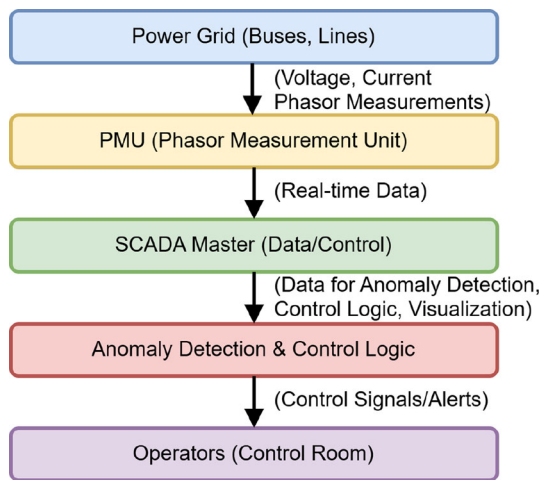


Fig. 8 Integration of PMU Measurements with SCADA Systems. Depicts the use of Phasor Measurement Units for real-time grid monitoring and enhanced situational awareness

To visually conceptualize the role of Phasor Measurement Units (PMUs) within a broader Cyber-Physical System, Fig. 8 has been structured following the architectural logic of Hybrid Cyber-Physical Systems (H-CPS) as presented by Sztipanovits et al. in [75, figure 1]. The design draws a clear parallel between the layered structure of H-CPS and the data flow in modern smart grid monitoring systems.

Each block in the figure corresponds to a specific H-CPS layer. The physical layer includes the power grid infrastructure and PMUs, which generate real-time synchronized phasor measurements. These measurements are transmitted through a networking layer—not explicitly labeled but implicit in the data flow arrows—to the service platform layer, represented by the SCADA Master, which collects and distributes the data. The application layer encompasses the anomaly detection and control logic, which interprets the data, detects threats or failures, and triggers automated responses. Finally, the human layer is embodied by the control room operators who interact with the system, validate alerts, and can intervene manually if necessary.

This layered mapping enables a structured understanding of how monitoring, control, and human decision-making coexist in a secure and resilient smart grid, and how PMUs contribute to the critical CPS-H-CPS interface.

This diverse collection of sensor data reflects the broad applications of sensor data in monitoring and managing energy systems, production environments, and computing infrastructure across various sectors.

External Weather and Environmental Data

Because renewables (e.g., solar PV and wind) are inherently weather-dependent, each Renewable Energy Source (RES) defines *standard or ambient conditions*—for example, solar PV performance is benchmarked at an irradiance of 1000 W m^{-2} and a cell temperature of 25°C , while wind

turbines reference standard wind speeds. Such reference conditions allow anomaly-detection systems to distinguish normal, weather-driven variations from genuine faults or attacks.

Our literature review shows a spectrum of specificity in the use of external weather data. Some studies explicitly model or measure detailed parameters: Ye et al., [22] consider module temperature and irradiance; Yan et al., [26] include ambient temperature, pressure, and humidity; Al Homoud et al., [32] source temperature and irradiance from the Qatar Environment and Energy Research Institute (QEERI); and Johannesen et al., [40] combine DryBulb and WetBulb temperatures to infer water-vapor content, enthalpy, and wind speed.

Other works mention weather information only in general terms—e.g., Czekster et al., [33] use it to optimize energy management without specifying variables—or merely imply its inclusion, as in [37, 38, 44]; Santorsola et al., [23] model solar-energy production on real solar-exposure profiles with added noise to emulate weather variation rather than modeling specific meteorological variables; Mylrea et al., [28] likewise refer only to ambient conditions generically, without specifying meteorological variables. Several papers, such as [25, 27], make no explicit reference to external environmental data. Explicit tracking of standardized weather parameters is therefore essential for accurately interpreting RES performance and isolating true anomalies from normal, weather-induced fluctuations.

Smart meter data (energy consumption, power quality)

Based on the extracted results, we outline the use of smart meter data by various researchers for different purposes, primarily focusing on monitoring and managing energy systems; these smart meters generally operate on the final consumer side:

Commonly Cited Uses of Smart Meter Data: Energy consumption and power quality are frequently mentioned as primary uses of smart meter data. This data is crucial for managing energy distribution effectively and monitoring quality [23, 28, 33, 35, 40, 44, 46]. Additionally, real-time data utilization is emphasized by several studies, highlighting the importance of real-time data from end-users and through Advanced Metering Infrastructure (AMI) for various applications, including data sharing and monitoring [25, 29].

Specific Applications and Implications: Energy production tracking is a key application noted by [22], focusing on the production side of energy management. Energy management involves an efficient balance between energy production and consumption, which is crucial for grid stability.

Additionally, smart meter data is implied in detecting false data injection (FDI) attacks as discussed by [27]. Furthermore, [37, 38] highlight the role of Advanced Metering Infrastructure (AMI) and monitoring systems for situational

awareness (SA). Moreover, load and generation monitoring is the focus of [31], which monitors both load and generation at network nodes.

This reflects varied approaches to addressing user roles in energy consumption and production across different studies.

System Integration and Advanced Uses: The collection of energy consumption data from various loads is mentioned by [32]. Additionally, [36] discuss gathering energy consumption data specifically from the external communication environment of an Energy Aware Smart Home (EASH) system. Furthermore, [42] utilize smart meter data for state estimation processes and anomaly detection.

Authors Not Covering or Not Specifying Use: Several studies, including [26, 34, 41, 43], do not cover or explicitly mention the use of smart meter data in their studies.

Consumers vs. Prosumers data (differentiation and examples) Based on the extracted results, we explore how different studies address the data related to consumers and prosumers within their studies, with varying degrees of differentiation and focus:

Explicit Differentiation: Both [23, 24] clearly address data concerning both consumers and prosumers. Additionally, [31, 33] explicitly differentiate between consumer and prosumer data in their research.

Implied Differentiation: Several studies imply a differentiation between consumer and prosumer data without explicitly detailing it. These include [22, 27, 37, 40, 44, 45].

Focus on Consumers: The study by [36] specifically focuses on consumer data, without mentioning prosumers.

Not Differentiated or Not Mentioned: Several studies, including [25, 26, 28–30], do not differentiate between consumers and prosumers in their presented data. Similarly, [39, 41, 43, 46] do not provide any discussion or details regarding differentiation between consumer and prosumer data.

This consumers and prosumers differentiation data summary, highlights that while some researchers make a clear distinction between consumers and prosumers, others imply differentiation and a significant number do not specify or differentiate at all.

Other type of data sources (System logs and maintenance records) Based on the extracted results, we explore various studies and the types of “Other Data Sources” they use in their research, with a particular emphasis on system logs, maintenance records, and operational data:

Commonly Used Data Types: Most studies extensively use system logs and maintenance records, which are often applied to simulate various scenarios, including cybersecurity incidents [22–24, 27, 28]. Additionally, operational logs or operational technology data are the focus of studies such as [26, 34], sometimes supplemented with specific methodologies like Monte Carlo simulations [32] for enhanced analysis.

Specialized Data Sources: Communication data, including network traffic, is specifically mentioned as being used

by [36, 42], which is crucial for monitoring and security analysis. Furthermore, data related to credible cyber-attacks and naturally occurring faults are utilized by [38], while [46] focus on system logs that include attack scenarios. Additionally, [31, 35] employ syslog events, system call logs, and traffic data for system monitoring and troubleshooting purposes.

Authors Not Mentioning Specific Data Types: Several studies, including [25, 40, 41, 45], do not specify or mention the use of these types of data explicitly.

3.1.3 Distinguishing data sources

We further differentiate the data sources based on their relevance to operational failures versus cybersecurity incidents. Operational anomalies often arise from equipment malfunctions or environmental conditions, requiring data from sensors and maintenance logs. In contrast, cybersecurity-related anomalies necessitate the analysis of system logs and network traffic to identify potential attacks.

Dataset Fidelity Tiers Beyond the type of data source, the realism of the evaluation data strongly influences how confidently a reported anomaly-detection result can be interpreted. To make this more explicit, the reviewed studies can be grouped into four fidelity tiers: **T1** real operational data collected from live environments; **T2** hardware-in-the-loop or hybrid settings with real devices and partially simulated infrastructure; **T3** high-fidelity simulations such as RTDS, GridLAB-D, MATLAB/OpenDSS, or similar co-simulation environments; and **T4** synthetic or purely software-generated datasets. Across the 25 reviewed studies, only 2 can be placed in T1, 3 in T2, 12 in T3, and 8 in T4. This means that 80% of the reviewed evidence base is validated without direct exposure to fully operational DER-rich conditions.

This distribution has important implications for generalization. Models trained and evaluated on T3–T4 data may report optimistic performance because measurement noise, sensor drift, communication loss, topology reconfiguration, weather-induced intermittency, maintenance artifacts, and operator workarounds are often simplified or absent. Consequently, strong results on simulated data remain useful for methodological comparison, but they should not be interpreted as equivalent to field-validated operational robustness. This issue is especially relevant for cyber-physical distinction, because the ambiguity between malicious and non-malicious events typically becomes more difficult—not less—as data realism increases.

Anomaly data utilization and sources for operational failures and safety faults (types, sources, examples)

Different data sources are used in various studies to analyze and detect operational failures: **Common Data Types:** Sensor data and smart meters are widely used by studies such as [23, 24, 35, 43] for detecting various types of operational

failures. Additionally, system logs are mentioned as crucial for analyzing failures by studies like [28, 33, 35, 38, 44].

Specific Operational Issues: Voltage and power changes, including variations in voltage, current, and power, as well as related outages, are the focus of [25, 31]. Furthermore, regulator tap changes and abnormal outputs are analyzed by [32].

Advanced Techniques and Simulations: High-fidelity simulations and high-dimensional data analysis are employed by [26, 34], who use advanced simulation techniques and time-series analysis to predict and study operational failures. Additionally, network and communication data are investigated by [36, 39], focusing on sensor nodes, communication channels, and SCADA network data, respectively.

Cybersecurity and Disturbances: Data related to cybersecurity threats, such as false data injection (FDI) and denial of service (DoS) attacks, as well as natural disturbances, are considered by [37, 41, 42].

Detailed Power System Monitoring: Detailed data from load profiles, SCADA systems, and phasor measurement units are utilized by [40, 46] for comprehensive monitoring and failure detection.

Anomaly data utilization and sources for cybersecurity incidents and attacks (types, sources, examples) Our examination provides insights into the diverse cybersecurity incident data sources used by various studies to study and detect cybersecurity threats:

Common Data Types for Cybersecurity Analysis: System logs are frequently utilized by studies such as [23, 24, 33, 35] for tracking and analyzing cyber incidents. Additionally, network and communication data are highlighted as essential for monitoring network vulnerabilities and attacks by studies like [29, 35, 36].

Specific Cybersecurity Threats and Their Data Sources: False Data Injection (FDI) and Denial of Service (DoS) attacks are addressed by [22, 30, 34, 41, 42], and others who employ advanced detection and simulation techniques. Moreover, Network Intrusion Detection Systems (NIDS) are utilized by [31], which focuses on agent-based detection systems.

Advanced Simulations and Scenario-Based Studies: High-fidelity simulations are used by [26, 37] to model and study the impact of simulated cyber-attacks. Additionally, scenario-based data for Distributed Denial of Service (DDoS) and Man-In-The-Middle (MITM) attack scenarios are employed by [39] and [42] to understand and counter these specific threats.

Physical and Operational Data Integration: Phasor Measurement Unit (PMU) and Supervisory Control and Data Acquisition (SCADA) data are incorporated by [40], [44], and [46] for a comprehensive analysis of cybersecurity vulnerabilities.

Anomaly Detection in Network Traffic: Unusual network traffic patterns are analyzed by [43, 44] to detect and respond to potential cyber threats effectively.

Diverse Cyber Threats and Manipulation: Spoofing, jamming, and manipulated historical measurements are addressed by [45] and [27], showcasing the need for robust security measures against a range of cyber manipulation techniques.

Mixed operational data or feature engineering for security and safety (methods, examples)

Drawing upon our extracted findings, we outline how various studies employ mixed data feature engineering techniques in their research to enhance the detection and analysis of cybersecurity and operational anomalies:

Integration of Data Types: studies such as [23], [28], and [44] combine operational and cybersecurity data from different systems (OT and IT) to create robust feature sets. Additionally, Appiah-Kubi et al. [31] and [46] correlate physical data with communication data and integrate data from Phasor Measurement Units (PMUs), Supervisory Control and Data Acquisition (SCADA) systems, and relays, respectively.

Advanced Machine Learning and Statistical Techniques: Studies by [22] and [30] utilize Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and other machine learning algorithms for cyber-physical threat detection. Furthermore, [25], [41], and [40] apply Principal Component Analysis (PCA), k-Nearest Neighbors (kNN), Isolation Forest, and Local Outlier Factor techniques to optimize feature selection and enhance anomaly detection capabilities.

Specialized Feature Engineering Approaches: [27] and [39] employ autoencoders and Minimum Redundancy Maximum Relevance (MrMR) methods, respectively, focusing on enhancing detection capabilities and improving model performance. Additionally, [33] and [35] discuss behavior-based and knowledge-based monitoring techniques for anomaly detection.

Utilization of Temporal and Sequential Techniques: Researchers such as [34] and [37] use sliding window techniques and Sequential Hypothesis Testing to handle time-series and sequence data, enabling real-time anomaly detection and improving the responsiveness of detection systems.

Comprehensive and Optimized Data Utilization: Studies by [29, 43, 45] focus on feature selection, optimization, and normalization to manage comprehensive datasets effectively, thereby ensuring accurate and reliable anomaly detection.

Real-Time and Historical Data Analysis: [38] combines real-time and historical data to enhance the analytical depth and reliability of detection systems, leveraging the strengths of both data types to improve overall system performance and threat identification accuracy.

3.1.4 Data temporal nature (real-time vs. historical data)

Understanding the temporal nature of data-distinguishing between real-time and historical datasets-is vital for timely anomaly detection and long-term trend analysis. Real-time data enables immediate response to anomalies, while historical data supports the development of predictive models and the refinement of detection algorithms.

Utilization of real-time data (definition, examples, and usage)

The literature reveals how various studies utilize real-time data in their research, predominantly for monitoring, anomaly detection, and immediate decision-making:

Anomaly and Intrusion Detection: Several studies emphasize the crucial role of real-time data in the immediate detection of anomalies. Notably, [23], [25, 26], [34], and [32] highlight the importance of real-time data for promptly identifying irregularities. Additionally, [27, 43, 44] focus on using real-time data to detect specific types of threats, such as false data injection (FDI) attacks and other intrusions.

Monitoring and Response: Real-time data plays an essential role in enabling immediate monitoring and response mechanisms. Studies by [24], [28], [29], and [30] underscore the necessity of real-time data for effective surveillance and swift reaction to incidents. Furthermore, [35] details the use of real-time data for monitoring cyber-physical systems operations, while [36] introduces a framework for monitoring energy consumption in real-time.

Real-Time Decision-Making: The utilization of real-time data is pivotal for facilitating immediate decision-making processes, particularly through automated systems like Advanced Metering Infrastructure (AMI). [33], [37], and [45] discuss how real-time data supports instantaneous decisions in energy management and cybersecurity contexts. Additionally, [39] and [40] mention the collection of real-time data for network monitoring and the application of statistical methods, such as correlation coefficients, for detection purposes.

Specialized Applications: Specialized applications of real-time data include high-precision grid monitoring and cyber threat identification. [41] utilizes real-time monitoring with Phasor Measurement Units (PMUs) to achieve precise grid oversight, while [42] employs real-time cloud analysis to identify and mitigate cyber threats effectively.

General Importance: Continuous data collection is highlighted as crucial for making immediate and informed decisions. Specifically, [46] emphasizes the importance of ongoing data acquisition to support timely decision-making processes in dynamic environments.

Utilization of historical data (definition, examples, and usage)

Drawing from the reviewed literature, we summarize how various studies use historical data in their research, predominantly for model training, trend analysis, and prediction purposes:

Model Training and Machine Learning: A significant number of studies, including [22, 27, 29, 30, 32, 34, 36, 42–46], utilize historical data primarily for training machine learning models. These models are employed for various purposes such as anomaly detection, predictive maintenance, and optimization.

Trend Analysis and Pattern Recognition: Studies by [23, 24, 28, 33] focus on using historical data for trend analysis and recognizing patterns over time, which helps in understanding long-term changes and planning future strategies.

Specific Applications: [25] and [40] use historical data specifically for prediction and anomaly detection, respectively. Appiah-Kubi et al. [31] employs historical data for offline planning, providing a basis for making informed decisions without real-time pressure. [37] applies historical data to optimize power dispatch, which is critical for efficient energy management. Additionally, [38, 39] use historical data for monitoring key systems and analyzing network traffic data, respectively.

Longitudinal Monitoring: [35] utilizes historical data for longitudinal monitoring, focusing on observing trends over an extended period to gauge changes and developments within critical physical systems.

3.1.5 Model application levels (purdue models vs. CPS levels)

Finally, the application levels of models, such as the Purdue Model and Cyber-Physical System (CPS) levels, are examined. These frameworks help in structuring the data analysis process and in deploying anomaly detection mechanisms at appropriate layers within the grid infrastructure.

Application of Purdue models (explanation, levels, examples) Based on our extracted results, we review the application of the Purdue Model by various authors, indicating a general lack of explicit mention or specification in most studies, with a couple of exceptions:

Not Specified or Not Mentioned: The majority of studies, such as [23], [22], and [24], either do not specify or do not mention the application of the Purdue Model in their research. This suggests either a lack of relevance to their specific studies or an oversight in detailing their methodological frameworks.

Exceptions in Application: However, [38] describes the use of hierarchical data flow and security measures, which aligns with the principles of the Purdue Model by focusing on structured, layered approaches to industrial control system

security. Similarly, [44] implements a hierarchical approach that is similar to the Purdue Model, indicating the use of layered security or organizational frameworks that resemble the model's structure.

Application of CPS Levels

Based on our extracted results, we summarize how various studies apply Cyber-Physical System (CPS) concepts across different levels and aspects of their research, highlighting integration and security within cyber-physical frameworks:

Integration of Physical and Cyber Elements: studies such as [23], [24], and [37] focus on integrating physical and cyber aspects within CPS, emphasizing comprehensive system unity. Additionally, [26, 29] discuss applications specifically within control systems and grid infrastructure.

Security and Resilience in CPS: Studies by [25] and [30] address CPS in the context of security, particularly at device and network levels. Furthermore, [28] and [33] explore methodologies for enhancing resilience and real-time monitoring in SG.

Advanced CPS Frameworks and Technologies: Researchers like [31] and [32] implement CPS frameworks for decentralized control and hierarchical data collection. Additionally, [39] and [41] incorporate SCADA systems and edge-based subsystems within CPS structures.

CPS Methodologies in Monitoring and Detection: Studies by [35], [38], and [46] utilize integrated monitoring, digital twins, and modeling across various levels for anomaly detection and simulation. Moreover, [34, 44] apply machine learning and co-simulation frameworks within CPS to enhance cybersecurity.

Simulation and System Integration: studies such as [40, 43] focus on integrating various data sources like sensors and smart meters, and using digital twins for real-time simulations within CPS. Additionally, [42] and [45] discuss cross-layered integration in SG and hierarchical, distributed intrusion detection systems (IDS) within CPS.

A final critical observation is that explicit linkage between detection methods and operational system levels remains weaker than the importance of the topic would suggest. Many papers mention CPS or layered architectures conceptually, but relatively few tailor the detector logic, data model, latency budget, and explanation needs to a clearly defined layer such as field device, substation, SCADA network, or supervisory application. As a result, the literature often discusses anomaly detection in a system-wide sense without fully specifying where, in architectural terms, the detector is meant to operate. Stronger layer-aware design would likely improve both deployability and the ability to separate cyber anomalies from safety faults in operational practice.

3.1.6 Conclusion

This comprehensive examination of data utilization and sources sets the baseline for understanding how to effectively detect anomalies in modern SG, ensuring their reliability, efficiency, and security. In summary, effective anomaly detection in DER-rich SG necessitates a holistic approach that leverages a variety of data types:

- **Operational Data:** Provides real-time insights into the physical state of the grid.
- **Environmental Data:** Helps anticipate and explain anomalies due to weather and environmental factors.
- **Smart Meter Data:** Offers detailed information on consumption and generation at the consumer level.
- **Consumer and Prosumer Activities:** Enhances understanding of energy flow dynamics within the grid.
- **System Logs and Maintenance Records:** Essential for cybersecurity monitoring and historical analysis.

The sourcing of this data involves deploying sensors, smart meters, environmental monitoring equipment, and robust data logging systems throughout the grid infrastructure. Processing this data requires advanced analytical tools, machine learning algorithms, and feature engineering techniques that can handle both real-time and historical data. By distinguishing between data relevant to operational failures and cybersecurity incidents, and integrating them through advanced analytics, the anomaly detection system becomes more accurate and comprehensive. The application of CPS levels further enhances the system's ability to integrate and process data across different layers of the grid.

Key Takeaways:

- **Holistic Data Integration:** Across the 25 reviewed studies, the most effective anomaly detection approaches utilise diverse data types—sensor, environmental, smart meter, and system log data—confirming that comprehensive data integration strengthens detection capability.
- **Advanced Analytics:** The reviewed literature demonstrates that machine learning and feature engineering are widely applied for processing the complex, high-dimensional data characteristic of DER-rich Smart Grids.
- **Anomaly Distinction:** Among the small subset of studies that attempt it (6/25), differentiating between operational and cybersecurity anomalies is shown to enhance detection accuracy, though evidence remains limited.
- **CPS Levels Utilization:** Several studies apply CPS-level thinking to structure data integration across grid layers, but explicit linkage between detection methods and operational system levels remains weaker than expected.

- **Enhanced Grid Stability:** The evidence suggests that comprehensive anomaly detection systems contribute to prompt detection and effective response, though most evaluations are conducted in simulated rather than operational environments.

In summary, the comprehensive utilization of diverse data types and advanced analytics lays the foundation for effective anomaly detection in modern DER-rich SG. This approach not only enhances the grid's ability to detect and respond to anomalies promptly but also ensures high levels of reliability, efficiency, and security. As the integration of Distributed Energy Resources continues to grow, such robust anomaly detection systems will become increasingly vital for the stability and resilience of SG.

3.2 AI/ML methodologies

3.2.1 Introduction

The escalating complexity of SG, driven by the proliferation of Distributed Energy Resources (DERs), necessitates advanced methodologies for effective anomaly detection. As DERs grow in number and variety (e.g., solar panels, wind turbines, battery systems), grid operators face an explosion of high-dimensional, real-time data that traditional monitoring tools struggle to manage. Furthermore, the increasing digitalization of control processes exposes new cyber-attack surfaces, demanding rapid, intelligent detection methods. Ensuring the reliability and security of these grids is paramount, since undetected faults or malicious intrusions can lead to cascading blackouts or compromised infrastructure. Because AI/ML algorithms excel at spotting subtle, non-linear patterns and anomalies in vast data streams—far beyond what conventional rule-based systems can manage—these techniques have emerged as critical tools in this endeavor, even though key gaps remain in their effective implementation across real-time, large-scale, and highly diverse grid conditions. Following Fig. 9 structure, we will discuss finding in more details.

To reduce conceptual overlap in the discussion that follows, we explicitly treat **Artificial Intelligence (AI)** as the umbrella category, **Machine Learning (ML)** as the data-driven subset of AI, and **Deep Learning (DL)** as the neural-network-based subset of ML. This distinction is important because several reviewed papers use the terms interchangeably even when they describe materially different modeling assumptions, training requirements, and deployment implications. In the revised synthesis, supervised and unsupervised learning methods are treated separately from deep learning architectures, while graph-based or formal AI methods, explainable AI (XAI), system identification



Fig. 9 Topics mapping under artificial intelligence/machine learning category/group 2 shown in Fig. 6

paradigms, and operational system-level linkage are discussed as complementary but analytically distinct topics.

3.2.2 Supervised and unsupervised learning methods

We focus on the specific gaps and demonstrated effectiveness of the ML and AI techniques that are most promising

for accurately detecting anomalies in DER-rich smart grid environments.

Supervised learning models such as Random Forests and Support Vector Machines (SVMs) are explored for their ability to classify grid behaviors based on labeled data, making them effective in scenarios where historical anomaly data is available. Unsupervised learning models, including K-means clustering and autoencoders, are highlighted for their capacity to identify patterns and detect anomalies without prior labeling, which is particularly useful in the ever-evolving landscape of SG. Reinforcement learning models are also examined, showcasing how they learn optimal detection strategies through interactions with the grid environment.

Supervised Machine Learning Models Our review provides insights into the use of supervised machine learning models by various authors, specifically highlighting their effectiveness in different contexts.

Random Forest learning method is found effective for feature selection and intrusion detection, particularly in smart grid cybersecurity [30]. It is praised for its high accuracy and stability, especially in handling imbalanced datasets [29] and detecting DDoS attacks [39]. Additionally, Random Forest is used to differentiate between cyber-attacks and natural disturbances [40] and is noted for its robustness against overfitting and its capability to handle large datasets [44].

Support Vector Machines (SVM) are combined with PCA and found effective in fog computing environments [24]. They are highlighted for performing well in intrusion detection scenarios [29, 30] and are used for detecting non-technical losses and inaccuracies in load forecasting.

Decision Trees and their variants, such as Isolation Forest, are effective in identifying outliers [25]. They are used for classifying grid events and detecting DDoS attacks [39]. Additionally, Decision Trees along with Ada Boost and Extra Trees are utilized for effective multi-class classification [44].

Other supervised models include Extreme Learning Machines (ELMs), which are employed for efficient training and good generalization performance in Cyber-Physical Systems (CPS) [26]. Logistic Regression and Linear Discriminant Analysis (LDA) are also used for multi-class classification problems, demonstrating their effectiveness in handling complex data distributions [44].

Unsupervised Machine Learning Models Autoencoders are effective for detecting subtle anomalies in system behaviors, making them suitable for nuanced and complex anomaly detection [22]. They are utilized specifically for detecting False Data Injection (FDI) attacks due to their ability to reconstruct input data and identify deviations [27]. Additionally, autoencoders are used for dimensionality reduction, aiding in anomaly detection by simplifying the input features [30]. Linear Autoencoders are employed for dimensionality reduction and detecting cyberattacks on transformer protective relays, enhancing the ability to process and analyze large

datasets [34]. Deep Auto-Encoders capture deep features of data, proving effective in detecting complex anomalies [41].

K-means clustering analyzes unknown attack patterns, effectively grouping similar data points to identify unusual clusters that may indicate attacks [30].

Other unsupervised models include Stacked Denoising Autoencoders (SDAEs), which learn useful representations from noisy data, improving the clarity and usability of data under real-world conditions [26]. The Local Outlier Factor (LOF) is used for unsupervised, density-based local anomaly detection: each point is scored by its local density deviation from its k -nearest neighbours, and points whose score exceeds a threshold are flagged as anomalies [32]. Additionally, 1D-CNNs are utilized to recognize localized anomalies, highlighting their capability to handle spatial data within one-dimensional inputs [34]. High-fidelity Digital Twin models distinguish between normal operations and attack scenarios, enhancing security measures by simulating different conditions [38].

Reinforcement Learning Models Santorsola et al., [23] uses tabular Q-learning with a decayed- ϵ -greedy exploration policy. Distinctively, it is the only study in the corpus to apply reinforcement learning *offensively* rather than as a detector: operating at the Modbus protocol level, the agent searches for sequences of *individually legitimate* control commands (breaker-open coil writes) that, taken together, drive a simulated DER-rich grid into a safety-critical state in which prioritized consumers are disconnected, with a reward proportional to the number of newly compromised consumers. The discovered command sequences are then arranged into a weighted directed graph of grid states, ranked by a severity score (the ratio of accumulated node-relevance weights to path depth), and rated for impact using NIST CVSS metrics. The contribution is thus the systematic *discovery and severity-ranking of harmful command patterns*, not their runtime detection—a framing whose relevance to the review's central question is taken up in Sect. 3.7.

[30], a review, discusses reinforcement learning as a futuristic, self-learning approach to anomaly detection rather than implementing one itself (its own case study is a deep-neural-network controller). The appeal it highlights is reinforcement learning's capacity to continuously adapt and improve detection strategies from new data, suiting dynamic environments where attack patterns evolve.

Appiah-Kubi et al. [31] uses the RL model named Q-learning to learn and predict an attacker's next move without prior knowledge. This model is particularly useful in scenarios where attackers are constantly changing their tactics. By learning from each interaction, the model can anticipate future actions without needing explicit prior knowledge of attack methods.

Other Machine Learning Models Enhanced Detection and Data Integrity are achieved by [24], who utilizes Blockchain

to ensure data integrity and security, crucial for reliable anomaly detection. [25] applies PCA to improve the performance of Isolation Forest by identifying the most relevant features. Yan et al., [26] integrates machine learning with physics-based modeling to enhance detection accuracy, particularly in complex environments.

Advanced Learning Models include [29], which implements ensemble learning by combining CNN and MLP to improve classification accuracy. Jahromi et al. [34] uses LSTM to handle time series data effectively, particularly adept at detecting non-local, long-term anomalies.

Decentralized and Privacy-Preserving Approaches are employed by [31], which utilizes Multi-agent Systems to correlate attacks decentralizedly, enhancing the robustness of predictions. Both [38, 41] apply Federated Learning to train models collaboratively without exchanging raw data, thus preserving privacy.

Hybrid and Specialized Techniques include [37], which develops a Hybrid ML approach (Sequential Hypothesis Testing) that minimizes decision errors in detecting data integrity attacks. [39, 40] use ensemble methods and local outlier factors respectively to improve prediction performance and detect anomalies by analyzing data density.

Anomaly Detection in Complex and Dynamic Environments is addressed by [42], which utilizes ECD-AS, adapting to real-time Smart Grid data for anomaly detection using the Mahalanobis distance. [45] implements Online nonparametric anomaly detection (ODIT), suitable for high-dimensional, dynamic environments.

Enhanced Classification Techniques are demonstrated by [43], which enhances K-Nearest Neighbor (EKN) to handle complex data with heterogeneous views effectively, and by [44], which applies K-Neighbors (KNN) for classification tasks involving non-linear data distributions.

Reading Table 4 column by column, a precise recount shows that 19 of the 25 studies report no single comparable headline value (NR), two further studies (Agnew 2022 [42] and Hyder 2023 [44]) report only qualitative, scenario-dependent characterizations, and four studies report a directly extractable quantitative score. A *partial comparison* is therefore possible on this quantitative subset: Shukla et al. [24] reach 89% (fog) and 78% (cloud) accuracy with a blockchain-supported linear SVM; Reddy Shabad et al. [25] report 98.74% accuracy with Isolation Forest and PCA; Amritha et al. [27] report 98% accuracy for autoencoder-based FDI detection; and Naeem et al. [29] report up to 99% accuracy with deep ensemble learning. Even within this small subset, the values are computed on different datasets, anomaly definitions, and class balances, so they should be read as evidence that high within-study accuracy is attainable, not as a ranking between methods. Two clarifications temper the interpretation of the NR count. First, NR refers to the absence of a directly comparable headline value, not to

the absence of evaluation: several NR-marked studies report metric *types* (e.g., EER/ROC in Shen et al. [37], F1/ROC-AUC in Oyucu et al. [39]) without a single extractable score for the targeted anomaly class. Second, quantitative results are common in the adjacent literature outside the strict six-group intersection, so the pattern observed here is best framed as a *comparability and reporting-consistency* problem within the focused corpus—heterogeneous datasets, metrics, and reporting granularity preventing cross-study benchmarking—rather than as a field-wide absence of metrics; we treat this comparability gap, and the resulting need for shared benchmarks for the attack-versus-fault task in particular, as a finding of the review in its own right (see Sect. 3.7).

A critical cross-study comparison nevertheless reveals several recurring trade-offs. Random Forest and related ensemble methods are repeatedly valued for robustness to noisy and imbalanced data, whereas SVM-based approaches remain attractive in lower-dimensional or better-curated settings. Autoencoder-style unsupervised methods are especially useful when attack labels are scarce, but they often trade interpretability for flexibility and can be sensitive to the realism of the training distribution. Reinforcement learning contributes adaptability, yet only in a small number of papers and usually within tightly controlled environments. Most importantly, many so-called “hybrid” studies still use physics mainly to generate data rather than to constrain the model itself. This suggests that the literature is still short of truly integrated cyber–physical detection architectures in which domain equations, topology constraints, or protection logic are embedded directly into the learning process.

3.2.3 Graph-based and formal AI methods

The integration of AI techniques such as graph methods, Petri-Nets, and Bayesian Networks provides formal frameworks that, when coupled with AI, enhance the modeling of complex grid dynamics and anomaly detection processes. The utilization of Explainable AI (XAI) techniques like LIME and SHAP addresses the critical need for transparency and interpretability in AI models, fostering trust and facilitating better decision-making by grid operators.

Graph Methods Our review indicates that none of the surveyed authors have covered or utilized mixed AI formal techniques involving graph methods in their research. This absence suggests that graph methods or their integration with other AI techniques may not have been relevant or explored within the specific contexts of these authors’ studies. It also indicates a potential area for future research where graph-based AI methods could be explored for their applicability and effectiveness in various domains.

Petri-Nets Our review indicates that none of the surveyed authors have covered or utilized Petri-Nets in conjunction

Table 4 Cross-study comparison of AI/ML techniques, targets, evaluation environments, and reported outcomes

Study	Main AI/ML technique	Target anomaly	Dataset/environment	Primary metric	Best reported score
Santorsola 2023	Q-learning RL	Safety-critical states	Simulated smart-grid ICS	Safety-critical state discovery/reward	NR
Ye 2022	Survey of RF/CNN/LSTM etc	PV cyber-physical attacks	Reported PV case studies/testbeds	Review synthesis	NR
Shukla 2022	Linear SVM + blockchain	Network anomalies	iFogSim/fog-cloud setup	Accuracy	89% fog/78% cloud
Reddy 2021	Isolation Forest + PCA	Grid anomalies	Smart-grid dataset/testbed	Accuracy	98.74%
Yan 2019	ELM/SDAE/physics-aided ML	Malicious actions	High-fidelity gas-turbine setup	Detection accuracy	NR
Amritha 2022	Autoencoder	FDI attacks	Power-system simulation	Accuracy/precision/recall	98% accuracy
MyIrea 2021a	Digital twin + XAI	Process/cyber anomalies	Biopharma digital twin	Detection effectiveness	NR
Naeem 2024	Deep ensemble learning	Intrusion attacks	Smart power-grid intrusion data	Accuracy	up to 99%
Suprabhath 2023	CNN/RNN/LSTM/GRU survey	Microgrid cyber-attacks	Reported lab microgrid cases	Review synthesis	NR
Appiah-Kubi 2023	Multi-agent ML/RL	Coordinated cyber-attacks	IEEE feeder simulation	Detection reliability	NR
Al Homoud 2021	LOF (local detection)	Coordinated attacks/anomalies	Simulated distribution network	F1/local accuracy	NR
Czekster 2022	Analytics/simulation roadmap	Building cyber risks	Active-building CPS scenarios	Review synthesis	NR
Jahromi 2022	LSTM/autoencoders (linear, FC, ID-conv)	Transformer-relay anomalies	IEEE benchmark system	Precision/recall (PR curves)	NR
Fournaris 2019	Runtime secure-CPS monitoring	Runtime discrepancies	Secure CPS test scenarios	Error balancing	NR
Tertytchny 2019	Supervised ML	Attack-fault differentiation	Smart-home testbed	Accuracy, kappa statistic	NR
Shen 2023	Hybrid ML + SHT + DT	Data-integrity attacks	Hybrid microgrid simulation	Detection rate, FPR (SHT)	NR
MyIrea 2021b	AI-driven Digital Ghost	Cyber-physical threats	Critical-infrastructure DT	Detection effectiveness	NR
Oyucu 2024	Ensemble learning	DDoS in SCADA	SDN-based SCADA simulation	Accuracy/F1/sensitivity/specificity	NR
Johannesen 2022	kNN/iForest/LOF	Grid anomalies	Real smart-meter scenarios	Comparative anomaly score	NR
Husnoo 2023	Federated learning + DAE	Disturbance vs. cyberattack	PMU/RTDS environment	MSE/discrimination accuracy	NR
Agnew 2022	SDN + ML + state estimation	Cross-layer threats	IEEE-system simulation	Precision/recall/F1	high, scenario-dependent
Padmapriya 2024	EKNN + digital twin	IoT cyber-attacks	IoT/digital twin environment	Detection accuracy	NR
Hyder 2023	Comparative ML on high-fidelity data	Cyber-physical anomalies	High-fidelity co-simulation	Accuracy/precision/recall/F1	high, scenario-dependent
Yilmaz 2017	ODIT/hierarchical IDS	IoT cyber-attacks	Smart-grid IDS framework	Detection delay/false-alarm rate	NR
Chhokra 2021	Power-attack simulation chain	Power-system attacks	Attack simulation tool-chain	–	NR

The table is intended as an analytical comparison aid rather than a universal ranking, because the studies differ substantially in anomaly definitions, datasets, realism tiers, and performance-reporting practices. NR (“not reported”) indicates that the primary study does not state a single comparable headline value for the listed metric in the reviewed text; it does not imply that the study performed no evaluation

with AI techniques in their research. This absence suggests that Petri-Nets, known for their use in modeling discrete event systems and processes, may not have been deemed relevant or necessary within the specific domains or objectives of these authors' studies. This might indicate a gap or an area of potential exploration for future research in integrating Petri-Nets with AI to enhance modeling and analysis capabilities in relevant fields.

Bayesian Networks Our review indicates that none of the surveyed authors have covered or utilized Bayesian Networks in conjunction with AI techniques in their research. This absence suggests that Bayesian Networks, which are typically used for probabilistic inference and decision-making under uncertainty, may not have been relevant or explored within the specific contexts of these authors' studies. This highlights a potential area for future research where Bayesian Networks could be integrated with AI to enhance predictive analytics and decision-making processes in various fields.

Other Formal Methods Our review indicates that none of the surveyed authors have covered or utilized other formal methods in conjunction with AI techniques in their research. This suggests that such methods, which might include various mathematical modeling techniques, logic-based approaches, or complex algorithms outside the typical AI spectrum, have either not been relevant, recognized, or explored within the specific studies and contexts of these authors. This could highlight a research opportunity for integrating formal methods with AI to potentially enhance rigor, precision, and the scope of analyses in various disciplines.

Utilization of Explainable AI (XAI) Techniques XAI Utilization is not directly discussed in [23], but the use of reinforcement learning suggests a potential area where explainable AI techniques could help interpret model decisions and actions, enhancing transparency and trustworthiness in future studies.

[28] generally uses XAI techniques. These techniques improve understanding of AI decisions within cybersecurity contexts, helping stakeholders and users comprehend how and why certain decisions are made by AI systems. This is crucial in cybersecurity to validate and trust the defensive measures being proposed by the AI.

[29] uses the XAI technique SHAP (SHapley Additive exPlanations). SHAP is used to interpret and evaluate the decisions of deep learning models, particularly valuable for understanding the contribution of each feature in the output decision. This method enhances model transparency and aids in fine-tuning the model by understanding feature impacts.

A critical observation across this subsection is not simply that graph methods, Petri nets, and Bayesian networks are absent, but that this absence removes entire classes of relational and causality-aware reasoning from the smart-grid anomaly-detection literature. Smart grids are naturally structured systems with topology, dependencies, communication

routes, protection logic, and propagation effects. Methods that explicitly represent relations—rather than only feature vectors—could therefore be particularly valuable for distinguishing cyber-induced anomalies from physics-driven disturbances. The current literature is still dominated by vectorized data-driven formulations, which limits the explicit modeling of causality, event propagation, and attack paths.

3.2.4 Explainable AI (XAI) techniques: LIME and SHAP

Explainability is not merely an optional usability feature in DER-rich smart grids; it is directly tied to whether a detection result can be trusted, validated, and acted upon by operators working under time pressure. In practical grid settings, an anomaly flag often triggers operational judgment, and that judgment requires more than a probability score. Tools such as **LIME** and **SHAP** therefore matter because they can expose the local feature contributions, directional influence, or marginal impact associated with a model's decision. This is particularly relevant when the same observed anomaly could plausibly stem from a cyber-attack, a control misconfiguration, a sensor malfunction, or a genuine safety fault.

Among the reviewed papers, explicit use of XAI remains limited. [29] uses SHAP to interpret a deep-ensemble intrusion model; [28] discusses explainable reasoning at a higher level within a digital-twin cybersecurity context; and several other studies implicitly motivate explainability by stressing operator trust, system accountability, or validation needs. However, there is still no widespread practice of reporting explanation fidelity, explanation stability over time, or operator-oriented explanation usability in the evaluated smart-grid scenarios.

The challenges of deploying XAI in operational environments are more substantial than a brief model-interpretation add-on might suggest. First, **computational overhead** can be significant when explanation methods must run close to real time. Second, **temporal instability** arises because local explanations may vary sharply across adjacent time windows even when the underlying grid condition evolves smoothly. Third, there is often a **domain mismatch** between generic tabular XAI tooling and multivariate time-series power-system data, especially when topology, control interactions, or protection-state dependencies matter. Fourth, there remains an **operator comprehension gap**: even when a feature-importance plot is technically correct, it may not translate naturally into actionable reasoning for DSO/TSO staff or protection engineers.

For these reasons, a promising direction is not only to reuse general-purpose XAI tools, but also to develop *physics-grounded XAI* in which explanations are expressed through variables and constraints meaningful to grid operation, such as feeder state, relay logic, topology consistency, or power-flow imbalance. In other words, explainability is likely to

become practically valuable only when it is aligned simultaneously with machine-learning transparency and with the semantics of electrical-system operation.

A final critical comparison can therefore be made between post-hoc XAI and intrinsically structured models. Post-hoc tools such as SHAP and LIME are attractive because they can be applied to existing black-box models, but they inherit the limitations of those models and may provide only local or unstable interpretations. By contrast, grey-box or physics-constrained models may sacrifice some flexibility yet offer explanations that are closer to engineering reasoning from the outset. This trade-off is especially important in high-risk smart-grid use cases, where “why” an alert occurred can be almost as important as whether the alert was correct.

In this sense, although the literature often frames these approaches under the label of explainability, the methods identified in the reviewed studies do not yet provide full explainability in a strict operational or epistemic sense; rather, they function more as transparency-support mechanisms that offer partial insight into model behavior without fully clarifying the underlying causal reasoning behind an alert. As argued by Rudin [76], particularly in high-stakes settings, post-hoc explanations for black-box models should not be mistaken for genuine interpretability, because they may provide only an incomplete or misleading account of the true decision process. Lipton [77] similarly notes that transparency, interpretability, and post-hoc explanation are often conflated in the literature, despite their important conceptual differences. From this perspective, the XAI methods found in DER-rich smart-grid studies remain closer to transparency-oriented tools than to true explainability, because they reveal selected feature influences or local decision patterns without fully capturing the physical, causal, or engineering rationale behind why an alert was produced and their “explainable” results remain vulnerable and foolable such as demonstrated by [78].

3.2.5 Deep learning architectures

Deep Learning models have gained traction due to their ability to handle large-scale and high-dimensional data inherent in SG. Convolutional Neural Networks (CNNs) are effective in capturing spatial correlations in grid data, while Recurrent Neural Networks (RNNs) and Long Short-Term Memory Networks (LSTMs) excel in modeling temporal dependencies, crucial for time-series data analysis. Graph Neural Networks (GNNs) are particularly pertinent for SG, as they naturally model the grid’s network topology, enhancing the detection of anomalies that propagate through the grid’s interconnected nodes.

Convolutional Neural Networks (CNN) [22] reviews CNN–LSTM combinations, which provide robust feature extraction and are effective in learning temporal sequences, making

them suitable for applications where both spatial and temporal patterns must be recognized.

[30] surveys deep architectures such as CNN, RNN, LSTM, and GRU for cyber-physical security. These models are particularly effective in identifying intricate patterns and dependencies in data, useful for complex anomaly detection scenarios.

Jahromi et al. [34] utilizes 1D-CNN, an efficient model for localized anomaly detection, enabling precise identification of anomalies in one-dimensional data streams, such as time series from sensors or network traffic.

Recurrent Neural Networks (RNN) Recurrent Neural Networks (RNN) and their variants receive limited coverage, with explicit use documented in only one of the surveyed studies.

Jahromi et al. [34] is the one corpus study to employ a recurrent model: it benchmarks an LSTM neural network against three separate autoencoder variants—linear, fully connected, and 1D-convolutional—for detecting cyberattacks on transformer differential protective relays. The LSTM, which processes the temporal relay-measurement sequences, performs best against cyberattacks targeting asymmetrical-fault data, whereas the autoencoders (including the 1D-convolutional one) perform better on symmetrical-fault cases. These are distinct, benchmarked models rather than a single combined architecture; the deep auto-encoder of [41] is likewise reconstruction-based rather than recurrent (cf. Table 3), and no surveyed study documents an explicit combined CNN–LSTM architecture.

Graph Neural Networks (GNN) Our review indicates that none of the surveyed authors have covered or utilized Graph Neural Networks (GNN) in their research, as per the provided information. This lack of mention suggests that either GNNs were not relevant to the specific areas of study conducted by these authors, or that their potential applications in these research areas have yet to be explored or documented within the scope of this particular survey.

Long Short-Term Memory Networks (LSTM) Long Short-Term Memory Networks (LSTM) likewise receive limited coverage among the surveyed authors, with only one noted application.

Jahromi et al. [34] utilizes LSTM for detecting asymmetrical faults. LSTMs are particularly effective in this context due to their ability to learn and remember over long sequences, which is crucial for recognizing patterns that lead to asymmetrical faults in systems over time.

This sparse mention suggests that while LSTM has proven applications in scenarios requiring the analysis of time-series data or where pattern recognition over time is critical, it may not have been broadly explored or deemed necessary by other authors in the scope of their respective research topics.

A critical comparison of the deep-learning evidence suggests that CNNs are typically favored when local spatial

patterns or compressed feature extraction dominate, whereas LSTM-style models are selected when temporal dependence, state evolution, or delayed effects are central. CNN–LSTM hybrids are therefore appealing for cyber–physical grids because they can absorb both local structure and temporal dynamics. At the same time, the literature reveals a notable blind spot: despite the graph-structured nature of power grids, explicit GNN adoption is absent from the reviewed set. This absence matters, because topology-aware modeling may be especially relevant when one seeks to distinguish abnormal measurements caused by network attacks from those caused by true propagation of physical faults across connected components.

Other Deep Learning Models Our review indicates that none of the surveyed authors have covered or utilized other advanced deep learning models such as Generative Adversarial Networks (GANs) or Transformers in their research. This absence suggests that either these models were not relevant to the specific domains of study conducted by these authors, or their potential applications in these areas have yet to be explored or documented within the scope of this particular survey.

3.2.6 System identification paradigms (black-box, grey-box, white-box)

Identification techniques are also a focal point, with blackbox models offering simplicity and ease of use, greybox models providing a balance between interpretability and complexity, and whitebox models delivering complete transparency at the expense of computational intensity. Understanding the strengths and limitations of each aids in selecting the appropriate approach for specific grid scenarios.

Blackbox Our review indicates that none of the surveyed authors have covered or utilized blackbox identification techniques in their research. This absence suggests that such techniques, which involve modeling a system solely based on input–output observations without insight into its internal workings, were either not relevant or explored within the specific contexts of these studies. This might point to an unexplored area where blackbox identification could potentially be applied to enhance understanding and modeling of complex systems in future research.

Greybox Our review indicates that none of the surveyed authors have covered or utilized greybox identification techniques in their research. Greybox identification, which combines both system knowledge (structural or first-principles) and data-driven methods to model systems, has not been explored within the specific contexts of these studies. This absence highlights a potential area for future research where greybox identification could be employed to enhance modeling accuracy by leveraging partial knowledge of the system’s internal mechanisms along with observational data.

Whitebox Whitebox identification techniques are used by only one author, Santorsola et al., [23]; most other studies do not cover this approach. [23] details whitebox identification as the modeling of the Smart Grid environment, including power flow, node roles, and communication protocols. This comprehensive approach allows for systematic exploration of safety-critical states through simulations and reinforcement learning (RL) techniques, providing a thorough understanding of the system’s behavior under various conditions.

Concerning the other studies, the absence of whitebox identification techniques in their research implies that this method, which relies on detailed knowledge of the system’s internal workings, might not have been relevant or feasible within the scope of their studies. This could indicate a potential area for future exploration where detailed system modeling could enhance the understanding and control of complex environments.

From a comparative standpoint, black-box models remain attractive because they are fast to deploy and can leverage large empirical datasets, but they offer limited guarantees when operating conditions shift. White-box approaches are the most interpretable, yet they require extensive physical knowledge and are costly to maintain in highly heterogeneous DER-rich settings. Grey-box paradigms occupy a particularly promising middle ground because they can incorporate partial physical knowledge while remaining data-adaptive. This is one reason why physics-informed neural networks and related constrained-learning approaches deserve more attention in future work: they can be understood as a modern extension of grey-box thinking in which governing equations and data-driven learning coexist rather than compete.

3.2.7 Linkage to operational system levels

Lastly, the application of hierarchical models like the Purdue Model and Cyber-Physical System (CPS) levels is discussed in relation to anomaly detection. These models help structure and segregate the grid into different levels or layers, allowing for targeted application of AI/ML techniques and more effective anomaly detection across the system.

Purdue Model Link Our review indicates that none of the surveyed authors have applied Purdue models in their research with relevance to anomaly detection. This absence suggests that the Purdue Model, a hierarchical framework commonly used in industrial control systems to organize and secure different levels of operations, has not been explored or deemed necessary within the context of their studies focused on anomaly detection. This could highlight a potential area for future research, where integrating Purdue models might enhance the structure and efficacy of anomaly detection systems.

Cyber-Physical System (CPS) Levels Link Our review indicates that none of the surveyed authors have applied Cyber-Physical System (CPS) levels in their research with relevance to anomaly detection. This absence suggests that CPS levels, which involve the integration and interaction of computational (cyber) and physical processes within engineered systems, have not been utilized or considered necessary within the specific context of anomaly detection by these authors. This highlights a potential area for future exploration where incorporating CPS levels might enhance the capability and precision of anomaly detection systems.

3.2.8 Conclusion

This section has examined various AI and ML methodologies and their practical applications within the context of DER-rich SG, aiming to elucidate the reasons for their effectiveness in enhancing grid reliability, efficiency, and security amid growing complexity.

Reasons for Effectiveness:

- **Handling High-Dimensional and Complex Data:** Advanced models like CNNs and LSTMs effectively manage large-scale data, capturing intricate patterns in grid operations.
- **Adaptive Learning in Dynamic Environments:** Reinforcement learning models adapt to new anomalies and evolving threats, crucial for dynamic grid environments.
- **Improved Accuracy and Robustness:** Supervised models such as Random Forests and SVMs provide high accuracy in anomaly classification when sufficient labeled data is available, while unsupervised models detect unknown anomalies without prior labeling.
- **Transparency and Interpretability:** Explainable AI (XAI) techniques ensure that AI decisions are transparent and understandable, increasing trust among grid operators and aiding regulatory compliance.
- **Integration of Cyber and Physical Data:** Combining data from both domains enhances the detection of anomalies that may not be apparent when considering one domain alone.
- **Support for Real-Time Monitoring:** These methodologies facilitate immediate detection and response to anomalies, which is crucial for maintaining grid stability.

By leveraging these advanced AI/ML methodologies, SG can effectively manage the challenges posed by the integration of Distributed Energy Resources (DERs) and the increasing complexity of grid operations. Their effectiveness stems from their ability to process vast and intricate data, adapt to new patterns or anomalies, enhance grid security

by detecting cyber threats, and support real-time monitoring and response.

Key Takeaways:

- **Advanced Data Handling:** The reviewed studies confirm that AI/ML techniques are widely adopted for processing the high-dimensional and complex data inherent in DER-rich SG, with Random Forest, SVM, and autoencoder-based methods being most prevalent.
- **Dynamic Adaptability:** Reinforcement learning models are applied in 3 of 25 studies, demonstrating potential for adaptive threat detection, though evidence remains limited.
- **Enhanced Accuracy:** The combined use of supervised and unsupervised models is shown to improve detection accuracy in several studies, although direct comparison on identical datasets is lacking.
- **Trust through Transparency:** XAI techniques (SHAP, LIME) are discussed or applied in only 2 of 25 studies (8%), representing a significant gap given the safety-critical nature of grid operations.
- **Holistic Anomaly Detection:** Studies that integrate cyber and physical data domains report more comprehensive detection, but only 6/25 explicitly attempt the cyber-physical distinction.
- **Improved Grid Stability and Security:** Real-time monitoring is the most common operational context (>20/25 studies), though most evaluations are simulation-based rather than field-deployed.

In conclusion, the application of sophisticated AI and ML techniques in SG significantly contributes to the enhancement of reliability, efficiency, and security. As the complexity of grid operations continues to grow with the integration of DERs, these methodologies become increasingly vital for the stability and resilience of modern energy systems.

3.3 Application domains

3.3.1 Introduction

The evolution of electrical power systems into Distributed Energy Resource (DER)-rich SG has introduced unprecedented operational complexities. Anomaly detection systems have become essential tools in navigating these complexities, ensuring the stability, efficiency, and security of the grid. This section will detail the findings under group 3 as shown in Fig. 10. The operational contexts in which anomaly detection systems are most commonly applied will be examined as well as grids environments, the entities involved, and the outcomes achieved.

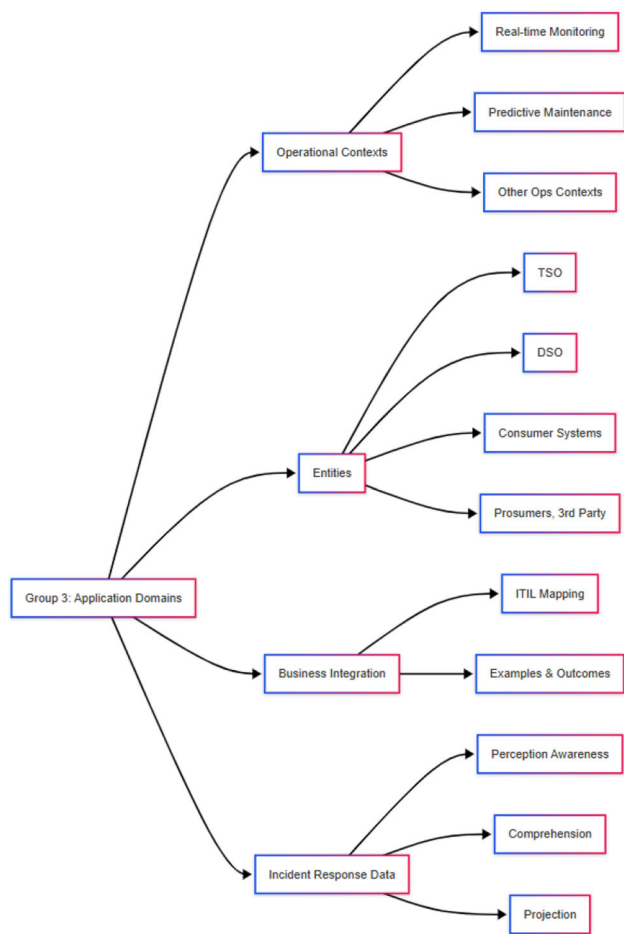


Fig. 10 Topics mapping under application domains category/group 3 shown in Fig. 6

3.3.2 Operational contexts

One of the primary operational contexts for anomaly detection is **real-time monitoring**. Real-time monitoring involves the continuous observation of grid parameters—such as voltage, current, and frequency—to detect irregularities instantaneously. For example, real-time detection of voltage fluctuations can prompt immediate corrective actions, preventing equipment damage and power outages. The effectiveness of real-time monitoring lies in its ability to provide instantaneous feedback, enabling operators to maintain grid stability amid fluctuating demand and supply conditions inherent in DER integration.

Another critical context is **predictive maintenance**. Predictive maintenance leverages historical and real-time data to forecast potential equipment failures before they occur. For instance, analyzing the thermal signatures and vibration patterns of transformers can predict degradation, allowing for maintenance to be scheduled proactively. This approach not only reduces downtime but also extends the lifespan of crit-

ical infrastructure, proving highly effective in cost savings and reliability enhancement.

Beyond these, anomaly detection systems are applied in other operational contexts such as **fault detection** and **load balancing**. Fault detection focuses on identifying and isolating faults within the grid to prevent cascading failures. For example, detecting a line fault promptly can isolate the issue and reroute power, minimizing the impact on consumers. Load balancing ensures the equitable distribution of electricity across the network, which is particularly challenging with the intermittent nature of renewable energy sources like solar and wind. Effective anomaly detection in these areas leads to improved grid resilience and optimized performance.

Cross-study synthesis of operational contexts Rather than enumerating each study per context, the per-study mapping is consolidated in Table 5, and only the cross-study pattern is analyzed here. Three observations stand out. First, *real-time monitoring is universal*: all 25 studies operate in, or explicitly target, a real-time or near-real-time detection loop, whether on grid telemetry (e.g., [25, 32, 42]), on metering and AMI data (e.g., [37, 40]), on industrial process signals (e.g., [22, 26]), or through digital-twin and co-simulation platforms (e.g., [28, 38, 43, 44]). Second, *predictive maintenance is secondary and mostly implied*: 14 of the 25 studies support it explicitly or by strong implication—typically as a by-product of anomaly detection, forecasting, or digital-twin process-integrity monitoring—while 11 studies do not address it at all (Table 5, “—” entries); where it appears, it complements rather than replaces established contingency criteria such as $(n - 1)/(n - 2)$ protection and routine or fault-based maintenance. Third, *other operational contexts cluster around fault detection and, less often, load management*: fault detection is touched by nearly every study, usually folded into the same detector that targets attacks, whereas load balancing appears only as an indirect benefit in a minority of works, and business-process or maintenance-workflow integration is essentially absent.

The analytical significance of this pattern is twofold. On the one hand, the universality of real-time monitoring shows that the field has converged on *detection latency* as the dominant design objective: virtually every contribution is built to flag an anomaly quickly. On the other hand, the same pattern exposes a blind spot directly relevant to this review’s central question: detection is not attribution. A real-time alarm states that something is wrong, but the operational follow-up—dispatching a maintenance crew for a physical fault versus activating an incident-response process for a cyber-attack—requires knowing *why* it is wrong. The thin coverage of predictive maintenance, diagnosis-oriented workflows, and response integration indicates that most pipelines stop at the alarm, which helps explain why only 6 of the 25 studies attempt the attack-versus-fault distinction (Sect. 3.7): the dominant real-time-detection framing simply does not force

Table 5 Operational contexts

Refs.	Real-time (RT) Monitoring	Predictive maintenance	Other operational contexts
[23]	Offline RL agent probes a monitoring environment that ranks loads by power-availability scores in real time	Reveals command sequences that precede safety-critical states, informing proactive prevention	Offensive attack-path discovery via reinforcement learning
[24]	Linear-SVM anomaly detector in fog layer for smart-grid metrics	Early-warning anomalies support preventive actions	Secured communication aids fault detection and load balancing
[25]	ML models track voltage/current continuously	Predicts future grid parameters to schedule service	Fault isolation with implied load balancing
[32]	Live asset monitoring to flag incidents	Schedules repairs ahead of failure windows	Fault detection plus continuous load balancing
[33]	Real-time DER analytics in active buildings	Advanced analytics feed predictive maintenance	Cyber-physical security roadmap (risk assessment, intrusion detection, threat modelling)
[45]	MIAMI-DIL framework detects, isolates and localises attacks instantly	-	Cyber-attack detection, isolation, and localization
[27]	Autoencoder flags FDI attacks in real time	-	FDI-attack detection (correlation-based)
[28]	Digital-twin “BioSecure” system gives continuous oversight	Supports predictive analytics for integrity management	Fault and cyber-attack detection (process integrity)
[29]	Deep-ensemble model for live cyber-attack recognition	-	Intrusion-detection focus
[31]	Multi-agent, decentralised real-time attack monitoring	Strengthens resilience (supports predictive functions)	Coordinated cyber-attack detection and distribution reconfiguration
[41]	Federated learning monitors power-system telemetry in real time	-	Distinguishes genuine faults from cyber-attacks
[42]	SDN + ML platform mitigates attacks in real time	Techniques feed maintenance analytics	State-estimation-based cyber-attack detection (FDI/DoS/MITM)
[22]	Sensor network tracks PV-farm performance continuously	Failure-prediction capability for scheduled service	Detects cyber-physical attacks/equipment faults
[26]	Live surveillance of gas-turbine control signals	-	Malicious-activity detection
[30]	Real-time deep-learning control of DC-DC converters	-	Mitigates stealth local covert attacks (attack detection and mitigation)
[35]	Continuous CPS performance/security assessment	Monitoring data feed predictive analytics	Cyberattack/anomaly detection via run-time monitoring (discrepancy analysis)
[36]	Implied RT monitoring of system-state deviations	-	Supervised ML differentiates cyber-attacks from faults on the communication channel
[40]	Evaluates RT smart-meter anomaly algorithms	Forecasts disruptions for maintenance scheduling	Anomaly detection across grid scenarios (cyber-attacks and natural disturbances)
[43]	Digital-twin simulation with live anomaly signals	Predicts failures and security breaches	Process-aware cyber-attack simulation and EKNN detection
[44]	Co-simulation platform monitors cyber-physical dynamics in real time	-	Simulates attacks and physical faults

Table 5 continued

Refs.	Real-time (RT) Monitoring	Predictive maintenance	Other operational contexts
[46]	RT attack simulation exposes vulnerabilities	-	Simulates protection-layer cyber-attacks and grounding faults
[37]	AMI enables live detection of false-data injections	-	Data-integrity attack detection via sequential hypothesis testing
[39]	Detects DDoS in SDN-SCADA architecture in real time	-	Security-centric (no explicit maintenance/load)
[34]	Continuous transformer-relay data monitoring	ML anticipates relay anomalies for preventive work	Cyberattack (tampering) detection on protective relays
[38]	“Digital Ghost”twin provides real-time asset visibility	Forecasting/anomaly detection enable maintenance	Detects and localizes cyber-physical attacks (distinguished from naturally-occurring faults)

the classification question that operators face after the alarm fires.

3.3.3 Entities

Several key entities play pivotal roles in the application of anomaly detection systems:

Transmission System Operators (TSOs) are responsible for the high-voltage transmission networks. They utilize anomaly detection to monitor the grid’s backbone, ensuring the efficient transfer of electricity over long distances. For example, TSOs may employ these systems to detect and mitigate oscillations in the transmission lines, maintaining system stability. Moreover, TSOs are also critical in monitoring grid frequency and the stability of interconnections, which are essential for preventing blackouts and ensuring the reliable operation of the entire power system.

Distribution System Operators (DSOs) manage the medium to low-voltage distribution networks that deliver electricity to end-users. DSOs apply anomaly detection to manage the complexities introduced by DERs at the distribution level, such as voltage regulation issues caused by rooftop solar panels feeding energy back into the grid.

Consumer System Applications involve end-users who can be both consumers and producers (prosumers) of energy. Smart homes equipped with anomaly detection can monitor energy consumption patterns, detect unusual usage that may indicate appliance failures, or optimize energy usage to reduce costs.

Other Entities, such as third-party service providers, offer specialized anomaly detection solutions or services. They play a role in aggregating data across multiple grids or regions, providing insights that individual operators may not achieve independently. Additionally, market operators can leverage these aggregated insights to optimize electric-

ity trading, adjust pricing strategies, and enhance overall grid stability by effectively balancing supply and demand.

TSOs and DSOs This section explores how different studies engage with the roles and significance of Transmission System Operators (TSOs) and Distribution System Operators (DSOs), especially in relation to grid management, anomaly detection, and cybersecurity efforts.

Inferred or general support for TSOs and DSOs is observed in several studies. Santorsola et al., [23] infers roles related to managing energy transmission and distribution, while [24] indirectly supports grid integrity through anomaly detection. [25] benefits grid management and operation efficiency, and [28] enhances monitoring and management of the electrical grid. [30] is key in managing grid stability and reliability amidst cyber-attacks, and [31] safeguards infrastructure from cyber-attacks and operational disruptions. [32] uses anomaly detection to maintain grid stability and security, while [33] manages energy flows and maintains grid stability. [35] focuses on monitoring and maintaining grid stability and security, and [37] incorporates anomaly detection to enhance grid reliability and security. [38] protects infrastructures with real-time threat detection and response capabilities, [40] enhances grid operations through anomaly detection for monitoring and efficiency, [42] uses enhanced cybersecurity measures and anomaly detection to maintain system integrity against cyber-attacks, and [46] utilizes simulations to detect and respond to anomalies in the transmission network.

However, many studies do not explicitly discuss or detail the roles of TSOs and DSOs in their research. These include [22, 26, 27, 29, 34, 36, 39, 41, 43–46]. These studies focus more on technological implementations and anomaly detection mechanisms without delving into the specific operational roles of TSOs and DSOs.

Additionally, broader or indirect contributions are noted in some studies. For instance, [22] focuses on cyber-physical

attack detection without explicit mention of TSOs/DSOs, [29] emphasizes real-time monitoring for cyber-attack detection without specific mention of TSOs/DSOs, [41] focuses on technological implementation rather than operational specifics of TSOs/DSOs, Padmapriya and Srivenkatesh [43] focuses on IoT networks without direct mention of TSOs/DSOs roles, [44] focuses on broader system infrastructure protection against cyber-attacks, and [45] focuses on overall smart grid infrastructure without detailing specific roles of TSOs or DSOs.

Consumer System Applications Energy management and efficiency are central themes in consumer system applications. Santorsola et al., [23] manages the interaction between consumption and production, enhancing stability. [28] focuses on energy management at the consumer level with enhanced security against cyber threats, while [32] manages consumption patterns to improve energy efficiency. [33] manages energy production and consumption, enhancing grid resilience and energy efficiency, and [35] optimizes energy usage through home energy management systems. In addition, regulators benefit from these insights by leveraging consumer data to monitor compliance with grid standards and to implement policies that ensure fair pricing and reliable service.

In terms of smart metering and data monitoring, [24] provides secure and reliable electricity data for consumers, [25] uses smart meters to monitor consumption for accurate billing, and [40] monitors smart meter data to identify unusual consumption patterns or meter failures. Additionally, [45] reports data from smart meters and IoT devices within consumer systems back to the utility control center (TSO/DSO).

Cybersecurity and anomaly detection are also integral to consumer system applications. [28] enhances security against cyber threats at the consumer level, and [37] involves end-use consumers benefiting from the security provided by anomaly detection.

General or indirect references to consumer applications are seen in [29], which involves end-users of electricity data, Appiah-Kubi et al. [31], which indirectly references end-nodes potentially impacted by cybersecurity measures, Padmapriya and Srivenkatesh [43], which focuses on IoT devices and systems within consumer applications, and [42], which has a broader impact on the smart grid system, including consumer-related applications.

However, several studies do not explicitly cover consumer system applications in detail. These include [22, 26, 27, 30, 34, 36, 38, 39, 41, 44–46].

Other Entities Various stakeholders are involved in grid operations, energy management, and security. [30] involves various stakeholders such as researchers and system operators, in Europe transmission owners are system operators, Appiah-Kubi et al. [31] focuses on the roles in security and

resilience of the grid through distributed agent-based systems, while [32] integrates distributed energy resources and optimizes energy usage. [33] engages in energy trading and demand response initiatives that support grid operations. [35] focuses on monitoring and managing energy production and consumption efficiently.

General or indirect references to other entities are seen in [46], which involves overall grid operations and security without specifying particular entities.

However, many studies do not specifically discuss or detail the involvement of other entities. These include [22–29, 34, 36–45], and [46]. These studies focus more on technological implementations and anomaly detection mechanisms without delving into the specific roles of other entities.

3.3.4 Business integration

Other Entities Mapping to IT Service Management Frameworks Integrating anomaly detection systems within business processes is enhanced through frameworks like **ITILv4** (Information Technology Infrastructure Library version 4). ITILv4 provides a set of best practices for IT service management that can be mapped to the needs of Smart Grid operations. By aligning anomaly detection with ITILv4 practices, organizations can improve incident management, change control, and continuous improvement processes. For instance, integrating anomaly detection alerts with service desks allows for streamlined response protocols, improving outcomes and operational effectiveness.

Our review provides an overview of the integration of IT Service Management (ITSM) frameworks, such as ITILv4, into various research contexts, with a focus on business integration and operational efficiency:

Implied or Potential Integration: [24] implies a structured approach, though no explicit ITILv4 integration is mentioned. [28] suggests that integration could enhance incident response, aligning with ITILv4 principles. [32] implies that integration is important for cyber-physical security and operational efficiency. [33] aligns cybersecurity practices with IT service management practices, and [35] mentions that security-by-design principles can align with IT frameworks.

Not Discussed or Detailed: The majority of studies, including [22, 23, 25–27, 29–31, 34, 36–46] do not discuss the integration of ITSM frameworks, such as ITILv4, in their research.

3.3.5 Incident response data

Effective incident response relies on different levels of **situational awareness data**:

Awareness data for perception involves the initial detection of anomalies. This includes raw data and immediate

indicators that something within the grid is not operating as expected. For example, a sudden spike in current flow detected by sensors triggers an alert. These proactive control methods enable operators to quickly detect potential issues and initiate timely preventative measures.

Awareness data for comprehension goes a step further to interpret what the anomalies mean. This involves data analytics to understand the context and potential impact. For instance, correlating the spike in current with maintenance schedules may reveal that it is due to a scheduled test rather than a fault. This correlation with routine operations and maintenance processes ensures that the response is well-informed and minimizes unnecessary interventions.

Awareness data for projection uses the information gathered to predict future states of the grid. By modeling potential scenarios, operators can prepare for possible outcomes, such as increased load during peak hours or the impact of a forecasted storm on grid stability.

Utilizing these levels of awareness data enhances the ability of operators to respond effectively to incidents, minimizing negative outcomes and improving overall grid reliability.

Use of Incident Response Data, Specifically Focusing on Aspects of Perception, Comprehension, and Projection Our review highlights the use of incident response data, specifically focusing on aspects of perception, comprehension, and projection, across various authors' research:

Comprehensive and Explicit Use: Santorsola et al., [23] comprehensively uses data in grid operations, indirectly covering aspects of perception, comprehension, and projection. [28] enhances situational awareness through its BioSecure Digital Twin and OODA-loop (BOLF) framework, improving perception and comprehension. Appiah-Kubi et al. [31] uses data to perceive, comprehend, and project threats, while [32] employs machine learning to enhance perception and analysis capabilities. [33] utilizes data analytics and monitoring for situational awareness, and [46] defines and uses data from PMUs and SCADA for detection, with simulation outputs for understanding states.

Implicit or Indirect Use: [24] incorporates real-time awareness without specific definitions, and [30] mentions simulation and real-time testing. [35] implies the collection and analysis of data, [42] implies processing of awareness data through anomaly detection, Padmapriya and Srivenkatesh [43] applies IoT and Digital Twins for security monitoring, [44] discusses network traffic data analysis implicitly, and [45] addresses continuous monitoring implicitly through anomaly detection.

Not Discussed or Detailed: Several authors, including [22, 25–27, 29, 34, 36–41] do not specifically discuss the use of incident response data related to perception, comprehension, and projection.

3.3.6 Conclusion

Anomaly detection systems are essential for the operation of DER-rich SG, applied across various contexts such as real-time monitoring, predictive maintenance, fault detection, load balancing, and cybersecurity. By enabling immediate detection and response to anomalies, predicting potential equipment failures, ensuring equitable load distribution, and safeguarding against cyber threats, these systems significantly enhance grid stability, operational efficiency, and security.

Collaboration among Transmission System Operators (TSOs), Distribution System Operators (DSOs), consumers, and service providers amplifies the effectiveness of these systems. Integrating anomaly detection with business frameworks like ITILv4 and leveraging comprehensive incident response data further enhances operational outcomes. This underscores the significance of anomaly detection in maintaining the robustness, efficiency, and security of modern SG, ultimately contributing to a sustainable and reliable energy future.

What the pattern means for cyber–physical distinction. Read together, the application-domain findings are not neutral descriptions; they reveal three structural blind spots of the field. First, because nearly every pipeline is framed as real-time *detection* while diagnosis, maintenance triage, and incident-response integration are thin, the literature optimizes for raising alarms rather than for the decision that follows the alarm – precisely the decision (maintenance crew versus security response) that requires distinguishing a safety fault from a cyber-attack. Second, the roles of TSOs and DSOs are mostly inferred rather than designed for: few studies state at which operator, voltage level, or control-room workflow their detector would sit, which makes it difficult to assess whether a proposed method could realistically support an operator's attribution decision under DER-rich conditions. Third, the near-absence of links to service-management and incident-response frameworks (e.g., ITILv4-style processes) means that even accurate detectors deliver an unrouted signal: without an explicit hand-off to either a maintenance or a security process, the cyber-versus-safety question is deferred to humans without machine support. These three blind spots jointly explain, at the application level, why so few primary studies attempt the distinction itself, and they motivate the focused comparison of those that do (Sect. 3.7).

Key Takeaways:

- **Improved Grid Reliability:** Proactive detection and management of anomalies prevent disruptions and ensure continuous service.

- **Enhanced Efficiency:** Optimization of grid operations through predictive insights and real-time adjustments increases overall efficiency.
- **Increased Security:** Protection against physical faults and cyber threats strengthens grid resilience and security.
- **Cost Reduction:** Preventative measures and efficient resource management lead to significant cost savings.
- **Support for Renewable Integration:** Effective management of renewable energy variability facilitates their incorporation into the grid.

In summary, anomaly detection systems play a pivotal role in maintaining the robustness and reliability of modern SG. Their effective application, combined with stakeholder collaboration and integration into business frameworks, is instrumental in advancing towards a sustainable and reliable energy future.

3.4 Evaluation strategies

3.4.1 Introduction

The integration of Distributed Energy Resources (DERs) into SG has significantly increased the complexity of managing and maintaining these advanced electrical networks. Anomaly detection systems play a crucial role in ensuring the reliability, security, and efficiency of SG by identifying irregularities that could indicate faults, cyber-attacks, or system inefficiencies. Evaluating the effectiveness of these anomaly detection systems is paramount to optimize their performance and to ensure they meet the operational demands of modern grids.

Following Fig. 11, this discussion focuses on the benchmarks and performance metrics that are most effective in evaluating the success of anomaly detection systems within DER-rich SG. Understanding these metrics is essential for grid operators, engineers, and stakeholders to assess system performance, make informed decisions, and implement improvements where necessary. Moreover, a digital perspective underpins these evaluation strategies, leveraging advanced data analytics and real-time processing capabilities that are intrinsic to modern smart grid infrastructures.

3.4.2 Anomalies evaluation metrics

One of the fundamental metrics in evaluating anomaly detection systems is the **False Positive Rate (FPR)**. The FPR measures the proportion of normal events incorrectly identified as anomalies. A high FPR can lead to unnecessary investigations and operational inefficiencies. It is calculated by dividing the number of false positives by the total number of actual normal instances. Reducing the FPR is essential for

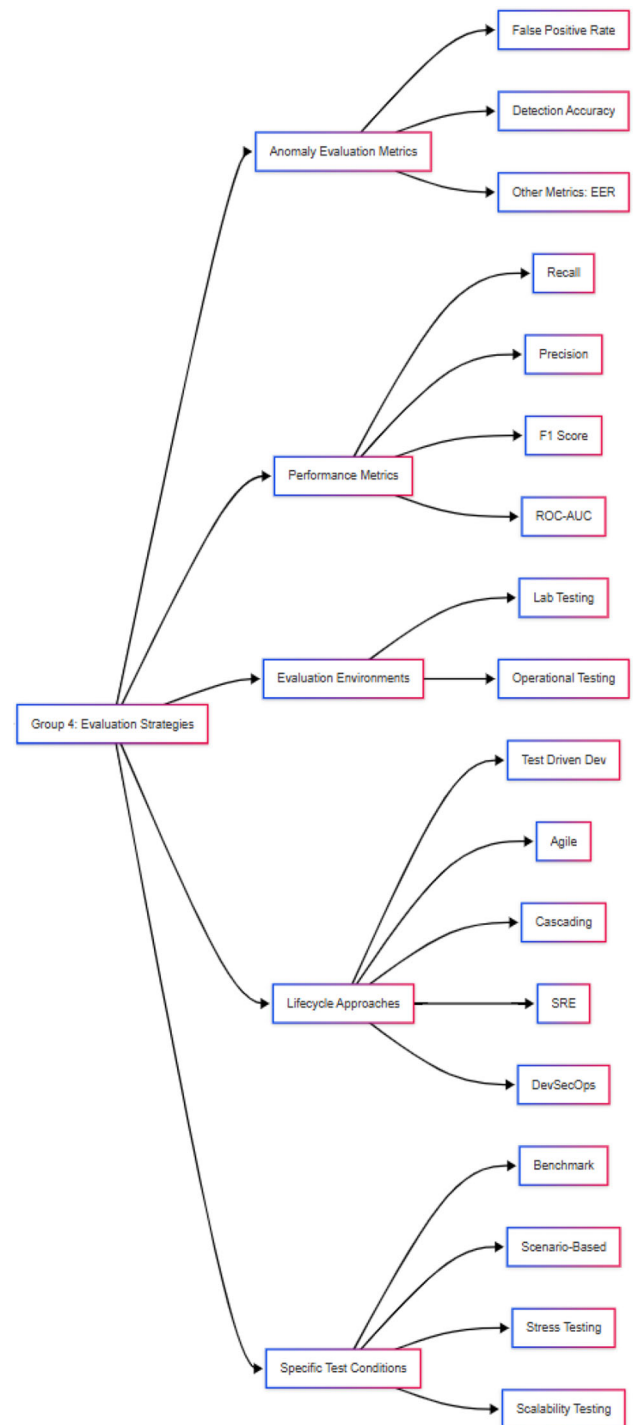


Fig. 11 Topics mapping under group evaluation strategies category/group 4 shown in Fig. 6

maintaining trust in the anomaly detection system and for avoiding alarm fatigue among operators.

Detection Accuracy is another critical metric, representing the overall ability of the system to correctly identify both normal and anomalous events. It is calculated by dividing the number of correct predictions by the total number of

instances evaluated. High detection accuracy indicates that the system is reliable and effective in distinguishing between normal operations and anomalies.

Other metrics, such as the **Equal Error Rate (EER)**, provide a balance between false positives and false negatives. The EER is the point on a receiver operating characteristic (ROC) curve where the false acceptance rate equals the false rejection rate. It is particularly useful when the cost of false positives and false negatives is equal, offering a single value to assess system performance.

Table 6 maps, per study, how False Positive Rate, Detection Accuracy, and further metrics are explicitly measured, implied, or absent. The analytically relevant pattern in the table is the asymmetry between the columns: detection accuracy is by far the most frequently quantified quantity (with the explicit values revisited in Table 4), whereas FPR—the metric that governs alarm fatigue and operator trust—is explicitly measured in only a minority of studies and is otherwise invoked qualitatively. Threshold-balancing measures such as EER appear only where a tunable decision procedure is central to the method (e.g., [37]). For operational deployment, this ordering is inverted relative to operators' needs, where sustained low false-alarm rates matter at least as much as headline accuracy.

3.4.3 Performance metrics

In addition to anomaly-specific metrics, standard performance metrics from the field of machine learning are employed:

- **Recall (Sensitivity)** measures the proportion of actual anomalies correctly identified by the system. It is calculated by dividing the number of true positives by the sum of true positives and false negatives. High recall is crucial for ensuring that most anomalies are detected, which is vital for grid security and reliability.
- **Precision** assesses the proportion of detected anomalies that are actual anomalies. It is calculated by dividing the number of true positives by the sum of true positives and false positives. High precision indicates that the system generates few false alarms, which is important for operational efficiency.
- The **F1 Score** is the harmonic mean of precision and recall, providing a single metric that balances both. It is particularly useful when there is an uneven class distribution between normal and anomalous events.
- Additional metrics like the **Area Under the ROC Curve (ROC-AUC)** and **Specificity** offer further insights. The ROC-AUC measures the ability of the system to distinguish between classes across all classification thresholds,

while specificity measures the proportion of normal events correctly identified.

Cross-study synthesis of standard metric usage The per-study mapping of recall, precision, F1, and additional metrics is consolidated in Table 7; here only the cross-study pattern is summarized. A consistent three-tier structure emerges across all four metric families. A small core of studies explicitly defines and reports the metrics—most consistently Amritha et al. [27], Naeem et al. [29], Shen et al. [37], and Agnew et al. [42] for recall/precision/F1, joined by Yan et al. [26], Jahromi et al. [34], Yilmaz et al. [45], and Chhokra et al. [46] for threshold-free measures such as ROC-AUC, precision–recall curves, or specificity. A second, larger tier invokes the metrics only by implication, asserting that high recall or low false-alarm rates matter without reporting values. A third tier—roughly ten studies for each metric family, dominated by the survey-type and framework-type contributions—does not engage with the metric at all. The pattern is informative in itself: explicit metric reporting tracks the maturity of the experimental setup rather than the choice of algorithm, and the harmonic-mean F1 perspective, which is the natural single-number summary for the imbalanced classes typical of grid anomalies, is reported by fewer than half of the studies. For the central question of this review, the most consequential absence is that none of the 25 studies reports class-confusion-level results for the attack-versus-fault decision (e.g., how often a genuine physical fault is misclassified as an attack or vice versa), which is precisely the quantity an operator would need to trust an automated attack-versus-fault distinction (Sect. 3.7).

3.4.4 Evaluation environments

Evaluating anomaly detection systems requires appropriate testing environments. **Lab Testing Environments** provide controlled settings where systems can be tested against pre-defined datasets and simulated conditions. This environment allows for repeatability and the isolation of variables, which is advantageous for initial testing and benchmarking. However, lab environments may not capture the full complexity of operational conditions, potentially limiting the applicability of the results. **Operational Testing Environments** involve deploying the system within a live grid or a realistic simulation of one. Testing in this environment provides insights into how the system performs under actual operating conditions, including unexpected events and noise. While offering more accurate evaluations, operational testing can be challenging due to safety concerns and the potential impact on grid performance.

Table 6 Anomalies evaluation metrics

	False positive rate (FPR)	Detection accuracy	Various metrics
Explicitly Mentioned or Measured	Measured as part of an intrusion–detection system evaluation [29]; <i>minimised through systematic tuning</i> [37]; defined as the proportion of normal instances misclassified [45]; <i>critical for minimising operational burden in practice</i> [38]; reflects legitimate observations misidentified as anomalies [39]; typically evaluated by monitoring incorrect benign activities [35]	Shukla <i>et al.</i> reported 89% accuracy in fog and 78% in cloud environments [24]; Reddy <i>et al.</i> achieved 98.74% with an Isolation Forest model [25]; Naeem <i>et al.</i> demonstrated up to 99% accuracy with deep ensemble learning [29]; detection accuracy proved crucial for identifying FDI attacks when using an Autoencoder [27]; Agnew <i>et al.</i> reported consistently high accuracy across diverse attack scenarios [42], while Hyder <i>et al.</i> achieved similarly high accuracy across diverse cyber-physical event classes [44]	a configurable detection rate with false-positive/false-negative rates from the sequential hypothesis test [37]; F1 score adopted as a balanced-performance measure [39]; Mean Squared Error (MSE) employed as a reconstruction-error metric [41]; precision, recall and F1 jointly reported to provide complementary insight [42]; the operating threshold is governed by a detection-delay versus false-alarm-rate trade-off rather than ROC/AUC [45]
Implied Importance or Indirect Reference	False-positive rate is indirectly highlighted [23]; implied as <i>critical</i> [25]; inferred but not explicitly measured [26]; likely assessed via ROC curves [27]; research efforts focus on <i>minimising false detections</i> [31, 34]; its critical nature is implied but not detailed [41]	Operational safety is repeatedly emphasised, though accuracy is not formally defined [23]; its importance is acknowledged in survey work [22]; effectiveness against cyber-attacks is underscored in defence-oriented studies [30, 31]; highlighted through <i>local</i> anomaly-detection schemes [32]; deemed essential for CPS reliability [33, 35]; stressed for resilient operation [28, 38]; linked to distinguishing normal from DDoS traffic [39]; noted as a key factor in simulation studies [41]; minimising overall errors remains a guiding principle [45]; consistently tied to Smart-Grid reliability [46]; operational <i>integrity</i> explicitly referenced [34]	Detection metrics are implied through diverse methods [22]; Fournaris <i>et al.</i> discuss balancing Type I and II errors without detailing specific metrics [35]; complementary metrics are suggested, but not exhaustively reported, in Autoencoder-based [27], cyber-defence [30], local anomaly-detection [32], CPS-oriented [33] and data-driven [34] studies. *

Table 6 continued

	False positive rate (FPR)	Detection accuracy	Various metrics
Methods or Approaches Highlighted	–	Detection accuracy evaluated under controlled test conditions [26]; sequential-hypothesis testing employed to refine decision thresholds [37]; accuracy scores often inferred from reconstruction or classification confidence [36]	–
Not Specifically Discussed or Provided	[22, 30, 32, 33, 42]. *	[40, 43]	[23, 25, 26, 29, 31, 38].*

Dash (–) indicates information not reported in the reviewed sources. * [24, 28, 36, 40, 43, 44]

Lab Testing Environments

Simulations and Virtualized Testbeds: Simulations and virtualized testbeds are commonly used in lab testing environments. Shukla et al. conducted simulations in iFogSim [24]. Yan et al. utilized high-fidelity simulations [26]. Appiah-Kubi et al. simulated environments using the IEEE Test Feeder [31]. Al Homoud et al., Naeem et al., and Tertychny et al. employed simulated environments [29, 32, 36]. Jahromi et al. used the IEEE Benchmark for simulations [34]. Shen et al. and Oyucu et al. conducted controlled simulations [37, 39]. Agnew et al. utilized simulations of the IEEE System [42]. Hyder et al. employed a co-simulation platform [44]. Mylrea et al. used a virtualized testbed in their studies [28, 38]. Yilmaz et al., Chhokra et al., and Husnoo et al. conducted experiments in controlled settings [41, 45, 46]. Fournaris et al. and Amritha et al. likely used simulations, although it was implied and not explicitly mentioned [27, 35].

Hardware-Based Testbeds: Reddy and Shabad employed hardware-based testbeds in their research [25].

Not Specifically Discussed: Several studies do not specifically discuss the use of lab testing environments in their research. Santorsola et al., Ye, Suprabhath and Koduru, Czekster et al., Johannesen et al., and Padmapriya et al. did not specifically discuss the use of lab testing environments [22, 23, 30, 33, 40, 43].

Operational Testing Environments

Real-World Data and Operational Settings: Several studies utilize real-world data and operational settings to evaluate anomaly detection systems. Yan et al. used real data from power plants [26]. Fournaris et al. conducted tests in real operational settings [35]. Tertychny et al. employed real-time testbeds [36]. Shen et al. and Oyucu et al. tested their systems under real-world conditions [37, 39]. Husnoo et al. conducted evaluations in real-world settings [41]. Yilmaz et

al. and Chhokra et al. performed real-world smart grid testing [45, 46].

Simulated Operational Settings: Some studies opted for simulated operational settings to evaluate their systems. Mylrea et al. simulated real operational settings in their research [28, 38]. Jahromi et al. used simulated operational scenarios [34].

Implied or Indirect Reference: Naeem et al. implied the use of real-world data in their evaluations [29].

Not Specifically Discussed or Defined: Several studies do not specifically discuss or define their operational testing environments. Santorsola et al., Ye, Shukla et al., Reddy and Shabad, Amritha et al., Suprabhath and Koduru, Appiah-Kubi et al., Al Homoud et al., Czekster et al., Johannesen et al., Agnew et al., Padmapriya et al., and Hyder et al. did not specifically discuss or define their operational testing environments [22–25, 27, 30–33, 40, 42–44].

3.4.5 Lifecycle evaluation testing

Implementing robust anomaly detection systems benefits from adopting comprehensive lifecycle evaluation methodologies:

Test Driven Development (TDD)

Test Driven Development (TDD) is a software development approach where tests are written before the code that needs to pass the tests. In the context of anomaly detection, TDD ensures that each component of the system is validated against specific criteria, leading to higher-quality software. In the research studies [37, 39] explicitly used TDD in their development processes. However, studies such as [22–36, 38, 40–46] do not discuss or mention the use of TDD in their research.

Table 7 Performance metrics employed in smart, grid anomaly detection studies

	Recall	Precision	F1 Score	Other metrics
Importance	High recall minimizes false negatives and safeguards grid security by ensuring true anomalies are captured	High precision limits false alarms, preserving operator trust and avoiding unnecessary control actions	Balances recall and precision, giving a single, class, imbalance, aware view of effectiveness	Metrics such as ROC, AUC and specificity provide threshold, independent and true, negative insights that complement single, point statistics
Explicitly Defined & Used	Explicitly calculated in classification reports [27]; evaluated in experimental studies [29]; formal definition TP/(TP+FN) stated and applied [37]; computed per, attack to expose scenario variance [42]; high recall values reported for FDI detection effectiveness [44]	Precisely reported alongside recall in Autoencoder results [27]; computed during ensemble evaluations to quantify false, alarm reduction [29]; formally defined as TP/(TP+FP) and leveraged for threshold tuning [37]; high values demonstrated across heterogeneous cyber, attacks [42]; very high precision achieved for FDI detection [44]	Listed in classification summaries as key index [27]; derived from precision and recall in ensemble learning [29]; adopted to balance errors in local anomaly detection [32]; used explicitly for trade, off analysis in CPS diagnosis [34]; combined precision and recall to calculate F1 Score [42, 45]; harmonic, mean formulation stressed in several grid, security works [35, 37, 39, 46]	Extensive ROC curves plotted for threshold analysis [26]; ROC, AUC reported with a precision/recall/F1 classification report in Autoencoder study [27]; PR curves preferred in class, imbalance settings [34]; ROC, AUC paired with specificity to expose TN behaviour [37, 45]; ROC, AUC highlighted as headline discriminator in smart, grid FDI work [46]
Implied or Indirect Importance	High TP rates in benchmarks indirectly underline recall relevance [26]; critical for false, negative reduction in mitigation frameworks [31]; highlighted for resilient local detection [32]; importance inferred in data, driven diagnosis [34]; repeatedly stressed in surveys and ensemble studies [35, 38, 39, 41]; singled out for FDI protection [44–46]	Model, performance discussions emphasise precision's role in false, alarm control [26, 31]; operational, efficiency arguments invoke precision without formal definition [32, 34, 35, 38]; surveys cite precision to discourage spurious alarms [39, 45, 46]	–	Detection frameworks imply use of ROC, based criteria in evaluating cyber, defence efficacy [22]; comprehensive simulation studies hint at multi, metric dashboards [28]; ensemble methods reference threshold, independent measures without explicit tables [29, 35, 41]
Not Discussed or Provided	[22, 28]	[22, 28, 41]	[22, 26, 28, 31, 38, 41, 44]	[31, 32, 39, 42, 44]

Dash (–) denotes information not reported in the reviewed sources. Common studies * [23–25, 30, 33, 36, 40, 43]

Table 8 Lifecycle evaluation testing

	Test driven development (TDD)	Agile methodologies	Cascading	Site reliability engineering (SRE)	DevSecOps	Other lifecycle evaluation approaches
Overview	Tests are specified <i>before</i> code is written	Iterative development with continuous feedback	Sequential chaining where the output of one detector feeds another	Blends software engineering and IT-operations to ensure scalable, reliable services	Injects security gates and checks across every stage of DevOps	Continuous updates/monitoring, improvement and testing throughout the system lifetime
Merit	Guarantees every module meets explicit requirements, yielding high-quality software	Rapidly adapts to evolving needs and evaluation insights	Potentially boosts accuracy by leveraging complementary models	Maintains performance under variable load, improving grid robustness	Builds security in from inception, reducing late-stage vulnerabilities	Sustains resiliency by continuously validating and refining detection performance
Demerit	Up-front effort to design and maintain comprehensive test suites can slow early progress	Requires disciplined scope management and sustained stakeholder engagement	Susceptible to error propagation and extra latency	Adds operational overhead and needs specialised SRE talent	Security gates may lengthen release cycles and increase pipeline complexity	Demands permanent resources for monitoring infrastructure and evaluation tooling
Mentioned	Unit tests precede detector code in a digital-twin prototype [37]; integration tests drive ensemble training [39]	Iterative sprints refine detection logic [37]; an agile backlog steers ensemble optimisation [39]	Sequential residuals boost accuracy through a cascading detector chain [37]	SLIs/SLOs and reliability budgets keep detection latency low [37]	Security gates plus automated scans embedded in the CI/CD pipeline [37]	Continuous drift monitoring [35]; automated retraining loops [37]; blue-green deployments for ensembles [39]; continuous fuzz testing for grid resilience [38]
Not Mentioned	[35, 38],*	[35, 38],*	[35, 38, 39],*	[35, 38, 39],*	[35, 38, 39],*	*

Common studies * [22–34, 36, 40–46]

Agile Methodologies

Agile Methodologies emphasize iterative development and continuous feedback. Applying agile principles allows for rapid adaptation to changing requirements and integration of new insights from evaluation metrics, improving system responsiveness and effectiveness. In the research studies [37, 39] emphasize iterative development. However, studies such as [22–36, 38, 40–46] do not discuss or mention the use of Agile Methodologies in their research.

Cascading

Cascading refers to a sequential approach where the output of one detection model feeds into another. This can enhance detection accuracy by combining multiple models, but it requires careful evaluation to prevent error propagation. In the research study [37] discusses the application of cascading through sequential algorithm application. However, studies such as [22–36, 39–46] do not discuss or mention cascading effects or methodologies in their research.

Site Reliability Engineering (SRE)

Site Reliability Engineering (SRE) combines software engineering and IT operations to build reliable and scalable systems. SRE practices in anomaly detection focus on maintaining system performance under varying loads and conditions, ensuring that detection systems remain effective as the grid evolves. In the research study [37] discusses the application of SRE by ensuring system robustness. However, studies such as [22–36, 38–46] do not discuss or mention Site Reliability Engineering (SRE) in their research.

DevSecOps

DevSecOps integrates security practices into the DevOps process, emphasizing the importance of security at every stage of development. For anomaly detection systems, DevSecOps ensures that security considerations are embedded from the outset, leading to more robust and secure solutions. In the research study [37] discusses the application of DevSecOps by integrating security checks into their development process. However, studies such as [22–36, 38–

[46] do not discuss or mention DevSecOps practices in their research.

Other Lifecycle Evaluation Approaches

Other lifecycle evaluation approaches include continuous updates and monitoring, continuous improvement, and continuous testing. In the research studies [35] discusses continuous updates and monitoring, [37] and [39] address continuous improvement, and [38] emphasizes continuous testing. However, studies such as [22–34, 36, 40–46] do not discuss or mention other lifecycle evaluation approaches in their research.

Table 8 is the summary of this section and as described in the section text indicates that while some authors discuss lifecycle evaluation approaches like continuous updates, monitoring, improvement, and testing, the majority do not address these approaches in their research.

3.4.6 Specific metrics and test conditions

Evaluating anomaly detection systems also involves testing under various conditions to ensure robustness:

- **Benchmark Conditions** involve testing systems against standard datasets or scenarios to compare performance objectively. Benchmarks provide a baseline for evaluating different systems or configurations.
- **Scenario-Based Testing** simulates specific situations that the system may encounter, such as cyber-attacks or equipment failures. This type of testing assesses how well the system can handle particular anomalies and adapts to complex events.
- **Stress Testing** subjects the system to extreme conditions, such as high data loads or rapid fluctuations, to evaluate its performance under pressure. Stress testing identifies potential weaknesses in system scalability and resilience.
- **Scalability Testing** examines how the system performs as the scale of the grid or the volume of data increases. This is crucial for ensuring that anomaly detection systems remain effective as the grid expands or as more DERs are integrated.

Discussion and Application of Specific Metrics and Test Conditions

Benchmark and Scenario-Based Testing: Implied benchmark and scenario-based testing are conducted by [22]. Benchmark (testbed) and scenario-based testing are employed by [25]. Additionally, benchmark and scenario-based testing are utilized by [26], [28], [29], [31], [32], [35], and [38]. Furthermore, benchmark, scenario, stress, and scalability testing are implemented by [37, 39], [45], and [46].

Scenario-Based Testing: Scenario-based testing is performed by [40], [42], and [44].

Implied Stress and Scalability Testing: Implied stress and scalability testing are suggested by [27] and [41], while [34] implies stress testing.

Not Discussed: Specific metrics or test conditions such as benchmark, scenario, stress, or scalability testing are not discussed in the research conducted by [23, 24, 30, 33, 36, 43].

3.4.7 Conclusion

Evaluating the success of anomaly detection systems in DER-rich SG requires a multifaceted approach that combines specific performance metrics, testing environments, and lifecycle methodologies. Key metrics such as detection accuracy, false positive rate (FPR), recall (sensitivity), precision, F1 Score, area under the ROC curve (ROC-AUC), and specificity provide quantitative measures that collectively assess system performance. These metrics balance the need to detect true anomalies with the minimization of false alarms, ensuring both reliability and efficiency.

By carefully selecting and applying these benchmarks within both laboratory and operational testing environments, and adopting robust lifecycle evaluation methodologies like Agile and DevSecOps, stakeholders can ensure that anomaly detection systems are robust, adaptable, and effective under real-world conditions. This comprehensive evaluation is essential for maintaining the integrity, security, and efficiency of SG in an era of increasing complexity and demand.

Key Takeaways:

- **Balanced Performance Metrics:** Employing a combination of metrics like F1 Score and ROC-AUC helps balance the trade-offs between detecting all anomalies and minimizing false positives.
- **Real-World Testing:** Validating systems in operational environments ensures performance aligns with actual grid conditions and challenges.
- **Lifecycle Methodologies:** Adopting approaches such as Agile and DevSecOps promotes continuous improvement in system performance and security.
- **Adaptability to Future Challenges:** Regular evaluation and updating of systems ensure they remain effective amid evolving grid complexities and demands.
- **Optimized Grid Operations:** A focus on these benchmarks enables grid operators and engineers to enhance the reliability and security of the energy infrastructure.

By concentrating on these critical aspects, stakeholders can optimize anomaly detection systems to meet the evolving demands of modern SG, ensuring a reliable, secure, and efficient energy infrastructure for the future.



Fig. 12 Topics mapping under group challenges and opportunities category/group 5 shown in Fig. 6

3.5 Challenges and opportunities

See Fig. 6.

3.5.1 Introduction

The integration of Machine Learning (ML) and Explainable Artificial Intelligence (XAI) into SG has opened new horizons for anomaly detection, enhancing the reliability and efficiency of power systems rich in Distributed Energy Resources (DERs). However, deploying these advanced technologies is not without its challenges. Technical and operational hurdles can impede their effectiveness, while also presenting opportunities for innovation and improvement. This subsection delves into the primary technical and operational challenges faced when implementing ML and XAI for anomaly detection in SG, and explores future opportunities to overcome these obstacles. A summary of

the major technical and operational challenges encountered in deploying ML/XAI techniques-along with corresponding opportunities-is depicted in Fig. 13. It highlights issues like data explainability, model accuracy, and standardization gaps.

ICT-Specific Challenges Scalable Pipeline Design: Real-time ingestion of streaming sensor data from thousands of grid nodes requires reliable, distributed message queues and data lakes that can scale horizontally. Quality Assurance and Data Governance: Ensuring label quality for supervised models is non-trivial. Missing or corrupted sensor readings can skew training sets, while inconsistent naming conventions across SCADA systems pose further integration hurdles.

Electrical Engineering-Specific Challenges Physical Grid Topology and DER Impacts: Designing robust monitoring strategies and protective relays that account for backfeeding from DERs demands power-flow analysis, short-circuit analysis, and novel protective coordination schemes. Fault Characterization and Grid Codes: Determining normal operating bounds versus anomalous conditions requires in-depth knowledge of IEC/IEEE standards for fault detection, such as standard guidelines for protective relay settings, inertial response from spinning machines, and low-voltage ride-through protocols for inverters.

Following Fig. 12 we will discuss finding in more details.

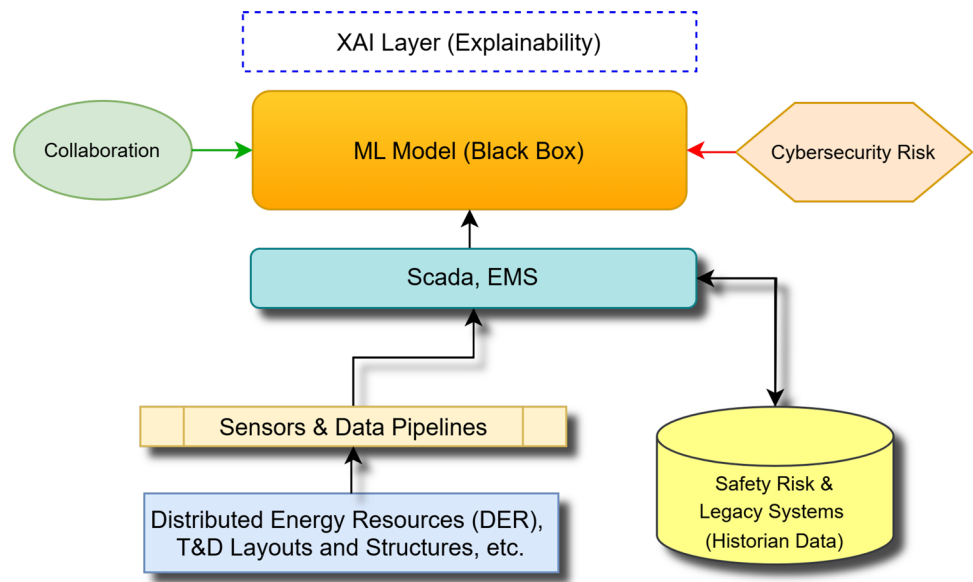
3.5.2 Technical challenges

One of the foremost technical challenges is **explainability**. ML models, particularly deep learning algorithms, often operate as “black boxes,” making it difficult for operators to understand how decisions are made. This lack of transparency can hinder trust and acceptance among stakeholders who require clear justifications for anomaly detection results. For instance, when an ML model flags a critical component for maintenance, operators need to comprehend the reasoning to validate the recommendation and take appropriate action.

Model accuracy and reliability are also significant concerns. Achieving high detection accuracy is essential to minimize **false positives** (incorrectly identifying normal behavior as anomalous) and **false negatives** (failing to detect actual anomalies). High false positive rates can lead to alarm fatigue, where operators become desensitized to alerts, potentially ignoring critical warnings. Conversely, false negatives can result in undetected faults or security breaches, posing severe risks to grid stability and security.

Another technical hurdle is **data quality and availability**. Effective ML models require large volumes of high-quality data, but SG often face issues with data sparsity, noise, and inconsistencies. Real-time data processing adds another layer of complexity, as systems must handle continuous data streams with low latency to provide timely anomaly detection. For example, sensor malfunctions or communication

Fig. 13 Key Technical and Operational Challenges and Opportunities (RQ5). This figure categorises the principal barriers to deploying ML and XAI-based anomaly detection in DER-rich Smart Grids—including explainability gaps, data quality issues, scalability constraints, and standardisation deficits—alongside corresponding opportunities for improvement such as federated learning, edge computing, and collaborative regulation



delays can introduce gaps or errors in the data, affecting model performance.

Explainability challenges

Explicitly Addressed Explainability Challenges:

[29] discusses XAI techniques to validate outputs of deep-stacked ensemble models, improving transparency and trustworthiness. [38] acknowledges the difficulty in making complex AI algorithms understandable and calls for advancements in XAI to improve trust and contextual understanding.

Implied or Indirect Reference to Explainability Challenges:

Santorsola et al., [23] does not directly discuss explainability challenges but mentions the importance of trust via safety-critical command pattern identification. [24] does not explicitly discuss explainability challenges; however, the focus on transparency and security through blockchain may affect trust. Similarly, [25] does not explicitly discuss explainability challenges but focuses on enhancing trust through security. [27] does not directly address explainability challenges; its autoencoder approach is inherently opaque, with an implied impact on trust. The complexity of models in studies by [32], [33], [34], [36], [37], [40], [41], [42], [43, 45], and [46] imply potential trust and deployment issues. [44] states that ML models in anomaly detection pose explainability challenges, and the lack of transparency can hinder trust and deployment.

Not Specifically Addressed:

The studies [22, 26], [28], [30], [31], [35], and [39] do not specifically address explainability challenges in their research.

Potential Impact on Trust and Deployment Due to Lack of Explainability:

The complexity of ML models may affect deployment and trust due to their “black box” nature, as discussed in [28], [30], [31], [35], and [39]. [44] explicitly mentions that lack of explainability may reduce trust and confidence among operators and decision-makers.

In operational smart-grid settings, these explainability challenges become even sharper. Detection outputs may need to be interpreted within tight operational time budgets, and in some communication contexts the overall control loop is constrained by the latency expectations associated with IEC 61850-based automation environments [79]. An explanation that is too slow, too unstable across adjacent time windows, or too abstract for control-room personnel provides limited practical value. In addition, the operator audience is heterogeneous: a DSO analyst, a protection engineer, and a cybersecurity responder may each require a different form of explanation. This creates a practical requirement for multi-level explainability rather than one generic feature-importance view. The issue is further reinforced by the regulatory environment, because high-impact grid AI systems increasingly fall under governance expectations related to transparency, human oversight, and auditable reasoning, including the high-risk framing introduced in Article 6 and Annex III of the EU AI Act [80].

Model accuracy and reliability

Explicitly Discussed Model Accuracy and Efforts to Minimize False Positives/Negatives:

[24] highlights an 89% accuracy with Linear SVM in fog environments, emphasizing the reduction of false positives and negatives. [25] reports a 98.74% accuracy using Isolation Forest, underscoring efforts to minimize false positives and negatives. [27] demonstrates high accuracy (98%) in FDI attack detection, suggesting minimal issues with false posi-

tives and negatives. [29] achieves high accuracy (up to 99%) in anomaly detection and significantly reduces misclassification rates. [32] emphasizes high accuracy to avoid false negatives and discusses the low incidence of false negatives. [34] highlights the effectiveness of LSTM models for detecting cyber-attacks and mentions potential high false detection rates. [37] focuses on high accuracy for detecting data integrity attacks using sequential hypothesis testing (SHT). [38] demonstrates high accuracy in neutralizing attacks with a Digital Twin system, noting ongoing challenges in maintaining accuracy across scenarios. [39] employs an ensemble learning approach to enhance accuracy and reliability through feature selection and hyperparameter tuning. [41] focuses on high accuracy in detecting cyber-attacks versus normal disturbances using DAE within federated learning. [42] utilizes advanced ML algorithms to improve accuracy and reliability in anomaly detection, reducing false positives and negatives. Padmapriya and Srivenkatesh [43] adopts an EKNN approach to improve accuracy and reliability in anomaly detection within IoT networks. [44] enhances accuracy and reliability through high-fidelity dataset generation for ML model training. [45] proposes a non-parametric framework designed to handle high-dimensional data, crucial for achieving high accuracy and reliability in anomaly detection. [46] aims to enhance anomaly detection accuracy by simulating realistic attack scenarios.

Implied or Indirect Focus on Accuracy and Reliability:

Santorsola et al., [23] focuses on using reinforcement learning for pattern detection, implying the need for high accuracy. [22] discusses the use of ML for cyberattack detection, implying a focus on minimizing false positives and negatives. Yan et al., [26] highlights difficulties in accurate modeling due to system complexity and the stealthy nature of attacks. [28] emphasizes high accuracy in detecting cybersecurity threats and focuses on reducing false positives and negatives. [30] emphasizes deep learning for enhanced detection, with the importance of reducing false positives and negatives implied. Appiah-Kubi et al. [31] makes a general mention of the importance of reliable detection, without detailing specifics. [35] discusses the importance of accurate threat detection, implying the critical impact of false detections. [36] focuses on accuracy in distinguishing faults from attacks, with high accuracy scores suggesting minimal impact from false positives and negatives. [40] evaluates various models for anomaly detection, highlighting their effectiveness in maintaining high accuracy and reliability.

Not Specifically Discussed:

[33] does not detail issues related to achieving high accuracy and reliability in anomaly detection within Active Building contexts.

Data quality and availability

Direct and Indirect Discussion on Data Quality and Availability: Santorsola et al. use simulations indicating

potential real-world data limitations, suggesting challenges with noise and real-time data processing [23]. Ye highlights challenges in data quality suggested by falsified sensor data in cyber-attacks and implies the need for robust real-time data processing [22]. Shukla integrates blockchain for data integrity and employs fog computing to address real-time data processing while reducing data sparsity and noise [24]. Reddy discusses challenges in feature extraction and optimization, using PCA to address data sparsity and noise, and emphasizes real-time processing [25]. Yan utilizes advanced feature generation techniques to improve data quality from noisy measurements and addresses data sparsity and noise through sophisticated data processing [26]. Amritha employs autoencoders that learn the data correlation structure and compress high-dimensional grid data, with denoising autoencoders noted as future work for cleaning corrupted data [27]. Mylrea addresses the complexity of securing high-quality data in biopharma and indirectly tackles data sparsity, noise, and real-time processing through Digital Twin [28]. Naeem uses publicly available datasets and Grey-Wolf metaheuristic optimization for feature selection to handle high-dimensional, noisy data [29]. Suprabhath implies challenges in obtaining high-quality data for training AI models, suggesting issues with data sparsity, noise, and real-time processing [30]. Al Homoud utilizes simulated data indicating challenges in obtaining real-world high-quality data, implying the handling of issues related to data sparsity, noise, and real-time processing [32]. Czekster implies challenges in obtaining high-quality data for robust cybersecurity and suggests the need for effective data management [33]. Jahromi uses a standard benchmark test system, highlighting challenges in data quality and representativeness, with advanced ML techniques handling high-dimensional and potentially noisy data [34]. Fournaris implies challenges in ensuring high-quality data through security-by-design processes and run-time security monitoring [35]. Tertychny implies challenges in data collection in real-time testbeds and simulations [36]. Shen implicitly addresses data quality through accurate detection of data integrity attacks [37]. Mylrea ensures data quality and availability is essential, identifying real-time data processing and noise as significant challenges [38]. Oyucu highlights challenges in obtaining realistic and representative data for DDoS attacks, implying the need for robust data collection and preprocessing methodologies [39]. Johannesen utilizes recorded smart meter data, implying the handling of data sparsity, noise, and real-time processing needs [40]. Husnoo uses federated learning to mitigate data quality and privacy issues by allowing data to remain local [41]. Agnew uses data from both power and communication networks, highlighting the importance of high-quality data for security analysis [42]. Padmapriya utilizes real-time and historical data in a digital twin framework, addressing data quality and sparsity through comprehensive

data integration [43]. Hyder addresses data quality by generating high-fidelity datasets using detailed simulations [44]. Yilmaz uses a hierarchical data collection infrastructure to improve data quality and comprehensiveness in smart grid systems [45]. Chhokra creates synthetic data through simulations to mimic a wide range of attack scenarios, providing comprehensive and noise-controlled data [46].

Not Specifically Addressed: Appiah-Kubi does not specifically address data quality and availability in the document [31].

3.5.3 Operational challenges

On the operational front, **integration with existing systems** poses significant challenges. SG comprise a heterogeneous mix of legacy infrastructure and modern technologies. Ensuring compatibility and interoperability between new ML/XAI systems and existing grid components is complex. For instance, integrating an AI-based anomaly detection system with traditional SCADA (Supervisory Control and Data Acquisition) systems requires careful planning to prevent disruptions. To address these challenges, potential solutions include developing standardized APIs and middleware that bridge legacy systems with new technologies, adopting open communication protocols, and designing modular interfaces that allow for incremental integration and testing without impacting ongoing operations. **Scalability** is another operational concern. As SG expand and incorporate more DERs, the volume of data and the complexity of the network increase exponentially. Scaling ML and XAI solutions to accommodate this growth demands substantial computational resources and efficient algorithms. Limitations in processing power or storage can hinder the deployment of these systems across large-scale grids.

Moreover, the incorporation of ML and XAI introduces new **cybersecurity risks**. These systems can become targets for malicious attacks aiming to exploit vulnerabilities in AI algorithms or data pipelines. For example, adversarial attacks might manipulate input data to deceive the anomaly detection system, causing it to miss critical anomalies or generate false alarms. Ensuring the security and resilience of ML and XAI systems is paramount to maintaining grid integrity.

Deployment Constraints Beyond generic operational difficulty, practical deployment is constrained by latency, footprint, interoperability, and scale. First, many primary studies report classification accuracy but do *not* report inference time, memory footprint, model update cost, or communication overhead. This omission is itself a reproducibility and deployability gap. Second, field deployment may require execution on edge devices, substation gateways, or embedded controllers with limited compute and storage budgets, which makes heavyweight deep-learning pipelines difficult to operationalize. Third, integration with existing SCADA

and EMS environments remains challenging because of protocol heterogeneity, vendor-specific data formats, and the need to interoperate with established automation workflows governed by standards such as IEC 61850, IEC 62351, and IEC 62443 [69, 79, 81]. Fourth, scalability evidence remains limited: federated approaches such as [41], hierarchical IDS strategies such as [45], and SDN-oriented cross-layer designs such as [42] are among the more scalable architectural directions in the reviewed corpus, yet none of the 25 studies evaluates detection on a full-scale grid model above the thousand-bus range. This means that deployment feasibility is still weaker than algorithmic promise in much of the literature.

Integration with Existing Systems

Discussed Complex Integration Efforts:

The integration of ML/XAI with existing smart grid systems is complex, addressing compatibility and interoperability challenges [23]. Additionally, integrating cybersecurity measures, including blockchain, within existing PV systems presents significant complexities [22]. The integration of blockchain and fog computing with existing grid infrastructures is also discussed [24]. Furthermore, the Isolation Forest model is integrated within a smart grid, focusing on real-time monitoring and anomaly detection [25]. ML-based AD schemes are integrated into CPS, especially in critical infrastructures like power plants [26]. The challenges of integrating complex security systems into existing infrastructures are highlighted [35]. A cross-layered cybersecurity framework is integrated within existing smart grid architectures, emphasizing system compatibility challenges [42]. The integration of EKNN with IoT and digital twin technologies presents complex challenges, including ensuring compatibility and functionality across systems [43]. Additionally, integrating ML models into existing grid infrastructure poses compatibility and interoperability challenges, particularly with legacy systems [44]. Finally, integrating sophisticated simulation tools into existing power grid systems poses challenges, especially regarding compatibility with existing infrastructure [46].

Implied or Indirect Integration Challenges:

The feasible integration of autoencoders in SG is implied, though specific challenges are not detailed [27]. Similarly, frameworks designed to integrate with existing biopharma systems are mentioned without detailing specific integration challenges [28]. Successful integration within the smart grid framework is indicated, but lacks detail on specific operational challenges [29]. Significant challenges in integrating advanced ML models into existing systems are indicated, though specifics on operational challenges are not detailed [30]. There is a general mention of ML implementation for cyberattack detection and mitigation [31]. Integration with existing smart grid operations is implied without detailing specific challenges [32]. Integration challenges with exist-

ing infrastructures are implied without detailing specific operational challenges [33]. The integration of ML models into existing cybersecurity frameworks is discussed, but specific challenges are not detailed [34]. Integration challenges within smart homes are implicitly suggested, though specifics are not detailed [36]. Integration efforts by applying SHT within hybrid microgrid systems are suggested, but specific challenges are not detailed [37]. Integration challenges are implied due to the sophisticated nature of Digital Twin systems within existing infrastructure [38]. Integration into existing SDN-based SCADA systems is addressed by tailoring models to specific system architectures, though specific challenges are not detailed [39]. Integration into existing grid infrastructures is implied, though specific challenges are not detailed [40]. Finally, integration is facilitated by adapting to local data environments without central data aggregation, but specific challenges are not detailed [41], and the framework integrates with existing smart grid infrastructures without detailing challenges such as compatibility with legacy systems [45].

Not Specifically Discussed:

No authors fall into this category, as all mention integration efforts to some extent.

Scalability

Indirectly Mentioned or Implied Scalability:

Scalability issues are indirectly mentioned, hinting at computational and resource limitations [23]. The application of blockchain and ML across distributed systems implies scalability issues, with computational demands hinted [22]. Scalability is implicitly considered through the use of Isolation Forest, highlighting computational efficiency [25]. Scalable ML techniques such as ELMs are used, applicable across various CPS, though specific scalability challenges are not detailed [26]. Deep ensemble learning is employed for scalability, but computational and resource limitations are not directly addressed [29]. Scalability challenges are touched upon, with high computational demands and resource limitations implied for deploying Digital Twins [38]. The scalability of ensemble models is implied, though specific challenges are not discussed, and inherent computational resource limitations are noted [39]. Scalability is tested across different grid scenarios, suggesting considerations for large-scale deployment [40]. Scalability is supported by distributing the computational load, with gradient quantization reducing communication overhead [41]. A scalable architecture using distributed controllers is proposed to address scalability concerns and mitigate computational bottlenecks [42]. Scalability across IoT environments is aimed for using a digital twin framework, employing resource-efficient methods [43]. Scalability in real-world environments might differ from simulations, with significant computational resources required [44]. Lastly, a solution designed to be scalable with

a non-parametric, distributed approach for high-dimensional data is presented [45].

Not Specifically Discussed but Implied through Techniques:

The unsupervised learning nature and handling of large datasets suggest good scalability [27]. Frameworks are suggested as scalable across different settings, though specific challenges are not elaborated [28]. Scalability is crucial across different microgrid setups, though specifics are not addressed [30]. Scalability is not explicitly discussed, but robust ML models like LSTM imply scalability considerations and computational demands [34]. The use of sophisticated systems across large-scale infrastructures implies scalability considerations, though specifics are not detailed [35]. The scalability of detection systems is not directly discussed, but sophisticated methods suggest computational demands [37]. Frameworks are designed for scalability across different scenarios, though inherent computational and resource limitations are not explicitly discussed [46].

Not Specifically Discussed:

Scalability is not specifically addressed in the deployment of fog computing nodes [24], and it is not specifically addressed in [31]. The scalability of approaches in larger or different configurations is not discussed [32]. Scalability issues are not directly addressed, though the expansion and sophistication of measures are implied [33]. Scalability is not specifically discussed, though the use of common ML algorithms implies scalability efforts [36].

Cybersecurity Risks

Directly Discussed Cybersecurity Risks:

The integration of IT/OT increases the attack surface, emphasizing the need to strengthen fault detection to improve system security and resilience against attacks [23]. Vulnerabilities to cyber-attacks are focused on, with strategies for detection and mitigation of cyber-physical attacks explored [22]. A secure blockchain system implies a focus on mitigating vulnerabilities, using blockchain and fog computing to enhance security and resilience against cyber-physical attacks [24]. Enhancing security and resilience against cyber-physical threats is the focus, with no specific discussion on new vulnerabilities [25]. CPS security is enhanced with advanced AD techniques, indirectly addressing resilience against sophisticated attacks [26]. The effectiveness of autoencoders in enhancing grid security against FDI attacks is highlighted, though broader cybersecurity risks are not detailed [27]. Security and resilience against cyber threats are enhanced, without discussing new vulnerabilities introduced by ML and XAI [28]. Intrusion detection and resilience against cyber-attacks are enhanced, without discussing new vulnerabilities introduced by ML/XAI [29]. ML techniques for cyberattack detection are generally mentioned, without specifically discussing new vulnerabilities [31]. ML is used to detect vulnerabilities, aiming to enhance cyber-

security without discussing new vulnerabilities introduced by ML itself [32]. The security of electrical substations is enhanced through ML, implying efforts to improve system resilience but not detailing specific challenges [34]. Security is designed into cyber-physical systems to enhance resilience, though specific challenges are not detailed [35]. The evolving threat landscape and potential exploitation of new vulnerabilities introduced by sophisticated attacks are discussed, emphasizing the need for adaptive and responsive systems [38]. Detection capabilities are enhanced through a framework designed to detect various cyber-attacks, improving the security and resilience of smart grid systems [42]. The integration of ML models into cybersecurity could introduce new vulnerabilities, while also enhancing resilience against a wide range of cyber threats in IoT environments [43]. The integration of ML into critical infrastructure could introduce new vulnerabilities, with a focus on enhancing anomaly detection and cybersecurity [44]. Enhancing simulation and detection of attacks must ensure that no new vulnerabilities are introduced or sensitive data exposed [46].

Implied or Indirectly Mentioned Cybersecurity Risks:

Potential new vulnerabilities introduced by AI models are not detailed, as the focus remains on enhancing security [30]. Potential new vulnerabilities introduced by ML and AI systems are not addressed, with an emphasis on robust cybersecurity frameworks [33]. Differentiating between faults and attacks aims to enhance system security and resilience, though specific strategies are not detailed [36]. Improving cybersecurity in hybrid microgrids is focused on, without detailing specific strategies [37]. Enhancing security and resilience against DDoS attacks in SCADA systems is focused on, without detailing specific challenges [39]. Enhancing the security and resilience of grid systems to cyber threats is indirectly addressed by improving anomaly detection capabilities [40]. Improving detection capabilities to enhance security and resilience is the focus, without detailing new vulnerabilities introduced by federated learning [41]. Enhancing the security and resilience of SG by improving anomaly detection and cyberattack response capabilities is aimed for [45].

Not Specifically Discussed Cybersecurity Risks:

Cybersecurity risks are not specifically discussed in [31], [32], [35, 38], [39], [40], [41, 45], and [46].

Overall, the summary indicates that while many studies focus on enhancing security and resilience against cyber threats, specific details on new vulnerabilities introduced by ML and AI models are often lacking. The integration of advanced ML models and cybersecurity measures within existing infrastructures is seen as essential for improving system security, but the challenges of managing and mitigating new vulnerabilities require further attention.

3.5.4 Assurance problems

Balancing **quantitative and qualitative approaches** presents additional challenges. While quantitative metrics like accuracy and F1 score are essential for evaluating model performance, they do not capture qualitative aspects such as interpretability and user trust. An overemphasis on numerical performance can overlook the importance of providing explanations that operators can understand and act upon.

Contextual considerations also play a crucial role. SG operate in diverse environments with varying operational contexts, regulatory requirements, and technological infrastructures. ML and XAI solutions must be adaptable to these different contexts, requiring customization to address specific needs. For instance, an anomaly detection model suitable for one region's grid topology may not perform well in another due to different DER integration levels or consumption patterns.

Quantitative vs. Qualitative Approaches

Discusses Customization and Adaptability to Specific Contexts: Many studies emphasize the need to customize detection technologies to address vulnerabilities and security needs specific to various systems, such as PV systems [22], SG [24–29], gas turbines in power plants [26], and Active Buildings [33]. Additionally, enhancements in transformer protection [34], cyber-physical systems [35], and smart home contexts within SG [36] demonstrate the practical applications of ML for security tailored to specific operational nuances. Studies contextualized within hybrid microgrids [37], different energy systems [38], SDN-based SCADA systems [39], and IoT networks through digital twin technology [43] further highlight the adaptability of ML techniques. Challenges in adapting ML models to specific operational contexts within SG are also discussed [44], as well as addressing specific IoT security threats [45] and tailoring simulations to ensure relevance and applicability [46].

Indirect or Implied Customization and Adaptability: Some studies indirectly highlight the importance of customization and adaptability. For instance, the effectiveness of AI in quantitative terms is emphasized, while qualitative aspects like user trust receive less attention [30]. The use of standard IEEE distribution network models addresses specific operational contexts [32], and there is a general mention of ML for enhancing cybersecurity without detailing specifics [31]. Additionally, the adaptation of ML models to specific grid scenarios is demonstrated without extensive discussion [40]. Notably, there is a lack of extensive discussion on how these models can be customized to address the unique operational, regulatory, and trust-related challenges inherent in diverse grid environments.

Not Specifically Addressed: Several studies do not explicitly define, measure, or discuss the customization and adaptability of ML and cybersecurity solutions within their

research. These include works by [31], [32], [33], [34, 35], [36], [37, 38], [39], [40], [41], [42], [43, 44], [45], and [46].

Overall, the summary indicates that most studies discuss or imply the importance of customizing and adapting ML and cybersecurity solutions to specific operational contexts, whether in SG, biopharma, Active Buildings, or IoT environments. This customization is crucial for addressing the unique challenges and ensuring the relevance and effectiveness of these technologies in varied settings.

Contextual Considerations

Balances Both Quantitative and Qualitative Aspects: The study by [29] effectively balances quantitative performance metrics with qualitative aspects through the use of Explainable AI (XAI).

Focus Primarily on Quantitative Metrics: Many studies concentrate on quantitative performance metrics such as F1 scores, precision, recall, accuracy, and kappa values. For example, [32] focuses on F1 scores without balancing these with qualitative aspects like interpretability or user trust. Similarly, [34] emphasizes precision and recall with less focus on user trust and model transparency. Other studies, including [36], [37, 39], [40], [41], [42], [43, 44], [45], and [46], similarly emphasize quantitative improvements in various aspects of ML and cybersecurity systems, often overlooking qualitative aspects such as user confidence, interpretability, system transparency, and user perception. It should be noted that the relative emphasis on quantitative metrics versus qualitative aspects may vary depending on the application context; for example, highly regulated environments or consumer-facing systems may require greater transparency and user trust than industrial applications where numerical performance is paramount.

Focus Primarily on Qualitative Aspects or Operational Contexts: The work by [30] highlights the need for adapting AI solutions to the specific operational context of microgrid systems, emphasizing qualitative considerations over purely quantitative metrics.

Not Specifically Addressed: Several studies do not explicitly balance quantitative and qualitative approaches within their cybersecurity strategies. These include [31], [33], and [35]. Additionally, [38] indicates the need for systems that perform well quantitatively and are qualitatively trusted and understood by human operators but does not detail the balance between these aspects.

Not Covered: The studies by [23], [22, 24], [25, 26], [27], and [28] do not specifically address the balance between quantitative and qualitative approaches.

Overall, most studies tend to focus on quantitative performance metrics, such as accuracy, precision, and recall, for evaluating ML and cybersecurity systems. However, there is a recognized need to also consider qualitative aspects like interpretability, user trust, and system transparency to ensure

these systems are effectively integrated and trusted by human operators.

3.5.5 Standardization

The lack of **standardization** in deploying ML and XAI for anomaly detection in SG is a significant impediment. Current gaps and inconsistencies in standards lead to fragmented approaches, making it difficult to achieve cross-system compatibility and benchmarking. Without common guidelines, each utility may develop proprietary solutions, hindering collaboration and knowledge sharing.

Critical Analysis of Relevant Standards A more critical reading of the standardisation problem shows that the relevant standards do not simply “exist in the background”; they actively shape what anomaly-detection systems must be able to do in practice. The IEC 62351 series constrains how cybersecurity mechanisms can interact with power-system communications and therefore affects where network-aware anomaly detectors can be inserted [69]. IEC 62443 introduces security levels, zones, conduits, and industrial hardening expectations that influence how detection components can be segmented and managed across operational technology environments [81]. IEC 61850, meanwhile, affects communication semantics, timing expectations, and integration constraints in substation automation and related supervisory contexts [79]. Yet none of the 25 reviewed studies explicitly validates its anomaly-detection design against these standard families. This is an important gap because it means that many academically promising methods have not been demonstrated under the interoperability, timing, segmentation, and assurance conditions expected in regulated operational deployments.

Regulatory Compliance Constraints Regulation introduces an additional filter on deployability. In North American settings, prescriptive requirements such as NERC CIP, especially CIP-007 on systems security management, can constrain which monitoring agents, update mechanisms, and data flows are permissible in production environments [82]. In the European context, the EU AI Act is also directly relevant. Article 6, read together with Annex III, frames critical-infrastructure-related AI uses as high-risk, which brings explicit expectations around transparency, human oversight, documentation, and conformity assessment [80]. As a result, opaque high-performing models may still be difficult to deploy if they cannot satisfy auditability or oversight expectations. These regulatory realities suggest that future anomaly-detection research should not evaluate performance only in terms of accuracy, but also in terms of compliance readiness, explainability readiness, and interoperability readiness.

However, there is progress in this area, with ongoing efforts to establish standards and best practices. Organiza-

tions like the IEEE and NIST are working on frameworks that could provide **future standards and guidelines**. These initiatives aim to facilitate interoperability, improve security protocols, and establish performance benchmarks, paving the way for more unified and effective deployments of ML and XAI in SG.

Issues in Standardization

Explicitly Discusses Standardization Challenges: [22] discusses challenges in standardizing cybersecurity measures and suggests the need for future standards. Similarly, [24] implies the need for standardized approaches without detailing specific impacts on deployment and compatibility.

Does Not Address Standardization Directly: [27] does not address standardization directly, and the deployment impact and cross-system compatibility are not discussed. [29] does not specifically address standardization gaps and inconsistencies, nor does [30]. Additionally, Appiah-Kubi et al. [31] does not specifically address standardization issues. The works of [32], [33], [34, 37], [35], [36], [39], and [40] do not discuss current gaps or inconsistencies in standards for ML and XAI in SG. Furthermore, [41] does not directly address standardization issues, although the use of widely applicable federated learning principles may help bridge standardization gaps. Similarly, [42] does not explicitly address standardization issues but uses widely adopted SDN standards and ML models to ensure compatibility and ease deployment challenges. Padmapriya and Srivenkatesh [43] does not specifically address gaps in standards for ML and XAI in SG but highlights the potential need for new standards with emerging technologies like digital twins and EKNN. Lastly, [44] discusses high-fidelity dataset and simulation development, which may highlight gaps in existing standards and guidelines for integrating ML in SG, while [45] does not discuss current standardization gaps or inconsistencies directly. [46] highlights the development of a simulation framework with a domain-specific language, underscoring the need for new standards in attack simulations and reporting.

Implied Need for Standards or Coherent Approaches: [38] does not directly address standardization gaps but implies the need for coherent standards to guide the deployment of these technologies. The application of standard ML models in [40] suggests efforts to align with current practices, potentially aiding in addressing standardization gaps. Similarly, [42] uses widely adopted SDN standards and ML models to ensure compatibility and ease deployment challenges. [44] emphasizes that standardized approaches in dataset generation, simulation, and ML model training are crucial for ensuring compatibility and facilitating deployment across systems. Additionally, Padmapriya and Srivenkatesh [43] suggests that their approach could influence future standardization efforts by demonstrating effective ML integration in cybersecurity.

Not Covered: The false positive rate in standardization is not covered by [23], [25, 26], or [28].

Overall, while a few studies explicitly discuss or imply the need for standardization in cybersecurity and ML applications in SG, many do not address the issue directly, indicating a significant gap in the literature regarding the standardization and compatibility of these technologies.

Progress and Future Directions

Discusses Future Directions and Implications for Standardization: [25] includes future work that involves further integration with SCADA systems, hinting at standardization efforts. Yan et al. [26] implies future work on expanding methods to other power plant components, which could influence standardization efforts. [28] contributes to standardization efforts in biopharma cybersecurity and suggests future standards and guidelines, though not in detail. [35] hints at ongoing developments in future research directions but does not specify potential future standards. [36] suggests refining differentiation methods in future research, potentially influencing standards. Lastly, [42] incorporates software-defined networking and advanced ML techniques as cutting-edge improvements, highlighting ongoing enhancements in algorithms and networking approaches.

Does Not Discuss Ongoing Efforts or Future Directions in Detail: [22, 24, 27, 29–34, 37–41, 43–46] do not discuss ongoing efforts or future directions in detail regarding standardization.

Implied Future Work or Indirect References to Standardization: [38] implies that standardization issues could affect deployment and compatibility, although specific details are not provided. [40] suggests potential improvements in anomaly detection models but does not explicitly discuss future standards or guidelines.

Not Addressed: Santorsola et al., [23] does not discuss standardization or future directions.

Overall, while a few studies hint at or suggest directions for future standardization efforts in their work on ML and cybersecurity in SG, many do not explicitly address ongoing standardization efforts such as Cybersecurity standardisation based on IEC 62351 and IEC 62443 for IEC 61850 Systems [83] or future implications, indicating an area for further development and attention in the literature.

3.5.6 Opportunities for improvement

Despite these challenges, there are numerous opportunities to enhance the deployment of ML and XAI in anomaly detection:

Technological advancements offer promising avenues. Emerging technologies such as edge computing can alleviate scalability issues by processing data closer to its source, reducing latency and bandwidth requirements. Advances in algorithms, like federated learning, enable models to learn

from distributed data without compromising privacy or security.

A particularly promising opportunity is the development of **hybrid physics–data-driven detection architectures**. The literature reviewed here contains several studies that use physics-based simulation to generate training or test data, but relatively few embed physical laws directly into the learning architecture itself. This distinction matters. If Kirchhoff-type constraints, power-flow consistency, relay logic, topology constraints, or other engineering priors are introduced directly into the learning objective, then the resulting model may be better equipped to distinguish truly malicious deviations from benign physical disturbances. In that sense, physics-informed neural networks can be interpreted as one forward path, while grey-box system identification offers a complementary framework rooted in a longer engineering tradition [56]. Both directions are attractive because they aim to combine the adaptability of data-driven learning with the interpretability and plausibility checks of power-system modeling.

Collaborative efforts between industry, academia, and government can accelerate innovation. Sharing datasets, methodologies, and findings can lead to more robust and generalizable solutions. Joint research initiatives and partnerships can pool resources and expertise, addressing common challenges more effectively.

Regulatory and policy support is also critical. Governments and regulatory bodies can facilitate the adoption of ML and XAI by developing supportive policies that encourage innovation while addressing security and privacy concerns. Incentives for adopting advanced anomaly detection systems can drive investment and development in this area.

Technological advancements

Discusses Specific Technological Advancements and Opportunities for Improvement:

Enhancements include blockchain and advanced ML methods like deep learning [22]. The integration of fog computing and blockchain is focused on for continual advancements [24]. Advanced ML techniques such as Isolation Forest are utilized, with suggestions for further refinement in algorithms and hardware [25]. Extreme Learning Machines (ELMs) and sophisticated feature generation are used, indicating ongoing improvements in ML applications in Cyber-Physical Systems (CPS) [26]. Advanced autoencoder techniques are employed with potential for further improvements in predictive accuracy and adaptability [27]. Emerging technologies like the Digital Twin are incorporated to enhance anomaly detection and system monitoring [28]. Deep ensemble learning and metaheuristic optimization are used, suggesting ongoing improvements [29]. Advanced deep learning models are utilized, indicating ongoing technological advancements [30]. Technologies such as Long Short-Term Memory (LSTM) and autoen-

coders are employed, implying ongoing improvements in model architectures and anomaly detection techniques [34]. The emphasis on security-by-design and run-time security monitoring suggests ongoing methodological advancements and potential for improved detection logic and data processing [35]. The adaptation of SHT for cybersecurity in microgrids represents an innovative approach, with further refinement of methodologies suggested [37]. The development of technologies like Digital Ghost points to a continuous need for innovation in AI and cybersecurity [38]. The potential of ensemble learning for enhancing DDoS detection is highlighted, with suggestions for continuous refinement of methods for improved performance [39]. The integration of privacy-preserving gradient quantization techniques represents a technological advancement, with continuous improvement of federated learning models suggested [41]. Advanced ML techniques for anomaly detection are used, indicating opportunities for further technological advancements and continuous improvements in algorithms, data processing techniques, and hardware [40].

Mentions General Use of Advanced Technologies without Specific Details:

The use of ML for improving cyberattack detection and mitigation is discussed [31]. Advanced cybersecurity technologies are suggested without detailing specific new or emerging technologies [33]. The potential for improved algorithms and data processing techniques to enhance detection accuracy and reduce false negatives is implied [32]. Engagement with current ML technologies is mentioned, with suggestions for improving algorithmic approaches and data analysis for future advancements [36].

Does Not Address Technological Advancements:

Technological advancements or opportunities for improvement are not specifically discussed by [43–46].

Implied Potential for Technological Advancements:

Collaborative efforts are involved, though specific partnerships are not detailed, implying potential for shared datasets and benchmarking [42].

Overall, while many studies discuss or imply the importance of technological advancements and opportunities for improvement in the context of ML and cybersecurity for SG, there is a variation in the level of detail provided, with some focusing on specific technologies and others making general suggestions.

Collaborative Efforts Opportunities for Improvement

Implies Opportunities for Collaboration:

Opportunities for collaboration are implied through advanced cybersecurity implementations and research needs [22]. Strong collaborative potential is indicated, supported by the United States Department of Energy [26]. Potential for collaboration between industry, academia, and government is implied via academic context and public datasets [29]. Collaboration between industry, academia, and government

is implied, with shared datasets and joint research needed [30]. Implicit potential for collaboration exists through an academic research setting, suggesting shared datasets and benchmarking opportunities [39]. Although not specifically mentioned, the academic research nature implies potential for further collaboration, with shared datasets suggesting benefits from increased collaborative research efforts [40].

Highlights Ongoing Collaboration or Collaborative Opportunities:

Ongoing collaboration with federal agencies is highlighted, suggesting a framework for further academic and industrial cooperation [25]. Collaboration opportunities between industry, academia, and government are highlighted to enhance cybersecurity in biopharma [28]. Collaboration is evident in the development of the Digital Twin system, involving partnerships with governmental bodies and industry stakeholders [38]. Collaboration is facilitated by allowing multiple entities to contribute to model training without sharing raw data, encouraging collaborative improvements and benchmarking [41]. The need for robust regulatory support is implied to ensure compliance and foster broader adoption in critical infrastructure [42].

Suggests Potential for Collaboration but Not Explicitly Discussed:

Potential for collaboration is given the research context, though not explicitly discussed [27]. An innovative approach suggests potential for collaborative research and implementation, though not specifically discussed [24]. Although opportunities for collaboration are not explicitly mentioned, the academic nature of the research suggests potential for future collaboration [32]. The framework suggests potential for collaboration between industry, academia, and government, though not explicitly discussed [33]. The research context suggests potential for collaboration between industry, academia, and government, with shared datasets and joint research opportunities implied but not detailed [34]. The complex nature of cyber-physical system security suggests significant potential for industry, academia, and government collaboration, though not explicitly discussed [35]. The academic nature implies potential for industry, academia, and government collaboration, with shared datasets and benchmarking suggested but not detailed [36]. Although not explicitly discussed, the research context implies potential for collaboration between industry, academia, and government [37].

Does Not Specifically Address Collaborative Efforts:

Collaborative efforts are not specifically addressed by [23, 31, 43–45].

In summary, while some studies explicitly highlight ongoing collaborations or collaborative opportunities, others imply potential for collaboration through the academic and research contexts of their work. Several studies do not

specifically address collaborative efforts but suggest potential benefits from such initiatives.

Regulatory and Policy Support

Opportunities for Improvement

Supports Development of Regulatory Frameworks:

The development of regulatory frameworks to enhance cybersecurity in biopharma manufacturing is supported [28].

Implied Need for Supportive Policies:

Regulatory issues and policies are not discussed; however, the critical nature of cybersecurity in microgrids suggests a need for supportive policies [30]. Although opportunities for regulatory frameworks or policies are not specifically mentioned, a need for supportive policies is implied [33]. Opportunities for regulatory frameworks or supportive policies are not addressed, but supportive policies are generally beneficial for advancing cybersecurity measures in SG [34]. Opportunities for regulatory frameworks or policies are not specifically mentioned; however, the strategic importance of cyber-physical system security suggests a need for supportive policies [35]. Supportive policies are implied to facilitate further innovation in smart home cybersecurity, though not addressed [36]. Opportunities for regulatory frameworks or supportive policies are not addressed; the focus on cybersecurity suggests a need for supportive policies [37]. There is a need for supportive regulatory frameworks to facilitate the adoption of advanced cybersecurity measures, though specific policies are not detailed [38]. Opportunities for regulatory frameworks or supportive policies are not directly discussed; however, the focus on cybersecurity in critical infrastructure suggests a need for supportive policies [39]. Improvement in anomaly detection capabilities implies a need for policies that support technological innovation and address related challenges, though opportunities for regulatory frameworks or supportive policies are not discussed [40]. Addressing privacy concerns through federated learning aligns with regulatory requirements for data protection and highlights the need for supportive policies to foster innovation and address cybersecurity challenges [41]. Supportive policies could facilitate innovation and address specific cybersecurity challenges in SG [42].

Not Specifically Addressed:

Opportunities for regulatory frameworks or policy support are not specifically addressed by [22, 24–27, 29, 31, 32, 43–46].

In summary, while several studies imply the need for supportive regulatory policies to advance cybersecurity measures, specific discussions on regulatory frameworks and policy support are generally sparse. Some studies emphasize the importance of regulatory alignment and the benefits of supportive policies for technological innovation and cybersecurity enhancements.

3.5.7 Conclusion

Deploying Machine Learning (ML) and Explainable Artificial Intelligence (XAI) for anomaly detection in DER-rich SG presents a complex interplay of technical and operational challenges. Issues such as explainability, model accuracy, data quality, system integration, scalability, and cybersecurity require careful consideration and innovative solutions. However, these challenges also offer significant opportunities for advancement through technological innovation, collaborative efforts, and supportive regulatory frameworks.

By:

- **Advancing Technology:** Embracing edge computing, federated learning, and XAI techniques to enhance model performance and interpretability.
- **Fostering Collaboration:** Encouraging partnerships among stakeholders to share knowledge and resources.
- **Developing Standards:** Working towards unified guidelines and policies that facilitate interoperability and innovation.

Stakeholders can overcome these hurdles, leading to more effective anomaly detection systems. This progress will enhance the reliability, security, and efficiency of SG, supporting the sustainable integration of renewable energy resources and contributing to a resilient and efficient energy future.

Key Takeaways:

- **Technical Challenges Require Innovation:** Improving explainability, accuracy, and data quality of ML models is essential to address technical hurdles.
- **Operational Challenges Demand Integration:** Prioritizing seamless integration with existing systems, scalability solutions, and robust cybersecurity measures is critical for operational effectiveness.
- **Opportunities Through Collaboration and Standards:** Leveraging technological innovations, fostering collaborative initiatives, and developing supportive regulatory frameworks can address challenges and drive progress.
- **Enhancing Grid Resilience:** Effective anomaly detection systems significantly contribute to the reliability and security of SG, enabling the sustainable integration of Distributed Energy Resources (DERs).

By addressing both the technical and operational challenges and capitalizing on future opportunities, the deployment of ML and XAI in Smart Grids can be significantly enhanced, ensuring a secure, reliable, and sustainable energy infrastructure.



Fig. 14 Topics mapping under group testbeds for evaluation and validation category/group 6 shown in Fig. 6

3.6 Testbeds for evaluation and validation

See Fig. 6.

3.6.1 Introduction

The increasing complexity of SG, enriched with Distributed Energy Resources (DERs), has amplified the need for sophisticated anomaly detection systems to maintain grid stability, security, and efficiency. Before deploying these systems in operational environments, it is crucial to evaluate and validate their performance under conditions that closely mimic real-world scenarios. Effective testbeds serve this purpose by providing controlled environments where anomaly detection systems can be rigorously tested and refined.

To evaluate anomaly detection systems realistically, an effective testbed must incorporate both physical hardware and simulated cyber infrastructure. Figure 15 showcases a reference architecture, including hardware-in-the-loop components, realistic threat scenarios, and integrated SOC functionalities.

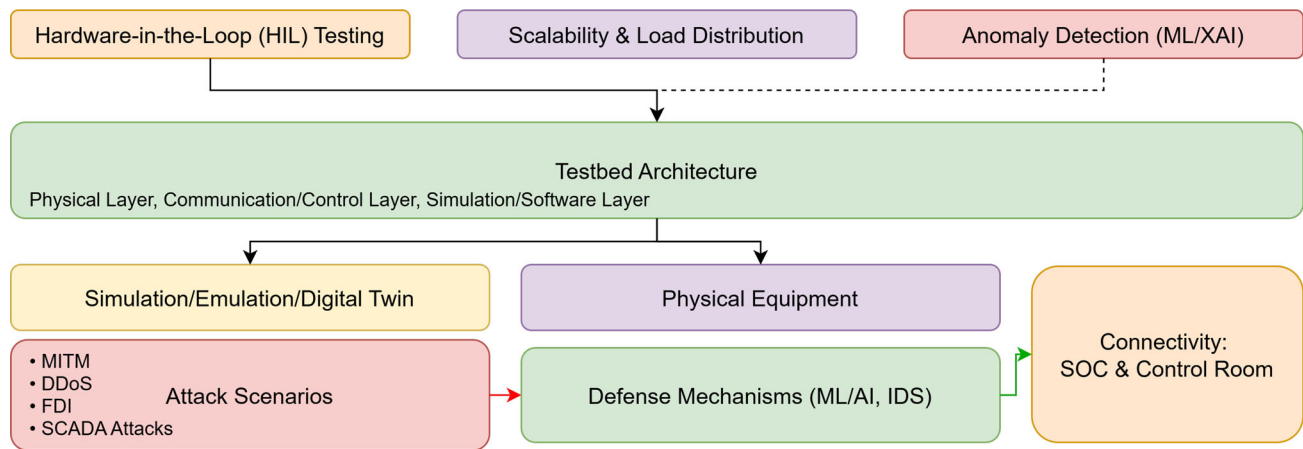


Fig. 15 Testbed Architecture Considerations Addressing RQ6. Highlights essential components-ranging from physical hardware to SOC integration-for realistic evaluation of anomaly detection systems

Following Fig. 14, this discussion examines the characteristics of effective testbeds designed for evaluating and validating anomaly detection systems in DER-rich SG.

3.6.2 Testbed purposes

We explore the various **purposes** these testbeds serve, such as **Hardware-in-the-Loop (HIL) testing**, which integrates real hardware components with simulated environments to assess system responses accurately. HIL testing is essential for verifying the interaction between physical devices and anomaly detection algorithms, ensuring that systems behave as expected when faced with real-world signals and disturbances.

Another critical purpose is **designing for scalability**, where testbeds simulate the expansion of the grid and the integration of additional DERs. This aspect is vital for testing how anomaly detection systems perform as the grid grows in size and complexity, highlighting potential bottlenecks or performance issues. **Load distribution** is also examined within testbeds to assess how systems manage varying demand levels and the impact on anomaly detection accuracy. **Testing HIL (Hardware in the Loop)**

Definition and Importance: Most studies emphasize the integration of real hardware with simulated environments to validate system performance under realistic conditions. For example, [30] discusses integrating physical hardware into simulations to test and validate performance and robustness. In contrast, some studies, such as [23] and [25], do not define or discuss the importance of HIL.

Examples and Applications: Specific applications include the use of Opal-RT real-time test beds for PV system performance validation as mentioned by [22], the development of an in-house HWIL simulator for gas turbine AD algorithms by [26], the description of a BioSecure Digital Twin for bio-

pharma production systems by Mylrea et al., and the use of MATLAB and OpenDSS for cybersecurity testing in power networks by [32]. General HIL applications for testing in industries such as power systems are also mentioned by [33].

Lack of Detail: A significant number of entries, like [24], [27], and [36], either do not provide definitions or applications or both. This indicates gaps or a lack of emphasis on HIL in their respective studies.

In summary, HIL is widely recognized for its utility in integrating real and simulated components to validate complex systems across various sectors, there is inconsistency in how well it is defined or utilized across different studies.

Designing for Scalability

Definition and Importance: Several studies highlight the ability of systems to expand and handle increased loads or complexity without compromising performance. Common themes include the capability of systems to manage growth, accommodate additional resources, and ensure consistent performance as demands increase. For instance, [22] and [28] stress the importance of systems handling increased loads efficiently.

Examples and Applications: Specific technologies and systems discussed by [22] include scalability in cybersecurity for grid-forming inverters, while [38] mention digital twin technology scaling from individual components to entire power plants. Additionally, [44] focus on a co-simulation platform that integrates large distribution grids. General applications are highlighted by studies such as [35], who mention modular designs in manufacturing systems that allow for easy expansion, and [41], who describe the use of federated learning to enable scalable attack detection in power grids. Furthermore, Padmapriya and Srivenkatesh [43] discuss scalability in cloud computing services, which support dynamic resource allocation.

Lack of Detail: A significant number of entries, such as [23], [25], and [36], do not provide specific definitions or examples. This suggests a gap in detailed discussion or application of scalability in their research.

Overall, the review demonstrates a focus on the essential role of scalability in ensuring that systems can accommodate growth and handle increased operational demands efficiently across various domains, including cybersecurity, cloud computing, and power grid management.

Load Distribution

Definition and Importance: Many studies define load distribution as the balancing of power, computational, or network loads across systems to optimize performance and prevent overloading. For example, Santorsola et al., [23] emphasize balancing power distribution for stability and efficiency. Similarly, [28] and [37] discuss evenly spreading loads to avoid bottlenecks and enhance system efficiency. Additionally, [39] and [42] highlight spreading workload across resources to enhance performance and reliability.

Examples and Applications: In power systems, Santorsola et al., [23] describe a monitoring system for distributing power based on the criticality of different facilities, and [32] use randomized load profiles to generate realistic scenarios for electrical load balancing. For computational and network systems, [38] employ AI algorithms to manage load distribution by predicting system demand changes, while [43] note the importance of spreading network traffic across servers in web services for high availability. In distributed systems, [41] utilize a federated learning framework to distribute training loads across multiple systems, reducing the central computational burden.

Lack of Specific Details: Several authors, such as [24], [25], and [45], either did not discuss load distribution or provided no specific examples, indicating a potential gap in the detailed application or study of load distribution in their research.

Anomaly Detection

Definition and Importance: All authors emphasize anomaly detection as the process of identifying patterns that deviate from normal operations, which is crucial for identifying faults, cyber-attacks, and other potential issues. Common themes include the identification of unexpected behaviors, deviations from expected data patterns, and unusual patterns that could indicate security breaches or system malfunctions.

Examples and Applications: In smart grids and cybersecurity, Santorsola et al., [23] use a Q-Learning RL agent, while [25] apply the Isolation Forest algorithm for smart grid data. [22] highlight the use of machine learning for security patches in grid systems. Advanced machine learning techniques are utilized by [27], who employ Autoencoders for detecting False Data Injection (FDI) attacks, and [39], who apply ensemble learning techniques to detect DDoS attacks in SCADA systems. Diverse fields and techniques include [37]

using SHT to detect DIAs on renewable energy sources and [41] using a Deep Auto-Encoder network to detect anomalies in power systems and cybersecurity. Comprehensive solutions are implemented by [38], who develop the Digital Ghost system using AI and ML for real-time anomaly detection, and [44], who create a machine learning-based system using high-fidelity datasets for detecting cyber-attacks in SG.

Coverage and Focus: Most entries detail specific methodologies or technologies used for anomaly detection, demonstrating a wide range of approaches from traditional machine learning models to cutting-edge deep learning and reinforcement learning applications. The applications span a variety of fields, emphasizing the versatility and critical importance of anomaly detection across sectors like power systems, cybersecurity, industrial control systems, and smart grid management.

Other Purposes

Cybersecurity Enhancements and Anomaly Detection: Santorsola et al., [23] use reinforcement learning to identify safety-critical state patterns. [22] incorporate hot patching for firmware updates and use blockchain for secure communications. [27] and [34] focus on anomaly detection, specifically targeting FDI attacks and evaluating various cyberattack detection models. Additionally, [36] and [37] enhance cybersecurity measures and analyze faults versus security attacks using machine learning algorithms.

System Optimization and Real-Time Simulation: [28] enhance workforce development through cyber wargaming and training. [30] test control strategies and the impact of cyber-attacks, validating simulations with real-time setups. Furthermore, [35] and [38] implement continuous monitoring and simulation of CPS via Digital Twins, enhancing decision-making and system response.

Smart Grids and Control Systems: [32] aim at detecting coordinated attacks on S. [44] focus on generating high-fidelity datasets for cybersecurity tools development. [41] validate the efficiency of federated learning frameworks in control systems with multiple SCADA subsystems. Additionally, [45] focus on mitigating IoT-based cyber-attacks using the MIAMI-DIL framework.

Data Analytics and Prediction Models: [33] discuss simulation-based approaches—controlled stress testing, sensitivity analysis, and prediction under uncertainty—for planning. [39] optimize detection models through feature selection and hyperparameter tuning. [42] validate cyber threat detection frameworks and improve state estimation using simulations.

Lack of Specific Details: Some studies, like [40], did not specify their testbed purposes or applications, indicating potential gaps or a broader focus not confined to specific applications.

3.6.3 Testbed architecture

We delve into the **architecture** of these testbeds, emphasizing the importance of a detailed and modular design that encompasses all relevant components of a Smart Grid. This includes the integration of physical hardware, software simulations, and communication networks to create a holistic environment. A well-architected testbed allows for comprehensive testing of anomaly detection systems across different layers and functionalities of the grid.

Detailed Architecture of Testbeds

Architecture Components: Most testbeds incorporate elements like sensors, actuators, controllers, communication networks, and specialized software. For example, a modular simulation framework with virtualized control components is used [23], and components like PV modules and inverters are included [22]. Advanced setups include layered architectures integrating data, control, and physical layers to manage complex network interactions and cyber-physical systems effectively [39, 44].

Purpose and Importance: The primary purpose of these architectures is to simulate real-world conditions to validate control algorithms, test cybersecurity measures, and ensure system robustness against attacks. For instance, robustness against cyber-attacks is evaluated [31], and microgrid (MG) operations are tested under various scenarios [37]. These testbeds are crucial for continuous testing, anomaly detection, and threat mitigation across critical infrastructures, utilizing high-resolution models and AI-driven anomaly detection algorithms [38].

Applications and Validation: Testbeds are applied in diverse scenarios, ranging from smart grid management and cybersecurity to anomaly detection and system performance validation. Anomaly detection models are evaluated on a hardware-based testbed [25], and real-time digital simulators are used for attack detection model training [41]. They support a wide range of applications, including intrusion detection [45], secure data transmission [24], and comprehensive system evaluations under varying operational conditions [43].

Technological Integration and Simulation: Several studies highlight the integration of digital twins and machine learning within testbed architectures. Sensors and actuators are integrated to create realistic test environments [35], and a combination of SDN emulation and quick data generation is employed for cybersecurity evaluations [42]. Additionally, the use of hybrid models that combine real hardware with simulated components is a common theme, enhancing the depth and realism of the testing environment and ensuring comprehensive evaluation across various system aspects.

In summary, the detailed architecture of these testbeds is designed to replicate complex real-world systems closely, enabling researchers and engineers to validate the perfor-

mance, security, and resilience of new technologies and control strategies in a controlled yet realistic setting.

3.6.4 Testbed scenarios

The discussion further covers the **scenarios** simulated within testbeds, particularly those involving **systems under attack**. By incorporating simulations, emulations, or digital twins, testbeds can replicate cyber-physical threats like **Man-in-the-Middle (MITM)** attacks, **Distributed Denial of Service (DDoS)** attacks, **False Data Injection (FDI)**, and **SCADA system breaches**. Testing with these scenarios enables the evaluation of anomaly detection systems' ability to identify and mitigate various attack vectors, enhancing grid cybersecurity.

Moreover, we address the role of **defense mechanisms** within testbeds, which are implemented to test the resilience and response strategies of anomaly detection systems. Incorporating both offensive and defensive elements in test scenarios ensures a robust evaluation of system capabilities.

Systems Under Attack - Simulation/Emulation/Digital Twin

Definitions and Importance: Simulation is generally defined as creating virtual models to mimic or study system behaviors. This approach is extensively used across various studies to test algorithms, predict system responses, and train models under controlled but realistic conditions [22, 24]. Emulation is often described as replicating system functions using different hardware or software, allowing researchers to evaluate system operations in a substitute environment. Digital twin is typically defined as real-time virtual replicas of physical systems, used for ongoing monitoring, testing, and optimization. Digital twins are crucial for continuous system evaluation and response under dynamic conditions [32, 33].

Applications: In the realm of cybersecurity and anomaly detection, many entries, such as [38] and [39], utilize these methods to test cybersecurity measures and anomaly detection systems. For example, [37] uses digital twin technology to simulate Direct Integrity Attacks (DIAs) and develop mitigation strategies. Regarding fault diagnostics and system monitoring, Padmapriya and Srivenkatesh [43] employs digital twins for real-time monitoring and predictive maintenance, enhancing system reliability and security. In training and validation, [41] relies on extensive simulations over publicly available industrial-control-system attack datasets, aiding in the validation of attack detection models.

Technological Integration: Several studies highlight the integration of advanced technologies with traditional simulation and emulation techniques to create more comprehensive and effective test environments. For instance, [42] uses Mininet for Software Defined Network (SDN) emulation combined with Simulink for real-time power grid simulations.

Evaluation of System Performance: Testbeds are used not just for security, but also for evaluating overall system performance under potential stress conditions or in the face of operational challenges, as seen in the works of [33] and [35].

In summary, the integration of simulation, emulation, and digital twin technologies in testbed scenarios plays a crucial role in advancing system development, ensuring security, and optimizing performance across various disciplines. This approach allows for safer, more controlled experimentation and development, reducing risks before implementation in real-world environments.

Systems Under Attack - Physical Equipment

Integration of Real Hardware: Several authors, such as [22, 26], rely on real-time hardware-in-the-loop platforms (an OPAL-RT real-time simulator with grid-tied inverters in [22] and an in-house HWIL threat simulator in [26]) to ensure the authenticity of their testing environments. These platforms allow real-time control-algorithm validation and the testing of systems under practical operational conditions [22, 26]. In the context of smart grid simulations, [25] and [26] utilize real hardware to simulate S, applying real-world data streams to analyze performance and validate anomaly detection schemes.

Applications and Validation: In biopharma and industrial systems, [28] integrates physical sensors with digital twins to validate biopharma production systems, whereas [35] tests PLCs and SCADA systems in industrial environments to validate security measures and system performance. For energy resources and cybersecurity, [37] utilizes real micro-turbines and renewable energy resources to validate their digital twin simulations, ensuring realistic testing conditions. Additionally, [39] employs Mininet VMs and Open vSwitch to create network scenarios for testing against TCP, UDP, and ICMP flood attacks.

Diverse Technological Implementations: Some studies adopt hybrid approaches, blending physical devices with virtual environments to enhance the depth and reliability of their simulations. For example, [30] uses DC-DC converters to validate control mechanisms, and [38] employs real gas turbines to test AI-driven anomaly detection systems. In the realm of network and control systems testing, Padmapriya and Srivenkatesh [43] and [44] focus on testing control strategies and network layers using real hardware such as workstations and routers to ensure that new strategies work effectively with existing physical devices.

Data Collection and Analysis: Real-world data usage is pivotal in these studies. For instance, [40] utilizes real-world data from smart meters to test anomaly detection algorithms, ensuring that the simulations reflect actual operational scenarios and can effectively identify anomalies.

Gaps in Hardware Detailing: Some entries, like [24] and [46], do not provide detailed descriptions or applica-

tions related to the use of physical equipment. This omission might indicate a focus on theoretical aspects or the preliminary stages of research where physical implementation has not yet been detailed.

Attack Mechanisms - Man-in-the-Middle (MITM)

Definition and Importance:

MITM attacks are generally defined as cyber-attacks where the attacker intercepts and potentially alters communications between two parties without their knowledge. This definition is consistent across multiple entries [22, 28, 33]. The importance of understanding and mitigating MITM attacks is emphasized due to their potential to compromise communication integrity and security [22, 35].

Applications and Examples:

In the context of **Blockchain and Anomaly Detection**, [22] discuss using blockchain-based frameworks to detect MITM attacks and ensure communication integrity. For **Smart Grid Security**, fournarisspsdesignsps2019 provide an example of intercepting communications between sensors and control units within S, illustrating the practical implications of MITM in critical infrastructure. Regarding **Network Security**, Padmapriya and Srivenkatesh [43] mention eavesdropping on network traffic and injecting false information, showcasing how MITM can be simulated in network environments. Additionally, [36] and [46] describe more detailed scenarios, including replay attacks, sinkhole attacks, message modification, and data attacks like biasing and scaling on consensus variables.

Technological Approaches to Detection:

Advanced detection methods incorporate **AI and Machine Learning** techniques, as highlighted by [38], who emphasize the use of AI systems to detect and respond to MITM attacks. Furthermore, [42] discuss using network indicators such as increased packet counts and latency to identify MITM activity, particularly in scenarios where ARP messages mislead network participants.

Gaps and Undefined Entries:

Several authors, such as [24] and [25], do not provide specific details on MITM scenarios, which might suggest either a focus on other areas of cybersecurity or an initial stage of research in this specific area.

Attack Mechanisms - Distributed Denial of Service (DDoS)

Definition and Importance:

DDoS attacks are consistently defined across several entries as malicious efforts to overwhelm a system with internet traffic, thereby causing disruptions or making the system unavailable to legitimate users [22, 28, 33]. The importance of understanding and mitigating DDoS attacks is highlighted due to their potential to disrupt critical services and infrastructure, as noted by multiple studies.

Applications and Examples:

Regarding **Security Measures and Strategies**, [22] discuss network segmentation, encryption, and real-time intrusion detection as countermeasures against DDoS attacks. Similarly, [39] focus on using machine learning models to detect DDoS attacks. In terms of **Specific Attack Scenarios and Simulations**, [38] use the Digital Ghost system to identify and mitigate DDoS attacks, maintaining system availability, while [42] simulate scenarios where malicious traffic increases to study the impacts on system performance. Concerning the **Impact on Specific Sectors**, [35] provide an example of flooding a control system in a power grid to disrupt operations, and [43] mention crashing a website by flooding it with requests.

Technological Approaches to Detection and Mitigation:

Several studies incorporate **Advanced Technologies** and methodologies to enhance detection and response capabilities against DDoS attacks. For instance, [39] and [44] imply the use of sophisticated frameworks and machine learning to anticipate and counteract such threats. Additionally, [45] discuss IoT botnets like Mirai as real-world examples of DDoS attacks, highlighting the relevance of these scenarios in testing and validating security systems within testbeds.

Gaps and Undefined Entries:

Some authors, such as [24] and [34], do not provide specific details or mention DDoS within their documented research, which may indicate a focus on other types of cybersecurity threats or a preliminary stage of research regarding DDoS.

Attack Mechanisms - False Data Injection (FDI)

Definition and Importance:

FDI attacks generally involve injecting incorrect or deceptive data into systems to mislead operations and decision-making processes. This concept is consistently highlighted across multiple entries, demonstrating its critical role in cybersecurity threats to various infrastructures like S, industrial control systems, and power systems [29, 33]. The importance of detecting and mitigating FDI attacks is emphasized due to their potential to compromise system integrity and functionality, leading to erroneous outcomes or system failures.

Applications and Examples:

In **S and Power Systems**, several studies such as [29] and [33] discuss FDI threats against smart power grids, focusing on dynamic state estimation and smart grid operations. Specific scenarios include altering power balance and manipulating voltage and current sensor data. Regarding **Industrial and Renewable Energy Systems**, [35, 37] address FDI attacks in the contexts of renewable energy sources and industrial sensor networks, respectively, showcasing the broad applicability of FDI threat scenarios across different sectors. Additionally, [38] and [44] implement

machine learning techniques to detect anomalies indicative of FDI attacks, reflecting the integration of advanced analytics in combatting such cyber threats.

Technological Approaches to Detection and Mitigation:

Sophisticated **Machine Learning and Algorithms** are employed to identify and respond to FDI attacks, as illustrated in the works of [37, 38], and [42]. Furthermore, [34, 43] provide **Scenario-Based Simulations** that simulate the impact of FDI on systems, testing the robustness of detection mechanisms and the effectiveness of response strategies.

Impact on Decision Making and System Responses:

FDI attacks can lead to serious consequences, such as misconfigurations, maloperations, and non-technical losses, by misleading the data-driven decision-making processes that control critical infrastructure operations.

Gaps and Undefined Entries:

While most entries provide detailed definitions and scenarios, a few like [36] and [39] do not specifically detail their approach to FDI, indicating potential areas for further research or a focus on other aspects of cybersecurity.

Attack Mechanisms - SCADA Attack

Definition and Importance:

SCADA attacks primarily target control systems that manage critical infrastructure, aiming to disrupt operations, compromise data integrity, and cause data loss. The importance of these attacks is consistently emphasized due to their potential impact on industrial and critical infrastructure security [22, 28, 39]. Specific studies define these attacks as direct threats to the stability and security of control systems, with references to disrupting operations and affecting critical system functionalities.

Applications and Examples:

In terms of **Security Enhancements and Simulations**, [28] simulate SCADA attacks to test and enhance security protocols and response mechanisms. Similarly, [38] use advanced AI to detect and neutralize SCADA attacks, ensuring operational continuity. Regarding **Historical and Real-World Contexts**, [35] provide a specific historical example with the Stuxnet attack targeting Iran's nuclear facilities, highlighting the real-world implications and potential damages of SCADA attacks. Additionally, [39] propose detection mechanisms tailored to SCADA systems, aiming to mitigate threats and prevent operational disruptions.

Technological Approaches to Detection and Mitigation:

Advanced technologies, particularly AI, are utilized to enhance detection capabilities and response strategies against SCADA attacks, as demonstrated by [38]. Furthermore, [43] and [46] discuss altering control commands and manipulating data as methods of attack, underscoring the need for robust security measures in industrial environments.

Gaps and Undefined Entries:

Several authors, such as [24] and [26], do not provide specific details on SCADA attacks, which may indicate a broader focus on other security aspects or preliminary stages of research concerning SCADA-specific threats.

Attack Mechanisms - Other Types of Attacks

Types of Attacks and Importance:

The research spans a wide array of cyber-attacks including Direct Integrity Attacks (DIAs), replay attacks, False Data Injection (FDI), Denial of Service (DoS), phishing, ransomware, and more sophisticated threats like Advanced Persistent Threats (APT) and synchronized cyber incidents. These attacks are noted for their potential to compromise data integrity, disrupt critical operations, and breach sensitive information, emphasizing the importance of robust cybersecurity measures in protecting critical infrastructure and sensitive data systems.

Applications and Examples:

In the realm of **AI and Machine Learning**, [22] employ AI/ML-based detection systems and blockchain technology to ensure secure data transactions and detect DIAs. Regarding **Simulating Attacks**, [28] provide a platform for simulating a range of cyber-physical attacks, enhancing the ability to detect and respond to diverse attack vectors. Additionally, [35] discuss specific threats such as phishing and ransomware, providing real-world examples like email scams targeting utility employees and encryption attacks on critical infrastructure.

Technological Approaches and Detection Strategies:

Advanced detection systems are a focus, with [30] describing Stealthy Local Covert Attacks (SLCA) and their detection, illustrating the sophistication in mimicking normal behavior to evade detection. Comprehensive frameworks are discussed by [38] and [42], which detect and respond to hybrid cyber-physical attacks, insider threats, and other complex cyber threats. Additionally, [41] utilize a federated learning framework and Deep Autoencoder (DAE) models to collaboratively detect anomalies and cyber-attacks while preserving data privacy.

Impact and Mitigation:

The research highlights the critical impact of these cyber-attacks on system stability, data integrity, and overall security, with a strong focus on developing effective mitigation strategies to protect against these threats. Examples such as [46]'s focus on attacks on protection system components like relays and breakers, and [44]'s exploration of data loss and delay attacks, further underscore the diversity of potential threats and the necessity for tailored security solutions.

Defense Mechanisms

Defense Mechanisms and Importance: The common theme across the research is the utilization of advanced technology like machine learning (ML), artificial intelligence (AI), and deep learning for detecting, mitigating, and responding to cyber threats. The importance of these defense

mechanisms is universally recognized for their critical role in maintaining the security, stability, and integrity of systems, particularly in environments such as S, critical infrastructure, and industrial control systems.

Applications and Examples: Many entries discuss the use of ML models and intrusion detection systems (IDS) for identifying and defending against anomalies and attacks [25, 27, 33]. For instance, [28] enhance biopharma manufacturing systems, [35] focus on control system protocols in CPS testbeds, and [37] apply SHT techniques for real-time attack mitigation in power systems.

Technological Integration: [39] optimize ML models to detect DDoS attacks, illustrating the effectiveness of ensemble learning in cybersecurity applications. [38] implement AI-driven anomaly detection, threat localization, and automated neutralization responses, demonstrating advanced capabilities for continuous security management. [44] emphasize the testbed's role in generating realistic datasets and simulating attack scenarios, which are crucial for developing and testing defense mechanisms.

Impact on System Resilience: Defense mechanisms are crucial for ensuring that systems can continue to operate securely despite attempts to compromise their functionality. This is evident in examples like the MIAMI-DIL framework, which integrates detection, isolation, and localization functionalities to minimize service interruptions [45].

Comprehensive Security Strategies: The research collectively underscores a move towards integrated, holistic security strategies that incorporate both preventive measures and active response protocols. This includes everything from blocking harmful traffic and encrypting data to employing decentralized multi-agent systems for network management and using federated learning for collaborative threat detection.

3.6.5 Connectivity

Lastly, the discussion examines the **connectivity** aspects of testbeds, such as the **integration or isolation of Security Operations Centers (SOC)** and control rooms. The decision to include or isolate these components affects the realism of the test environment and the applicability of results to operational settings. For instance, integrating an SOC within the testbed can provide insights into how anomaly detection alerts are managed and acted upon in real-time by security personnel.

Integration or Isolation of SOC

Definition and Importance:

Integration of SOC is highlighted by several studies as crucial for centralized monitoring and real-time response to cyber threats. For instance, integration enhances the capability to monitor security events and respond rapidly to threats [28]. Conversely, some studies discuss the benefits of isolat-

ing SOC's to reduce attack surfaces and limit access, thereby enhancing security by protecting against potential insider threats [33, 35].

Applications and Examples:

Integration with SOC's is essential for continuous visibility and rapid response to cyber threats, as demonstrated in the works of [38] and [44]. Examples include the integration of SOC capabilities with the Digital Ghost system for continuous monitoring and rapid response. Additionally, Appiah-Kubi et al. [31] describe how a central agent reconfigures the network in response to alerts from decentralized agents, ensuring coordinated defense mechanisms. Furthermore, [41] discuss the control center's role in aggregating and distributing model updates, ensuring efficient coordination among SCADA subsystems, which highlights the balance between integration for control and isolation for security.

Technological Integration:

Advanced detection systems are utilized in conjunction with SOC's to enhance threat detection and response capabilities, as mentioned by [38]. These studies highlight the use of AI/ML-based detection and event-triggering control. Real-time threat detection and response are facilitated through SOC integration, ensuring systems are continuously monitored and protected [38, 44].

Security Trade-offs:

The balance between integrating SOC's for comprehensive monitoring and isolating them to enhance security by limiting access points is a common theme. For example, Padmapriya and Srivenkatesh [43] highlight that integration is prevalent in large enterprises for centralized monitoring, while isolation is preferred in sensitive environments to protect against insider threats.

Gaps and Undefined Entries:

Several authors, including [24–26], do not provide specific discussions on the integration or isolation of SOC's, indicating a potential area for further research or a focus on different aspects of cybersecurity.

Integration or Isolation of the Control Room

Definition and Importance:

Integration of control rooms is emphasized for ensuring seamless operation, quick response to detected anomalies or attacks, and real-time decision-making and control. This facilitates centralized control and monitoring, improving operational efficiency and resilience. On the other hand, isolation is highlighted as a method to enhance security by reducing potential attack surfaces and restricting access to critical control systems, thereby providing an additional layer of protection against cyber threats.

Applications and Examples:

Integrated control mechanisms are vital for real-time communication and coordination, facilitating quick and effective responses to security incidents [22, 28]. [31] mention real-time mitigation strategies and system resilience, where inte-

grated control rooms allow immediate responses to attacks by reconfiguring network settings and applying new security thresholds. Additionally, [35, 43] address the balance between integration for operational efficiency and isolation for enhanced security. For example, critical infrastructure like power plants often use isolated systems to protect against cyber threats.

Technological Integration:

Advanced monitoring and control systems are employed to enhance the functionality of control rooms. [37] describe the Hybrid Microgrid Central Controller (HMGCC) managing testbed operations, ensuring performance and security by integrating control and monitoring functions for different operational modes. Furthermore, [44] discuss the control layer mimicking real utility control centers, providing a realistic environment for testing control strategies and responses without impacting live operations.

Impact on System Security and Efficiency:

Integrated control rooms equipped with advanced cybersecurity measures offer real-time insights and effective response capabilities while maintaining operational continuity [33, 38]. Additionally, [45] highlight the role of the control center in coordinating data analysis and mitigation efforts within hierarchical intrusion detection systems (IDS), using aggregated statistics from data aggregators and smart meters to identify and mitigate attacks.

Gaps and Undefined Entries:

Several authors, including [24], [25], and [26], do not provide specific discussions on the integration or isolation of control rooms, indicating areas for further exploration or different research focuses.

3.6.6 Conclusion

Effective testbeds are crucial for the development, evaluation, and validation of anomaly detection systems in DER-rich S. By closely simulating real-world conditions through purposeful design, detailed architecture, realistic scenario simulation, operational integration, use of physical equipment, and flexibility, these testbeds ensure that anomaly detection systems are robust, reliable, and effective when deployed in complex grid environments.

These testbeds achieve realism by:

- **Replicating Smart Grid Dynamics:** Utilizing Hardware-in-the-Loop (HIL) testing and comprehensive architectures to mirror actual grid operations.
- **Simulating Cyber Threats and Operational Anomalies:** Allowing thorough testing of detection and response capabilities against both malicious activities and system faults.

- **Incorporating Real Hardware and Data:** Providing authenticity to the testing environment with actual equipment and datasets.
- **Including Operational Components:** Integrating elements like Security Operation Centers (SOCs) and control rooms to reflect real-world workflows.

By testing under these realistic conditions, stakeholders can enhance grid resilience against anomalies arising from both operational issues and malicious activities. This contributes significantly to the reliability and security of SG as they integrate increasing levels of Distributed Energy Resources (DERs).

Key Takeaways:

- **Realism is Essential:** Effective testbeds closely mirror actual Smart Grid conditions, providing accurate assessments of system performance.
- **Comprehensive Simulation:** Covering a wide range of scenarios and incorporating various system components ensures thorough evaluation.
- **Flexibility and Adaptability:** The ability to adjust to different configurations and technologies maintains the testbed's relevance as grids evolve.
- **Operational Integration Enhances Validity:** Including elements like SOC and control rooms improves the applicability of test results to real-world operations.

By adhering to these characteristics, testbeds play a pivotal role in advancing and validating anomaly detection systems in DER-rich S. This enables stakeholders to bolster grid resilience against both operational anomalies and cyber threats, ultimately supporting a reliable and secure energy infrastructure.

3.7 Focused comparison: the six studies that attempt the cyber—physical distinction

Only 6 of the 25 included studies explicitly attempt to distinguish cyber-attacks from safety faults (or physical disturbances) within a single evaluation setting: Yan et al. [26], Tertytchny et al. [36], Shen et al. [37], Johannesen et al. [40], Husnoo et al. [41], and Hyder et al. [44]. Because this small set carries the direct evidence for the review's central question, this subsection compares the six studies along the dimensions that matter for the distinction task: the operational context and data, the features used, the architectural choices, the mechanism by which the attack-versus-fault decision is made, the performance reported for that decision, and where each study falls short. Table 9 consolidates the comparison. We count a study toward this set only when separating cyber-attacks from genuine physical or safety faults

is an explicit, evaluated aim, not when a multiclass intrusion classifier inherits a non-attack “natural” or “disturbance” class from a generic benchmark. Naeem et al. [29], for example, report strong three-class results (no-event, attack, natural disturbance) on the MSU–ORNL power-system benchmark and even quantify attack-versus-disturbance confusion, yet frame their task as malicious-versus-benign intrusion detection on a generic SCADA dataset rather than as the DER-rich attack-versus-safety-fault distinction; the study is therefore mapped as attack detection (Table 3) and not counted among the six.

Where the six studies agree Despite their heterogeneity, all six anchor the distinction in the *physical-process context* rather than in network telemetry alone: physics-guided features [26], digital-twin reference deviations [37], PMU disturbance signatures [41], communication-channel behavior of an energy-aware smart home [36], metering behavior [40], and co-simulated cyber-physical dynamics [44]. This convergence is itself a finding: the attack-versus-fault question is treated by all who attempt it as a *cross-layer inference problem* that cannot be resolved from IT-layer indicators alone, which retrospectively validates the review's requirement (criterion I2 and concept Group 4) that eligible studies be grounded in a system or application context.

Two methodological families The architectural choices split into two families. The first is *model- or reference-based consistency checking*: a trusted representation of how the physical process should behave (plant physics in [26], a digital twin in [37], high-fidelity co-simulation ground truth in [44]) and the attribution decision follows from whether an observed deviation is consistent with plausible physical behavior (suggesting a fault) or not (suggesting manipulation). The second is *label-driven data classification*: supervised or comparative learners trained on examples of both classes (supervised attack/fault classes in [36], federated disturbance-versus-attack discrimination in [41], comparative unsupervised scoring in [40]). The trade-off is clear: the first family needs an accurate process model or twin, which is costly and asset-specific; the second needs labeled examples of attacks and faults, which are scarce and of uncertain realism in public datasets. No study in the corpus combines the two systematically.

The evaluation gap on the distinction task itself The most consequential observation from Table 9 concerns performance reporting: none of the six studies reports the attack-versus-fault decision as a per-class confusion matrix in a directly comparable form—that is, how often a fault is mistaken for an attack and vice versa, under stated class balances. Reported figures concern detection of a specific attack type [37, 44], are stated qualitatively [41], or are not given as a single comparable value at all [26, 36, 40]. Consequently, although the distinction has been *demonstrated to be feasible* in narrow settings, the field currently offers no quan-

Table 9 Focused comparison of the six included studies that explicitly attempt the cyber-attack versus safety-fault distinction

Study	Context and data	Features	Architecture	Distinction mechanism	Reported performance (distinction task)	Key shortcoming
Yan 2019 [26]	Gas-turbine cyber-physical system; high-fidelity simulation and real plant data	Physics-guided/domain features from process measurements	Extreme Learning Machine and stacked denoising autoencoder variants, physics-aided ML	Physical-layer detection: physics-derived residual features (a fault-diagnosis methodology repurposed for security) flag malicious activity as deviations inconsistent with the asset's learned physical behavior, so attribution rests on physical-process consistency	No single comparable headline value reported (NR in Table 4)	Single-asset (turbine) scope; energy-sector transfer and grid-scale validity untested
Tertytchny 2019 [36]	Energy-aware smart home (EASH); simulated and real-time testbed data	Labeled behavior of the EASH internal communication channel (attacks and faults on component communication)	Supervised ML classifiers over component states	Explicit supervised classification of <i>attack</i> versus <i>fault</i> classes; the attack–fault differentiation problem is formally defined	Classification accuracy discussed; no comparable per-class headline value in reviewed text (NR)	Smart-home scale; label availability assumed; far from DSO/TSO operating conditions
Shen 2023 [37]	PV-based hybrid microgrid; IEEE test system simulation with AMI data	Deviation statistics between digital-twin reference behavior and measured production (binary samples for SHT)	Digital twin plus fast hybrid ML with sequential hypothesis testing (SHT)	An SHT statistic is compared against two thresholds to decide between three options (per the primary study); deviations inconsistent with the digital-twin reference are classified as data-integrity attacks, so the attack-versus-fault distinction is made indirectly via reference consistency	Detection rate and configurable false-positive/false-negative rates reported for data-integrity-attack detection; no attack-vs-fault confusion breakdown (NR)	Distinction is indirect (via twin consistency); attack surface limited to data-integrity attacks

Table 9 continued

Study	Context and data	Features	Architecture	Distinction mechanism	Reported performance (distinction task)	Key shortcoming
Johannesen 2022 [40]	Real smart-meter consumption data, urban and rural grid scenarios	Consumption/load patterns from metering data	Unsupervised detectors: kNN, Isolation Forest, Local Outlier Factor	Comparative behavior of unsupervised scores across anomaly types and grid scenarios is used to characterize anomaly origin	Comparative anomaly scores across detectors; no single comparable metric (NR)	Anomaly classes not labeled as attack vs. fault; unsupervised scores not directly comparable across methods
Husnoo 2023 [41]	Power-system disturbance vs. cyber-attack discrimination; public PMU-based ICS datasets	PMU measurement representations compressed by a deep autoencoder	Federated learning (FedDiSC) with DP-SIGNSGD for privacy and communication efficiency	Discrimination via the unsupervised deep autoencoder's reconstruction error (above-threshold MSE flags a cyber-attack rather than a disturbance), learned across federated clients	“Superior” detection accuracy claimed relative to baselines; no comparable numeric value extractable in reviewed text (NR)	Depends on labeled public datasets; realism of attack labels and cross-utility generalization unverified
Hyder 2023 [44]	Distribution-grid cyber-physical co-simulation; high-fidelity synthetic dataset	Statistically diverse cyber and physical signals incl. simulated attacks and faults	Comparative ML anomaly-detection pipeline trained on high-fidelity data	Both attack and fault events are generated and learned, enabling class separation on diverse test data	“High, scenario-dependent” accuracy/recall across the diverse event classes; no per-class confusion values	Self-described preliminary study; results not broken down per anomaly class in comparable form

NR carries the meaning defined for Table 4: no single comparable headline value is stated in the reviewed text for the listed task

titative basis for claiming how reliably it can be made, and no shared benchmark on which two of these approaches could be compared. This sharpens the metric-inconsistency finding reported alongside Table 4 into a specific research need: a public benchmark with labeled attack *and* fault classes, standard splits, and confusion-matrix-level reporting for the distinction task. This need is independently corroborated by the field's own broad reviews: Ye et al. [22], surveying photovoltaic cyber–physical security, explicitly list *discriminating between a cyberattack and a fault* among the open challenges and report a hardware-in-the-loop waveform study that separates normal operation, data-integrity attacks, replay attacks, and physical faults at the point of common coupling—yet, being a review, it surveys this capability without consolidating it on a shared, labelled benchmark for the distinction task, pairing it with a qualitative success-rate impact metric rather than class-confusion reporting.

Shared shortcomings and what they imply Finally, the six studies share three structural shortfalls. (i) *Scale*: the evaluation environments are a smart home [36], a single turbine [26], microgrid or feeder-level simulations [37, 44], metering data slices [40], and public PMU datasets [41]; none operates at the scale or operational tempo of a DSO or TSO control room. (ii) *Data dependence*: each approach presupposes either a trustworthy reference model or labeled examples of both classes, and 4 of the 6 rely fully or partly on simulated data (cf. Table 3, where 20 of the 25 studies—80%—sit in realism tiers T3–T4). (iii) *Operational integration*: none reports a connection to operator workflows, incident-response processes, or sector standards, so even a correct distinction would currently arrive as an unrouted signal (cf. the application-domain blind spots identified in the Application Domains discussion). Together with the small absolute number of attempts, these shortfalls substantiate the

gap-analysis framing of this review: the six studies are proofs of feasibility in constrained settings, not evidence of a mature solution space.

A complementary perspective: why the distinction is intrinsically hard One further study in the corpus does not classify anomalies, and is therefore not counted among the six above, yet it sharpens the motivation for the distinction task from the attacker's side. Santorsola et al. [23] use a reinforcement-learning agent to discover sequences of *individually legitimate* Modbus control commands that drive a simulated DER-rich grid into a safety-critical state in which prioritized loads (e.g., hospitals) are disconnected. The relevance to the present review is that the resulting end state is, by construction, indistinguishable at the message and outcome level from an accidental misoperation or a genuine protective trip: the malicious sequence contains no malformed packet, no injected measurement, and no per-message signature that a conventional detector could flag, yet it yields a fault-like disconnection. This illustrates from the adversary's perspective the same difficulty the six studies confront from the defender's—when an attack can be assembled entirely from valid control actions, the attack-versus-fault decision cannot rest on the legitimacy of individual messages and must instead reason over the physical-process consequences of command *sequences*. It thereby reinforces, rather than weakens, the conclusion that a credible distinction capability requires the consequence-aware, cross-layer reasoning and the class-confusion-level evaluation that the corpus does not yet provide.

4 Synthesis and conclusions

4.1 Overview

This section provides a comprehensive summary of the key findings from our exploration of anomaly detection systems in Distributed Energy Resource (DER)-rich SGs. We have delved into various aspects, including the importance of anomaly detection, the methods and techniques employed, performance metrics for evaluation, challenges and opportunities in deploying Machine Learning (ML) and Explainable Artificial Intelligence (XAI), and the characteristics of effective testbeds for evaluation and validation. Additionally, we present a consolidated Table 4.3 that encapsulates the critical elements discussed across these areas.

4.2 Key findings summary

Structural Patterns from the Taxonomy The added taxonomy reveals several cross-cutting structural patterns that are not immediately visible from a purely theme-by-theme narrative. First, studies focused on the network or com-

munication layer are predominantly cybersecurity-oriented and frequently employ ensemble or deep-learning classifiers tuned for intrusion or traffic-anomaly detection. Second, studies operating at the physical or sensor layer more often combine operational-fault reasoning with cyberattack awareness, especially when PMU, SCADA, relay, or process-state information is involved. Third, only 6 of the 25 reviewed studies (24%) explicitly attempt cyber-physical distinction as an end goal rather than treating one anomaly class in isolation, while several others address both domains only partially. This reinforces one of the review's central conclusions: the literature is rich in anomaly detection, but still comparatively sparse in architectures designed specifically to distinguish malicious cyber events from non-malicious physical faults under realistic DER-rich operating conditions.

To consolidate the findings across the six research-question groups in one place, Table 10 summarizes, for each group, the dominant pattern observed in the corpus, the key gap, and the implication for the central distinction objective. The table is intended as the single cross-study synthesis entry point requested for this review, complementing the per-dimension detail of Tables 4–9.

4.2.1 Significance of anomaly detection in smart grids

Anomaly detection is crucial for maintaining the reliability, security, and efficiency of S, especially as they become increasingly complex with the integration of DERs. Effective anomaly detection systems help in identifying and mitigating irregularities arising from operational issues or malicious activities, ensuring uninterrupted power supply and safeguarding against cyber threats.

4.2.2 Methods and techniques for anomaly detection

Various methods and techniques are employed for anomaly detection in S:

- **Machine Learning Algorithms:** Utilization of supervised and unsupervised learning methods such as Support Vector Machines (SVM), Neural Networks, Autoencoders, Isolation Forests, and Reinforcement Learning for pattern recognition and anomaly identification.
- **Explainable AI (XAI):** Implementation of XAI techniques to enhance the interpretability of ML models, enabling operators to understand and trust the system's decisions.
- **Blockchain and Fog Computing:** Adoption of blockchain for secure data transactions and fog computing for decentralized data processing to improve security and efficiency.

Table 10 Cross-group synthesis: dominant patterns, key gaps, and implications for the cyber-attack versus safety-fault distinction, consolidated across the six research-question groups

Group (RQ)	Dominant pattern in the 25 studies	Key gap	Implication for attack-vs-fault distinction
G1: Data utilization and sources	Operational sensor/telemetry data dominate (grid parameters, PMU, AMI, process signals)	Datasets that jointly label security incidents <i>and</i> safety faults are rare	Without mixed labeled data, supervised distinction cannot be trained or validated at scale
G2: AI/ML methodologies	Supervised, ensemble, and autoencoder-style detectors prevail; XAI in only 2/25 studies (8%); graph, Petri-net, and Bayesian formalisms absent	Relational and causality-aware methods are missing; physics enters mostly as a data generator, not a model constraint	Vectorized detectors flag deviations but cannot reason about propagation paths or root cause—the essence of attribution
G3: Application domains and contexts	Real-time monitoring is universal (25/25); predictive maintenance in 14/25; TSO/DSO roles mostly inferred; ITSM/incident-response integration essentially absent	Pipelines stop at the alarm; no designed hand-off to maintenance vs. security processes	Detection latency is optimized while the post-alarm routing decision—the distinction—is left to humans without machine support
G4: Evaluation strategies and metrics	Accuracy is the most quantified metric; 19/25 studies report no comparable headline value and 2 only qualitative ones (Table 4); FPR rarely measured	No shared benchmark; no class-confusion-level reporting anywhere in the corpus	Feasibility of the distinction is demonstrated but its reliability is unquantified and methods are mutually incomparable
G5: Challenges and opportunities	Explainability, data quality, system integration, and standardization recur as challenges; digital twins and federated learning as opportunities	Trust, regulatory, and interoperability barriers are described but rarely addressed by design	Operator trust in an automated attack-vs-fault verdict requires explainability and standards that the corpus does not yet provide
G6: Testbeds for evaluation	Simulation-dominated validation: 80% of studies in realism tiers T3–T4; HIL and real operational data in a minority	Few testbeds inject <i>both</i> realistic attacks and realistic faults under DER-rich dynamics	The distinction has never been demonstrated under operationally realistic conditions; testbeds generating both classes (e.g., [44]) point the way

4.2.3 Performance metrics for evaluation

Effective evaluation of anomaly detection systems relies on a combination of performance metrics:

- **Accuracy:** Measures the overall correctness of the system's predictions.
- **Precision and Recall:** Assess the system's ability to correctly identify true anomalies and minimize false alarms.
- **F1 Score:** The harmonic mean of precision and recall, providing a balance between the two.

- **False Positive Rate (FPR) and False Negative Rate (FNR):** Evaluate the system's error rates in terms of incorrectly identified anomalies.
- **Area Under the ROC Curve (ROC-AUC):** Summarizes the system's performance across all classification thresholds.

4.2.4 Challenges and opportunities in deploying ML and XAI

Technical Challenges

- **Explainability:** The "black box" nature of advanced ML models can hinder trust and acceptance among operators who require clear justifications for anomaly detection results.
- **Model Accuracy and Reliability:** Achieving high detection accuracy is essential to minimize false positives and negatives, which can lead to alarm fatigue or undetected anomalies.
- **Data Quality and Availability:** Issues with data sparsity, noise, and inconsistencies can affect model performance, emphasizing the need for high-quality, real-time data processing.

Operational Challenges

- **Integration with Existing Systems:** Ensuring compatibility and interoperability between new ML/XAI systems and existing grid components is complex due to the heterogeneous nature of S.
- **Scalability:** Managing the exponential increase in data volume and network complexity as more DERs are integrated into the grid.
- **Cybersecurity Risks:** New vulnerabilities may be introduced with ML and XAI, requiring robust security measures to protect against adversarial attacks and data manipulation.

Assurance Problems

- **Balancing Quantitative and Qualitative Approaches:** While quantitative metrics are essential, qualitative aspects such as interpretability and user trust are equally important for effective deployment.
- **Contextual Considerations:** ML and XAI solutions must be adaptable to diverse operational environments, requiring customization to address specific needs and regulatory requirements.

Standardization Challenges

- **Gaps in Standards:** Lack of unified protocols and guidelines leads to fragmented approaches, hindering cross-system compatibility and benchmarking.
- **Need for Future Guidelines:** Ongoing efforts by organizations like IEEE and NIST aim to establish standards that facilitate interoperability, security, and performance benchmarking.

Opportunities for Improvement

- **Technological Advancements:** Emerging technologies such as edge computing, federated learning, and advanced

algorithms offer promising solutions to technical challenges.

- **Collaborative Efforts:** Partnerships among industry, academia, and government can accelerate innovation through shared resources, datasets, and expertise.
- **Regulatory and Policy Support:** Supportive policies and incentives can facilitate the adoption of ML and XAI while addressing security and privacy concerns.

4.2.5 Characteristics of effective testbeds

Effective testbeds for evaluating and validating anomaly detection systems are characterized by:

Purposeful Design

- **Hardware-in-the-Loop (HIL) Testing:** Integrates real hardware components with simulated environments to assess system responses under realistic conditions.
- **Designing for Scalability:** Simulates grid expansion and DER integration to test system performance as the network grows.
- **Load Distribution:** Assesses how systems manage varying demand levels and the impact on anomaly detection accuracy.

Detailed and Modular Architecture

- **Comprehensive Integration:** Encompasses all relevant components, including physical hardware, software simulations, and communication networks.
- **Modular Design:** Allows for flexibility in testing individual components or subsystems independently or in combination.

Realistic Scenario Simulation

- **Systems Under Attack:** Simulates cyber-physical threats like MITM, DDoS, FDI, and SCADA attacks to evaluate system resilience.
- **Defense Mechanisms:** Implements response strategies to test how systems detect and mitigate attacks.

Connectivity and Operational Integration

- **Integration or Isolation of SOC and Control Rooms:** Decisions on integrating or isolating these components affect the realism and applicability of test results to operational settings.

Use of Physical Equipment and Real Data

- **Incorporation of Physical Hardware:** Uses actual devices to introduce real-world variabilities and test hardware interactions.
- **Use of Realistic Data:** Employs high-fidelity or real operational data to ensure accurate performance evaluation.

4.2.6 What the reviewed studies do not address

Alongside the findings above, several important gaps should be noted to avoid overgeneralisation. First, no reviewed study validates its anomaly-detection design against established security standards such as IEC 62351, IEC 62443, or NERC CIP, meaning that compliance readiness remains untested. Second, the explicit distinction of cybersecurity events from physical safety faults is attempted by only 6 of the 25 studies (24%), and none of these operates at full transmission- or distribution-grid scale. Third, XAI techniques are used by only 2 studies (8%), leaving the interpretability of detection outputs largely unexamined. Fourth, 80% of the evidence base relies on simulated or synthetic data (T3–T4 tiers), so reported accuracy values may not generalise to operational DER-rich environments with measurement noise, sensor drift, topology reconfiguration, and DER intermittency. Fifth, no study benchmarks supervised against unsupervised methods on the same dataset and attack scenario, preventing direct methodological comparison. These gaps collectively define the priority research agenda for advancing anomaly detection from experimental promise to operational deployment in DER-rich Smart Grids.

4.3 Consolidated summary table

Building on the thematic synthesis presented in Discussion for each group, Table 4.3 offers a concise, two-column snapshot of the main points discussed above. This table provides a unified reference that juxtaposes motivations, methodological choices, evaluation criteria, prevailing challenges, emerging opportunities, and defining test-bed attributes. This structured panorama offers researchers and practitioners a rapid cross-reference when designing or benchmarking anomaly-detection solutions in DER-rich S (Table 11).

4.4 Conclusion

The evidence base assembled in this review—25 studies published between 2017 and 2024—supports the following conclusions with appropriate confidence:

Supported by strong evidence (>15 studies): Sensor data (voltage, current, frequency) and supervised ML methods (Random Forest, SVM) dominate the reviewed literature. Real-time monitoring is the most common operational con-

Table 11 Summary of key elements in anomaly detection for DER-rich Smart Grids

Aspect	Key elements
Importance of Anomaly Detection	Ensures reliability, security, and efficiency; critical with increased DER integration
Methods and Techniques	Machine Learning algorithms (SVM, Neural Networks, Autoencoders, Isolation Forests, Reinforcement Learning); Explainable AI (XAI); Blockchain; Fog Computing
Performance Metrics	Accuracy; Precision; Recall; F1 Score; False Positive Rate (FPR); False Negative Rate (FNR); Area Under the ROC Curve (ROC-AUC)
Technical Challenges	Explainability; Model Accuracy and Reliability; Data Quality and Availability
Operational Challenges	Integration with Existing Systems; Scalability; Cybersecurity Risks
Assurance Problems	Balancing Quantitative and Qualitative Approaches; Contextual Considerations
Standardization Challenges	Gaps in Standards; Need for Future Guidelines
Opportunities for Improvement	Technological Advancements; Collaborative Efforts; Regulatory and Policy Support
Characteristics of Effective Testbeds	Purposeful Design (HIL Testing, Scalability, Load Distribution); Detailed and Modular Architecture; Realistic Scenario Simulation; Connectivity and Operational Integration; Use of Physical Equipment and Real Data

text. Performance metrics are inconsistently reported across studies.

Supported by moderate evidence (6–14 studies): Deep learning architectures (CNN-LSTM, DAE) offer competitive detection performance for complex anomaly patterns. Federated and distributed approaches improve scalability. Testbed realism is a significant and largely unaddressed challenge.

Supported by limited evidence (<6 studies, requiring caution in generalisation): The explicit distinction of cybersecurity events from physical safety faults using ML is technically feasible (demonstrated in [26, 36, 37, 40, 41, 44]), but remains largely unexplored at scale, in real-world DER-rich environments, and with validated compliance against security standards.

These conclusions should be interpreted in light of the limitations described in Sect. 2.13, particularly the dominance of simulated evaluation environments and the small number of studies directly addressing the cyber-physical distinction challenge. Addressing the technical and operational challenges through technological innovation, collaborative efforts, and supportive regulatory frameworks remains crucial. Effective testbeds play a vital role in simulating real-world conditions, enabling rigorous evaluation and validation of these systems before deployment.

Future Directions

Future research should focus on developing more explainable and robust ML models tailored for anomaly detection in SG. There is a significant need for standardized protocols and datasets that reflect the unique characteristics and constraints of DER-rich environments. Collaborative efforts between industry, academia, and government agencies can facilitate the creation of such datasets and the establishment of comprehensive standards and guidelines.

Future work should also prioritize **hybrid physics–data-driven detection architectures**. In particular, models that integrate power-system constraints, protection logic, and topology-aware reasoning directly into the learning process may offer a more principled route to distinguishing cybersecurity attacks from safety faults than purely black-box classifiers. This includes physics-informed neural approaches, grey-box system identification pipelines, and cyber–physical digital-twin workflows that use physics not only to simulate data but also to constrain inference and explanation. Such architectures are especially promising for DER-rich environments, where operational variability is high and purely data-driven correlations may be misleading when transferred across systems.

In line with these objectives, we are interested in building a testbed that enables the generation of safety and security datasets adapted to the constraints of DER-rich Smart Grids. This testbed would incorporate the essential characteristics identified for effective evaluation and validation environments, such as:

- **Hardware-in-the-Loop (HIL) Testing:** Integrating real hardware components to assess system responses under realistic conditions.
- **Realistic Scenario Simulation:** Including various cyber-physical threats and operational anomalies to test system resilience.
- **Modular and Scalable Architecture:** Allowing flexibility in testing different configurations and accommodating future expansions.
- **Use of Realistic Data:** Employing high-fidelity or real operational data to ensure accurate performance evaluation.

By developing such a testbed, we aim to provide a platform for generating comprehensive safety and security datasets. These datasets will be crucial for training and evaluating anomaly detection systems, ensuring they are effective in detecting irregularities specific to DER-rich S. This initiative will contribute to filling the existing gaps in standardized datasets and support the advancement of more resilient and secure Smart Grid infrastructures.

The synergy between information and communications technology engineering (with its focus on robust pipeline development, advanced data analytics, and scalable storage) and Electrical Engineering (with its foundational knowledge of DER integration, protective relay schemes, and fault characterization) is crucial. Both fields must collaborate to build an end-to-end anomaly detection platform that effectively addresses the unique technical and operational challenges of modern S.

Final Remarks

By leveraging these insights and focusing on future developments, stakeholders can enhance the resilience and efficiency of modern Smart Grids. The collaborative creation of advanced testbeds and datasets will play a pivotal role in this progress. Through continued innovation and cooperation, we can contribute to a more secure and sustainable energy future, effectively addressing the challenges posed by the increasing integration of Distributed Energy Resources.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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